

Real Analysis I

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Homework 1: Week 1 due Aug 30

Sun 21 Aug 2022 23:38

1. 4.H. If $\xi \in \mathbf{R}$ is irrational and $r \in \mathbf{R}, r \neq 0$, is rational, show that $r + \xi$ and $r\xi$ are irrational.

Proof. We use, without proving, the fact that the rational numbers $\mathbb{Q}$ forms a field. In particular, it is closed under multiplication (and division, by existence of multiplicative inverse) and addition (and subtraction, by existence of additive inverse).

We prove the contrapositive. Suppose $r + \xi$ is not irrational. Then $r + \xi$ is rational. Because r is rational, $\xi = (r + \xi) - r$ is rational, which means it's not irrational.

The same proof for $r\xi$. Modify the previous proof by exchanging addition and multiplication (subtraction and division). $\square$

2. 5.A. If $a, b \in \mathbf{R}$ and $a^2 + b^2 = 0$, show that $a = b = 0$.

Proof. Suppose for the sake of contradiction that either $a \neq 0$ or $b \neq 0$. By symmetry, it suffices to only prove the case for $a \neq 0$. By Theorem 5.5(a), $a^2 > 0$.

Now we want to prove $b^2 \geq 0$. If $b = 0$, it follows from Theorem 4.5(a) that $b^2 = 0 \cdot 0 = 0$. If $b \neq 0$, then $b^2 > 0$ by Theorem 5.5(a).

In the first case, $a^2 + b^2 = a^2 + 0 = a^2 > 0$. In the second case, $a^2 + b^2 > 0 + 0 = 0$ by Theorem 5.6(b). In both cases $a^2 + b^2 > 0$, but it is assumed that $a^2 + b^2 = 0$. $\square$

3. 5.C. If $a > -1, a \in \mathbf{R}$, show that $(1 + a)^n \geq 1 + na$ for all $n \in \mathbf{N}$. This inequality is called Bernoulli's Inequality. (Hint: use mathematical induction.)

Proof. Induct on n .

Base case: We need to show $(1 + a)^0 \geq 1 + 0 \cdot a$. This simplifies to $1 \geq 1$ which is trivial.

Induction step: We need to show $(1 + a)^{n+1} \geq 1 + (n + 1)a$.

$$(1 + a)^{n+1} = (1 + a)^n(1 + a)$$

By induction hypothesis, $(1 + a)^n \geq 1 + na$. Apply Theorem 5.6(c) (note that $1 + a > 0$):

$$\begin{aligned} &\geq (1 + na)(1 + a) \\ &= 1 + a + na + na^2 \end{aligned}$$

Since $a^2 \geq 0$:

$$\geq 1 + (n + 1)a.$$

□

4. 5.O. If x, y, z belong to $\mathbf{R}$, then $x \leq y \leq z$ **or** $z \leq y \leq x$ if and only if $|x - y| + |y - z| = |x - z|$.

Proof. We split the goal into two subgoals, which together imply the goal:

$$\begin{cases} z \leq y \leq x \Leftrightarrow |x - y| + |y - z| = |x - z| \wedge x \geq z \\ x \leq y \leq z \Leftrightarrow |x - y| + |y - z| = |x - z| \wedge x \leq z \end{cases}.$$

By symmetry, we only need to prove the first subgoal.

$\Rightarrow$: Since $y \geq z$, $y - z \geq 0$, hence $|y - z| = y - z$. Similarly $|x - y| = x - y$ and $|x - z| = x - z$. The goal now simplifies to $(x - y) + (y - z) = x - z$, which follows by simplification.

$\Leftarrow$: It suffices to show that both $y > x$ and $y < z$ cannot be true. By symmetry, we only show that $y > x$ is false.

Suppose for the sake of contradiction that $y > x$. By similar reasoning, $|x - y| + |y - z| = (y - x) + (y - z)$. By assumption, $x \geq z$, so $|x - z| = x - z$. Since $y > x$, $y - z > x - z$. Therefore, $|x - y| + |y - z| > (y - x) + (x - z) = (y - z) + |x - z| > |x - z|$, a contradiction.

□

5. 6.D. Give an example of a set of irrational numbers that has a rational supremum.

Proof. Consider the set

$$\{-\sqrt{2}/n : n \in \mathbb{N}\}.$$

□

6. 6.G. If S is a bounded set in $\mathbf{R}$ and if S_0 is a nonempty subset of S , then show that

$$\inf S \leq \inf S_0 \leq \sup S_0 \leq \sup S.$$

Sometimes it is more convenient to express this in another way. Let $D \neq \emptyset$ and let $f : D \rightarrow \mathbf{R}$ have bounded range. If D_0 is a non-empty subset of D , then

$$\inf\{f(x) : x \in D\} \leq \inf\{f(x) : x \in D_0\} \leq \sup\{f(x) : x \in D_0\} \leq \sup\{f(x) : x \in D\}.$$

Proof. The second inequality follows from definition, so we only need to prove the first and third inequalities. By symmetry, we only need to prove the first inequality.

It suffices to show that $\inf S$ is a lower bound of S_0 . By definition, for all $x \in S$, $x \geq \inf S$. Take $s \in S_0$. Since $S_0 \subset S$, $s \in S$. Therefore, $s \geq \inf S$. $\square$

7. 6.H. Let X and Y be non-empty sets and let $f : X \times Y \rightarrow \mathbf{R}$ have bounded range in $\mathbf{R}$. Let

$$f_1(x) = \sup\{f(x, y) : y \in Y\}, \quad f_2(y) = \sup\{f(x, y) : x \in X\}.$$

Establish the Principle of Iterated Suprema:

$$\begin{aligned} \sup\{f(x, y) : x \in X, y \in Y\} &= \sup\{f_1(x) : x \in X\} \\ &= \sup\{f_2(y) : y \in Y\}. \end{aligned}$$

We sometimes express this in symbols by:

$$\sup_{x, y} f(x, y) = \sup_x \sup_y f(x, y) = \sup_y \sup_x f(x, y).$$

Proof. By symmetry, it suffices to show

$$\sup\{f(x, y) : x \in X, y \in Y\} = \sup\{f_1(x) : x \in X\}.$$

We first show $\sup\{f(x, y) : x \in X, y \in Y\} \leq \sup\{f_1(x) : x \in X\}$. To show this, we need to show that $\sup\{f_1(x) : x \in X\}$ is an upper bound of $\{f(x, y) : x \in X, y \in Y\}$. For any $(x_0, y_0) \in X \times Y$, $f_1(x_0) = \sup\{f(x_0, y) : y \in Y\}$. Now $f(x_0, y_0) \in \{f(x_0, y) : y \in Y\}$, therefore $\sup\{f(x_0, y) : y \in Y\} \geq f(x_0, y_0)$.

Next we show $\sup\{f(x, y) : x \in X, y \in Y\} \geq \sup\{f_1(x) : x \in X\}$. To show this, we need to show that $\sup\{f(x, y) : x \in X, y \in Y\}$ is an upper bound of $\{f_1(x) : x \in X\}$. Take $x_0 \in X$. Now $f_1(x_0) = \sup\{f(x_0, y) : y \in Y\} \leq \sup\{f(x, y) : x, y \in X \times Y\}$, where the inequality comes from exercise 6G. $\square$

Homework 2: Week 2 Due Sep 9

Mon 29 Aug 2022 14:24

I use the following lemma to prove Exercise 3.J.

Lemma 1 (pigeonhole principle). Let Y be a finite set and X be finite with cardinality greater than $|Y|$. Then for any function $f : X \rightarrow Y$, there exists $x_1, x_2 \in X$ such that $f(x_1) = f(x_2)$.

Proof. Denote A_y by the set $\{x \in X : f(x) = y\}$. We must have $|A_y| < 2$ for all $y \in Y$, otherwise there will be more than one element from X mapped to the same element in Y . Hence $0 \leq |A_y| \leq 1$. We have

$$\left| \bigcup_{y \in Y} A_y \right| = \sum_{y \in Y} |A_y| \leq \sum_{y \in Y} 1 = |Y|.$$

The first equality comes from the fact that A_y are all disjoint and finite (since elements mapped to different targets must be different).

But for any $x \in X$, it is mapped to some element $y = f(x)$. In other words, $x \in A_{f(x)} \subset \bigcup_{y \in Y} A_y$. In the other direction, for any $x \in \bigcup_{y \in Y} A_y$, it is some element in X by definition. We have shown that $X = \bigcup_{y \in Y} A_y$ and therefore $|X| = \left| \bigcup_{y \in Y} A_y \right| \leq |Y|$ which is a contradiction to the assumption that $|X| > |Y|$. Therefore there must exist some y such that $|A_y| \geq 2$, which is exactly what we want to prove. $\square$

I prove this exercise in extra, because I used it in Exercise 7.I.

7.I. Show that the collection of "right hand" end points in $\mathbf{F}$ is denumerable. Show that if all these end points are deleted from $\mathbf{F}$, then what remains can be put onto one-one correspondence with all of $[0, 1)$. Conclude that the set $\mathbf{F}$ is not countable.

Proof. Each "right hand" end point is in the form $\frac{k+1}{3^n}$ where $k, n \in \mathbb{N}$. Therefore it is a rational number, and the set of rational numbers $\mathbb{Q}$ is denumerable. Because the set of right hand end points is a subset of rational numbers, it is denumerable.

Now we construct cascaded segmentation of the half-open interval $[0, 1)$. For each nonnegative integer n and $0 \leq k \leq n$, define A_{nk} to be

$$A_{nk} = \left[\frac{k}{2^n}, \frac{k+1}{2^n} \right).$$

We use induction to verify the following properties:

1. for all n there are exactly 2^n disjoint sets of the form A_{nk} .
2. (disjoint segmentation) for all n and all distinct k, l ,

$$A_{nk} \cap A_{nl} = \emptyset.$$

3. (filling) for all n ,

$$\bigcup_{0 \leq k \leq n} A_{nk} = [0, 1).$$

4. (branch split) for any n, k , $A_{n+1, l} \subset A_{nk}$ if and only if

$$2k \leq l \leq 2k + 1.$$

5. (cascading property) whenever $x \in A_{nk}$ and $x \in A_{ml}$ where $n \leq m$, we have $A_{ml} \subset A_{nk}$.

Now for each n , we can construct a bijection $G : \bigcup_{0 \leq k \leq n} A_{nk} \rightarrow F_n^*$, where F_n is as defined in the textbook, and F_n^* is F_n with right endpoints removed. There are exactly 2^n disjoint half-open intervals A_{nk} . There are also exactly 2^n disjoint half-open intervals in F_n^* . We map the k -th interval A_{nk} to the k -th

interval in F_n^* . Note that any two half-open intervals on $\mathbb{R}$ have a bijection between them.

Now we show that there exists a bijection $\mathcal{F} : [0, 1) \rightarrow F^*$ where F^* is the Cantor set with right endpoints removed. Take any $x \in [0, 1)$. For any $n \in \mathbb{N}$, let A_{nk} be the unique half-open interval in the construction that contains x . Now G maps A_{nk} to some half-interval B_n in the construction of F_n^* . Now $B_{n+1} \subset B_n$ and $B_n \neq \emptyset$. Denote b_n as the left endpoint of B_n . Now $y = \sup_{n \in \mathbb{N}} b_n$ is the unique element in $\bigcap_{n \in \mathbb{N}} B_n \subset F^*$.

Such map is injective: if $x_1 < x_2$ are distinct point in $[0, 1)$, then there exists n such that $\frac{1}{2^n} \leq \frac{1}{n} < x_2 - x_1$. Then $x_1 \in A_{nk}$ and $x_2 \in A_{nl}$ where $n \neq l$, hence they are mapped to disjoint half-open intervals B_1 and B_2 in F_n^* . Now $\mathcal{F}(x_1) \in B_1$ and $\mathcal{F}(x_2) \in B_2$, so they must not be equal.

Such map is surjective: if $y \in F^*$, then for any $n \in \mathbb{N}$, $y \in B_n \subset F_n^*$ where B_n is a half-open interval in the construction of F_n^* . $\mathcal{F}^{-1}$ then maps B_n to some A_n . Since $B_{n+1} \subset B_n$, $A_{n+1} \subset A_n$. Let a_n be the endpoint of A_n , and b_n the left endpoint of B_n . Now $x = \sup_{n \in \mathbb{N}} a_n$ is the unique element in $\bigcap_{n \in \mathbb{N}} A_n$. By definition $\mathcal{F}(x) = \sup_{n \in \mathbb{N}} b_n$. This is the unique element in $\bigcap_{n \in \mathbb{N}} B_n$. Since $y \in \bigcap_{n \in \mathbb{N}} B_n$, $\mathcal{F}(x) = y$. $\square$

6.M, 7.F, 7.G, 7.J, 7.L, 3.F, 3.J

1. 6.M. Modify the argument given in Theorem 6.8 to show that if $a > 0$, then the number

$$b = \sup \{y \in \mathbf{R} : 0 \leq y, \quad y^2 \leq a\}$$

exists and has the property that $b^2 = a$. This number will be denoted by $\sqrt{a}$ or $a^{1/2}$, and is called the positive square root of a .

Proof. Denote the set by S . We first show that the supremum exists. Clearly $0 \in S$, so the set is nonempty. Next we show that the set is bounded from above. We split into cases where $x \leq 1$ and $x > 1$. When $x \leq 1$, we claim that the set is bounded above by 1. Suppose not, then there exists some $s \in S$ such that $s > 1$, hence $s^2 > 1$ but we have $s^2 \leq a \leq 1$, a contradiction. When $x > 1$, we claim that the set is bounded above by a . Suppose not, then there exists some $s \in S$ such that $s > a$, hence $s^2 > a^2$ but we have $s^2 \leq a < a^2$ (the second inequality is due to $a > 1$), a contradiction. In both cases the set is bounded above. By supremum property of $\mathbb{R}$, $b = \sup S$ exists.

Now we show $b^2 = a$. Suppose not, then by trichotomy property of real ordering, we have either $b^2 < a$ or $b^2 > a$. When $b^2 < a$, we take $n \in \mathbb{N}$, such that $\frac{1}{n} < \frac{a-b^2}{2b+1}$ (using Corollary 6.7 (b) of Archimedean Property). In this case

$$(b + \frac{1}{n})^2 = b^2 + \frac{1}{n} \left(2b + \frac{1}{n}\right) \leq b^2 + \frac{1}{n} (2b + 1) < b^2 + (a - b^2) = a.$$

This means $b + \frac{1}{n} \in S$, contrary to the fact that b is an upper bound of S .

When $b^2 > a$, we take $n \in \mathbb{N}$ such that $\frac{1}{n} < \frac{b^2 - a}{2b}$ (such n exists by Corollary 6.7(b) of the Archimedean Property). Since $b = \sup S$, there exists an $s_0 \in S$ such that $b - \frac{1}{n} < s_0$. But this implies that

$$a < b^2 - 2\frac{b}{n} < b^2 - 2\frac{b}{n} + \frac{1}{n^2} = \left(n - \frac{1}{n}\right)^2 < s_0^2.$$

But this means $s_0^2 > a$, contrary to the fact that $s_0 \in S$.

We have excluded the possibilities of $b^2 < a$ and $b^2 > a$, so the only possibility left is $b^2 = a$. $\square$

2. 7.F. Let $J_n = (0, 1/n)$ for $n \in \mathbb{N}$. Show that this sequence of intervals is nested, but that there is no common point.

Proof. We first show the sequence is nested. Fix some $n \in \mathbb{N}$. It suffices to show that $J_{n+1} \subset J_n$. Take some $x \in J_{n+1}$, then by definition we have $0 < x < \frac{1}{n+1}$. But since $\frac{1}{n+1} < \frac{1}{n}$, we have $0 < x < \frac{1}{n}$, hence by definition again we have $x \in J_n$.

Now we show there is no common point. Fix any $x \in J_1 = (0, 1)$. By Corollary 6.7(b), there exists some $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < x$. In other words, $x \notin J_n$. Therefore, $\bigcap_{n \in \mathbb{N}} J_n = \emptyset$ since each element is not in at least one of the sets. $\square$

3. 7.G. If $I_n = [a_n, b_n]$, $n \in \mathbb{N}$, is a nested sequence of closed cells, show that

$$a_1 \leq a_2 \leq \cdots \leq a_n \leq \cdots \leq b_m \leq \cdots \leq b_2 \leq b_1.$$

If we put $\xi = \sup \{a_n : n \in \mathbb{N}\}$ and $\eta = \inf \{b_m : m \in \mathbb{N}\}$, show that $[\xi, \eta] = \bigcap_{n \in \mathbb{N}} I_n$.

Proof. Take $n, m \in \mathbb{N}$ where $n \leq m$. By assumption $I_m \subset I_n$. We need to show that $a_n \leq a_m$ and $b_m \leq b_n$. By definition of closed intervals, for all $x \in \mathbb{R}$, $x \in I_n \Leftrightarrow a_n \leq x \leq b_n$ and $x \in I_m \Leftrightarrow a_m \leq x \leq b_m$. By set inclusion, $x \in I_m \Rightarrow x \in I_n$. Therefore

$$a_m \leq x \leq b_m \Rightarrow a_n \leq x \leq b_n.$$

Suppose that $a_m < a_n$. Then there exists some x such that $a_m < x < a_n$, such that $a_n \leq x \leq b_n$ is not true, therefore the implication is not true for the specific x , a contradiction to the universally quantified statement. By trichotomy of real order, we have $a_m \geq a_n$. By the same reasoning we have $b_m \leq b_n$.

Correction

We still need to verify that $a_n \leq b_m$ for all $n, m \in \mathbb{N}$. To do this, take $N = \max(n, m)$. Now

$$a_n \leq a_N \leq b_N \leq b_m.$$

Now we show that $[\xi, \eta] = \bigcap_{n \in \mathbb{N}} I_n$. We first show that $[\xi, \eta] \subset \bigcap_{n \in \mathbb{N}} I_n$. Take $x \in [\xi, \eta]$, and take an arbitrary $n \in \mathbb{N}$. It suffices to show that $x \in$

I_n . Since ξ is a supremum, we must have $\xi \geq a_n$. Therefore $x \geq \xi \geq a_n$. Similarly we have $x \leq b_n$. By definition of I_n , we have $x \in I_n$.

Next we show that $[\xi, \eta] \supset \bigcap_{n \in \mathbb{N}} I_n$. Take $x \in \bigcap_{n \in \mathbb{N}} I_n$. Then $x \in I_n$ for all $n \in \mathbb{N}$. Suppose that $x < \xi$. Then there exists some $m \in \mathbb{N}$ such that $x < a_m \leq \xi$. Clearly $x \in I_m$ is not true, a contradiction. Therefore we must have $x \geq \xi$. By the same reasoning we have $x \leq \eta$, so $x \in [\xi, \eta]$. $\square$

4. 7.J. Every open interval (a, b) which contains a point of $\mathbf{F}$ also contains an entire "middle third" set which belongs to $\mathcal{C}(\mathbf{F})$. Hence $\mathbf{F}$ does not contain any non-void open interval.

Proof. Fix some interval $(a, b) \subset \mathbb{R}$ and some point $x \in \mathbf{F} \cap (a, b)$. Since $a < x < b$, we have $b - x > 0$ and $x - a > 0$. We can choose some $n \in \mathbb{N}$ such that

$$3^n > \max\left(\frac{1}{x-a}, \frac{1}{b-x}\right).$$

To justify this, note that by Bernoulli's Inequality

$$3^n = (1 + 2)^n \geq 1 + 2n > n$$

and by Archimedean property we can choose n such that $n > \max\left(\frac{1}{x-a}, \frac{1}{b-x}\right)$.

By Corollary 6.7(c) of the Archimedean property, there exists $k \in \mathbb{N}$ such that $k \leq 3^n x < k + 1$. Since $3^n x - 3^n a > 1$, we have $k > 3^n x - 1 > 3^n a$. Since $3^n b - 3^n x > 1$, we have $k + 1 \leq 3^n x + 1 < 3^n b$. Therefore $\left[\frac{k}{3^n}, \frac{k+1}{3^n}\right] \subset (a, b)$.

By definition and construction, $x \in \left[\frac{k}{3^n}, \frac{k+1}{3^n}\right] \subset F_n$. F_{n+1} takes the middle third from this interval, which is $\left[\frac{3k}{3^{n+1}}, \frac{3k+1}{3^{n+1}}\right] \cup \left[\frac{3k+2}{3^{n+1}}, \frac{3k+3}{3^{n+1}}\right] \subset F_{n+1}$. The middle-third $M = \left(\frac{3k+1}{3^{n+1}}, \frac{3k+2}{3^{n+1}}\right)$ satisfies the properties $M \subset F_n$ and $M \subset (a, b)$. $\square$

5. 7.L. Show that $\mathbf{F}$ is not the union of a countable collection of closed intervals.

Proof. Suppose $\mathbf{F} = \bigcup_{n \in \mathbb{N}} I_n$ where each $I_n = [a_n, b_n]$. Suppose there exists some $m \in \mathbb{N}$ such that $a_m < b_m$. Then the open interval (a_m, b_m) is nonempty, and has a nonempty intersection with $\mathbf{F}$. By Exercise 7.J, the interval (a_m, b_m) contains a "middle-third" set. This contradicts the fact that $[a_m, b_m] \subset \mathbf{F}$.

Therefore $a_n \geq b_n$ for all $n \in \mathbb{N}$. If $a_n > b_n$ then the interval is empty, so we have $I_n = \{x_n\} \subset \mathbf{F}$ for at most countable number of I_n , and the rest of them are empty. This is to say that the set $\mathbf{F}$ is at most countable. But we have concluded in Exercise 7.I that $\mathbf{F}$ is not countable. $\square$

6. 3.F. Use the fact that every infinite set has a denumerable subset to show that every infinite set can be put into one-one correspondence with a proper subset of itself.

Proof. Consider an infinite set X and a denumerable subset $A \subset X$. De-

numerate A by $a_1, a_2, \dots, a_i, \dots$, and define the map

$$F : X \longrightarrow X \setminus \{a_1\}$$

$$x \longmapsto F(x) = \begin{cases} x & x \notin A \\ a_{i+1} & x = a_i \in A \end{cases}.$$

F is a function because for every x either $x \notin A$, or $x \in A$, in this case there must exist $i \in \mathbb{N}$ such that $x = a_i$. F is injective since for any $x_1 \neq x_2$, either they both $\notin A$, in this case $F(x_1) = x_1 \neq x_2 = F(x_2)$; or they are both $\in A$, in this case $F(x_1) = F(a_i) = a_{i+1} \neq a_{j+1} = F(a_j) = F(x_2)$; or one of them is in A and the other is not, in this case the claim is obvious.

F is surjective because for any $x \in X \setminus A$, its preimage is itself; and for any $x = a_i \in A \setminus \{a_1\}$ where $i \neq 1$, its preimage is $a_{i-1} \in A \setminus \{a_1\}$. $\square$

7. 3.J. Prove that $\mathbb{N}$ cannot be put into one-one correspondence with any initial segment of $\mathbb{N}$.

Proof. Suppose for the sake of contradiction that there exists some $n \in \mathbb{N}$ such that there is a bijection $f : \mathbb{N} \rightarrow S_n$. Hence in particular f is injective. Then by restricting the domain, $f|_{S_{n+1}} : S_{n+1} \rightarrow S_n$ is injective. However this cannot be true since by the pigeonhole principle there must exist two elements in S_{n+1} that are mapped to the same element in S_n . $\square$

Homework 3: Week 3 due Sep 16

Fri 09 Sep 2022 16:50

8.F, 8.G, 8.M, 8.Q, 9.G, 9.H, 9.L, 9.Q(a,b,c,d)

1. 8.F. If $x = (x_1, x_2, \dots, x_p) \in \mathbf{R}^p$, define $\|x\|_1$ by

$$\|x\|_1 = |x_1| + |x_2| + \dots + |x_p|.$$

Prove that $x \mapsto \|x\|_1$ is a norm on $\mathbf{R}^p$.

Proof. (a) Take any $x = (x_1, \dots, x_p) \in \mathbb{R}^p$. For any $1 \leq i \leq p$, $|x_i| \geq 0$ by property of absolute value. Now

$$\|x\|_1 = |x_1| + \dots + |x_p| \geq 0 + \dots + 0 = 0.$$

If $x = 0$, then $x_1 = \dots = x_p = 0$, hence $|x_1| = |x_2| = \dots = |x_p| = 0$. Hence $\|x\|_1 = 0$. Conversely, if $\|x\|_1 = 0$, then any $|x_i|$ must be 0 which means $x = 0$. Otherwise, the sum $|x_1| + \dots + |x_p|$ will be greater than zero, a contradiction.

- (b) $ax = (ax_1, ax_2, \dots, ax_p)$ by definition. Now

$$\begin{aligned} \|ax\|_1 &= |ax_1| + \dots + |ax_p| \\ &= |a||x_1| + \dots + |a||x_p| \\ &= |a|(|x_1| + \dots + |x_p|) \\ &= |a|\|x\|_1. \end{aligned}$$

- (c) Recall the triangle inequality for absolute value: for any $a, b \in \mathbb{R}$,

$$|a + b| \leq |a| + |b|.$$

Now

$$\begin{aligned}
 \|x + y\|_1 &= |x_1 + y_1| + |x_2 + y_2| + \dots + |x_p + y_p| \\
 &\leq (|x_1| + |y_1|) + (|x_2| + |y_2|) + \dots + (|x_p| + |y_p|) \\
 &= (|x_1| + \dots + |x_p|) + (|y_1| + \dots + |y_p|) \\
 &= \|x\|_1 + \|y\|_1.
 \end{aligned}$$

We have verified all the norm properties in the definition, so $\|\cdot\|_1$ is a norm. $\square$

2. 8.G. If $x = (x_1, x_2, \dots, x_p) \in \mathbf{R}^p$, define $\|x\|_\infty$ by

$$\|x\|_\infty = \sup\{|x_1|, |x_2|, \dots, |x_p|\}.$$

Prove that $x \mapsto \|x\|_\infty$ is a norm on $\mathbf{R}^p$.

Proof. Consider $x = (x_1, x_2, \dots, x_p) \in \mathbf{R}^p$, and we verify each property in the definition.

- Since for all $1 \leq i \leq p$, $|x_i| \geq 0$, $\|x\|_\infty = \sup_i |x_i| \geq 0$.

If $x = 0$, then for all i , $x_i = 0$, hence $|x_i| = 0$. Therefore $\|x\|_\infty = \sup_i |x_i| = \sup_i 0 = 0$. Now suppose $\|x\|_\infty = \sup_i |x_i| = 0$. If $x \neq 0$, then there exists some $1 < j \leq p$ such that $x_j \neq 0$, hence $|x_j| > 0$. Since 0 is a supremum, we have $0 \geq |x_j| > 0$, a contradiction.

- Consider

$$\begin{aligned}
 \|ax\|_\infty &= \sup\{|ax_1|, |ax_2|, \dots, |ax_p|\} \\
 &= \sup\{|a||x_1|, \dots, |a||x_p|\}.
 \end{aligned}$$

If $|a| > 0$, we have $|x_i| \geq |x_j| \Leftrightarrow |a||x_i| \geq |a||x_j|$. Therefore

$$\begin{aligned}
 \|ax\|_\infty &= |a| \sup\{|x_1|, \dots, |x_p|\} \\
 &= |a| \|x\|.
 \end{aligned}$$

If $|a| = 0$, then $\|ax\|_\infty = \|0\| = 0 = |0| \|x\| = |a| \|x\|$.

- We need to show

$$\sup\{|x_1 + y_1|, \dots, |x_p + y_p|\} \leq \sup\{|x_1|, \dots, |x_p|\} + \sup\{|y_1|, \dots, |y_p|\}.$$

We first show that

$$\sup\{|x_1 + y_1|, \dots, |x_p + y_p|\} \leq \sup\{|x_1| + |y_1|, \dots, |x_p| + |y_p|\}$$

which comes from triangle inequality for absolute values. Since p is finite, there exists some $1 \leq i \leq p$ such that $\sup\{|x_1 + y_1|, \dots, |x_p + y_p|\} = |x_i + y_i|$. But $|x_i + y_i| \leq |x_i| + |y_i| \leq \sup\{|x_1| + |y_1|, \dots, |x_p| + |y_p|\}$.

Next we verify that

$$\sup\{|x_1| + |y_1|, \dots, |x_p| + |y_p|\} \leq \sup\{|x_1|, \dots, |x_p|\} + \sup\{|y_1|, \dots, |y_p|\}.$$

To demonstrate this, first notice that there exists $1 \leq i \leq p$ such that $\sup\{|x_1| + |y_1|, \dots, |x_p| + |y_p|\} = |x_i| + |y_i|$. Now $|x_i| \leq \sup\{|x_1|, \dots, |x_p|\}$ and $|y_i| \leq \sup\{|y_1|, \dots, |y_p|\}$, so the claim follows.

□

3. 8.M. If x, y belong to $\mathbf{R}^p$, is it true that

$$|x \cdot y| \leq \|x\|_1 \|y\|_1 \quad \text{and} \quad |x \cdot y| \leq \|x\|_\infty \|y\|_\infty?$$

Proof. To verify the first statement, consider

$$\begin{aligned} |x \cdot y| &= |x_1 y_1 + \dots + x_p y_p| \\ &\leq |x_1| |y_1| + \dots + |x_p| |y_p| \\ &\leq \sum_{i,j=1,\dots,p} |x_i| |y_j| \\ &= (|x_1| + \dots + |x_p|) (|y_1| + \dots + |y_p|) \\ &= \|x\|_1 \|y\|_1. \end{aligned}$$

The second statement is not true in general.

Consider $x, y \in \mathbb{R}^3$ where

$$x = (1, 1, 1), y = (1, 1, 1).$$

Now $x \cdot y = 1 \times 1 + 1 \times 1 + 1 \times 1 = 3$ and $\|x\|_\infty \|y\|_\infty = 1 \times 1 = 1$. We have a counterexample for the statement now. □

4. 8.Q. A subset K of $\mathbf{R}^p$ is said to be convex if, whenever x, y belong to K and t is a real number such that $0 \leq t \leq 1$, then the point

$$(1-t)x + ty = x + t(y-x)$$

also belongs to K . Interpret this condition geometrically and show that the subsets

$$K_1 = \{x \in \mathbf{R}^2 : \|x\| \leq 1\}$$

$$K_2 = \{(\xi, \eta) \in \mathbf{R}^2 : 0 < \xi < \eta\}$$

$$K_3 = \{(\xi, \eta) \in \mathbf{R}^2 : 0 \leq \eta \leq \xi \leq 1\}$$

are convex but that the subset

$$K_4 = \{x \in \mathbf{R}^2 : \|x\| = 1\}$$

is not convex.

Proof. The geometrical interpretation is: any line segment between elements of K is itself entirely contained in K .

(a) Take $x, y \in K_1$. And let

$$z = (1-t)x + ty$$

where t is arbitrary fixed real number between 0 and 1 inclusive. Now

$$\begin{aligned} \|z\| &= \|(1-t)x + ty\| \\ &\leq \|(1-t)x\| + \|ty\| \\ &= (1-t)\|x\| + t\|y\|. \end{aligned}$$

Without loss of generality we may assume $\|x\| \leq \|y\|$. Hence

$$\begin{aligned}\|z\| &= (1-t)\|x\| + t\|y\| \\ &\leq (1-t)\|y\| + t\|y\| \\ &= \|y\| \\ &\leq 1.\end{aligned}$$

Therefore $z \in \mathbf{K}_1$, as desired.

- (b) Take $x, y \in \mathbf{K}^2$ and $t \in [0, 1]$. Now define $z = (1-t)x + ty = (x_1 + y_1, x_2 + y_2)$. Since $x_1 > 0, x_2 > 0$, we have $z_1 = (1-t)x_1 + ty_1 > 0 + 0 = 0$. Also,

$$\begin{aligned}z_1 &= (1-t)x_1 + ty_1 \\ &< (1-t)x_1 + ty_2 && \text{since } y_1 < y_2 \\ &< (1-t)x_2 + ty_2 && \text{since } x_1 < x_2 \\ &= z_2.\end{aligned}$$

Therefore $z \in \mathbf{K}_2$.

- (c) Take $x, y \in \mathbf{K}^3$ and $t \in [0, 1]$. Now define $z = (1-t)x + ty = (x_1 + y_1, x_2 + y_2)$. Since $x_1 \geq 0, x_2 \geq 0$, we have $z_1 = (1-t)x_1 + ty_1 \geq 0 + 0 = 0$. Since $x_1 \leq 1, x_2 \leq 1$, we have $z_1 = (1-t)x_1 + ty_1 \leq (1-t) \times 1 + t \times 1 = 1$. Also,

$$\begin{aligned}z_1 &= (1-t)x_1 + ty_1 \\ &\geq (1-t)x_1 + ty_2 && \text{since } y_1 \geq y_2 \\ &\geq (1-t)x_2 + ty_2 && \text{since } x_1 \geq x_2 \\ &= z_2.\end{aligned}$$

Therefore $z \in \mathbf{K}_3$.

- (d) It suffices to provide a counterexample. Take $x = (1, 0)$ and $y = (-1, 0)$. Now they are both elements of $\mathbf{K}_4$ since $\|x\| = \|1\| = 1$ and $\|y\| = \|-1\| = 1$. Now take $t = \frac{1}{2}$ and note that $(1-t)x + ty = \frac{1}{2}x + \frac{1}{2}y = \frac{1}{2}(1-1) = 0$. However $\|0\| = 0$ which does not satisfy the condition for being an element of $\mathbf{K}_4$. Hence $\mathbf{K}_4$ is not convex. $\square$

5. 9.G. Show that a subset of $\mathbf{R}^p$ is open if and only if it is the union of a countable collection of open balls. (Hint: the set of all points in $\mathbf{R}^p$ all of whose coordinates are rational numbers is countable.)

Proof. We already know $\mathbb{Q}^p$ is countable. By property of countability, $\mathbb{Q}^p \times \mathbb{Q}$ is countable. Now consider the family of open balls with rational center and rational radius:

$$\mathcal{O} = \{B_x(r) : (x, r) \in \mathbb{Q}^p \times \mathbb{Q}\}.$$

Thus $\mathcal{O}$ is countable. Let X be an arbitrary open subset of $\mathbb{R}^p$. Take a subcollection $\mathcal{O}_X$ of $\mathcal{O}$:

$$\mathcal{O}_X = \{A \in \mathcal{O} : A \subset X\}.$$

Note that $\mathcal{O}_X$ is at most countable. We claim that $\cup \mathcal{O}_X = X$. First we show that $\cup \mathcal{O}_X \subset X$. Take $x \in \cup \mathcal{O}_X$. So $x \in A$ for some $a \in \mathcal{O}_X$. Since $A \subset X$, $x \in X$.

Now we show $X \subset \cup \mathcal{O}_X$. Take any $x \in X$. Since X is open, there exists some positive real a such that $B_a(x) \subset X$. Using the lemma below, we can choose two rational numbers x' and a' , where $a' > 0$, such that

$$x \in B_{a'}(x') \subset B_a(x).$$

Thus

$$x \in B_{a'}(x') \subset X.$$

Hence

$$x \in B_{a'}(x') \in \mathcal{O}_X.$$

Therefore

$$x \in \cup \mathcal{O}_X$$

which concludes the proof. Now we supplement the lemma and proof.

Lemma 2 (Density of rational points in Euclidean space). In Euclidean space $\mathbb{R}^p$, every open ball contains a point whose coordinates are all rational numbers, i.e., $q \in \mathbb{Q}^p$.

Proof. Since we are working with euclidean space, we assume that the norm is euclidean norm. By doing this we can avoid certain topological arguments.

Let $B_r(x)$ be the open ball. Define

$$d = r/\sqrt{p}.$$

Let B be the open box

$$B = \{y \in \mathbb{R}^p : \forall i \in \{1, \dots, p\}, |y_i - x_i| < d\}.$$

Now take any element $y \in B$. Its distance to x is

$$\begin{aligned} \|y - x\| &= \sqrt{\sum_{i=1}^p (y_i - x_i)^2} \\ &\leq \sqrt{\sum_{i=1}^p d^2} \\ &= \sqrt{pd^2} \\ &= r. \end{aligned}$$

Hence $B \subset B_r(x)$. Now for each $1 \leq i \leq p$, we use density of rational in $\mathbb{R}$ to choose some rational number q_i such that

$$x_i - d < q_i < x_i + d.$$

Such choice is always possible because d is positive. Now we define

$$q = (q_1, \dots, q_p).$$

Note that $q \in B$ by construction. Hence $q \in B_r(x)$.

□

Use the last lemma to choose some rational $x' \in B_{\frac{a}{3}}(x)$. Then use density of rational in $\mathbb{R}$ to choose rational a' such that

$$\frac{a}{3} < a' < \frac{2a}{3}.$$

Now $x \in B_{a'}(x')$ since

$$\|x - x'\| < \frac{a}{3} < a'.$$

$B_{a'}(x') \subset B_a(x)$ since for any $y \in B_{a'}(x')$,

$$\|y - x'\| \leq \|y - a'\| + \|a' - x'\| < a' + \frac{a}{3} < \frac{2a}{3} + \frac{a}{3} = a.$$

□

6. 9.H. Every open subset of $\mathbb{R}^p$ is the union of a countable collection of closed sets.

Proof. We already know $\mathbb{Q}^p$ is countable. By property of countability, $\mathbb{Q}^p \times \mathbb{Q}$ is countable. Now consider the family of open balls with rational center and rational radius:

$$\mathcal{O} = \{B_x(r) : (x, r) \in \mathbb{Q}^p \times \mathbb{Q}\}.$$

Thus $\mathcal{O}$ is countable. Let X be an arbitrary open subset of $\mathbb{R}^p$. Take a subcollection $\mathcal{O}_X$ of $\mathcal{O}$:

$$\mathcal{O}_X = \{A \in \mathcal{O} : A \subset X\}.$$

Note that $\mathcal{O}_X$ is at most countable. Now define $\overline{\mathcal{O}_X}$ as

$$\overline{\mathcal{O}_X} = \{\overline{B_x(\frac{r}{2})} : B_x(r) \in \mathcal{O}_X\}.$$

We claim that $\cup \overline{\mathcal{O}_X} = \cup_{A \in \overline{\mathcal{O}_X}} A = X$. First we show that $\cup \overline{\mathcal{O}_X} \subset X$. Take $x \in \cup \overline{\mathcal{O}_X}$. So $x \in A$ for some $A = \overline{B_x(\frac{r}{2})} \in \overline{\mathcal{O}_X}$. Hence $B_x(r) \in \mathcal{O}_X$ by definition of $\overline{\mathcal{O}_X}$. Also, $A \subset B_x(r) \in \mathcal{O}_X$ since all the points y in A satisfies $\|y - x\| \leq \frac{r}{2} < r$. Hence $A \subset X$, and in particular $x \in X$.

Now we show $X \subset \cup \overline{\mathcal{O}_X}$. Take any $x \in X$. Since X is open, there exists some positive real a such that $B_a(x) \subset X$. Now we can choose two rational numbers x' and a' , just like in the last exercise, where $a' > 0$, such that

$$x \in B_{a'}(x') \subset B_a(x).$$

Now consider the closed ball

$$A_x = \overline{B_{\frac{a'}{2}}(x')}.$$

Since both a' and x' are rational numbers, $B_{a'}(x') \in \mathcal{O}_X$. Hence $A_x \in \overline{\mathcal{O}_X}$. Hence $x \in \cup \overline{\mathcal{O}_X}$. □

7. 9.L. If A is any subset of $\mathbf{R}^p$, let A^- denote the intersection of all closed sets containing A ; the set A^- is called the closure of A . Note that A^- is a closed set; prove that it is the smallest closed set containing A . Prove that

$$\begin{aligned} A &\subseteq A^-, \quad (A^-)^- = A^- \\ (A \cup B)^- &= A^- \cup B^-, \quad \emptyset^- = \emptyset. \end{aligned}$$

Give an example to show that $(A \cap B)^- = A^- \cap B^-$ may not hold.

Proof. Define

$$\mathcal{C}(A) = \{B \subset \mathbf{R}^p : A \subset B, B \text{ closed}\}.$$

Now A^- is defined as

$$A^- = \cap \mathcal{C}(A).$$

- (a) (smallest) Suppose there is a set B such that B is closed, $A \subset B$, and $B \subsetneq A^-$. Now there exists $x \in A^- \setminus B$, and in particular $x \in A$. But since B is a closed set containing A , it's one of the intersecting sets defining A^- , i.e., $B \in \mathcal{C}(A)$. Therefore, $A^- \subset B$, and hence $x \in B$, a contradiction.
- (b) ($A \subset A^-$) Proof: Take any element $x \in A$. Take any $B \in \mathcal{C}(A)$. By definition $A \subset B$, so $x \in B$. Since this is true for all $B \in \mathcal{C}(A)$, $x \in \cup \mathcal{C}(A) = A^-$.
- (c) ($(A^-)^- = A^-$) Proof: $(A^-)^- \supset A^-$ by the previous property. We prove $(A^-)^- \subset A^-$. It suffices to show $\left((A^-)^-\right)^c \supset (A^-)^c$. First of all, note that $(A^-)^c$ is open since A^- is closed. For any $x \in (A^-)^c$, there exists a open neighborhood B such that $x \in B$ and $B \subset (A^-)^c$. Now the set B^c is closed, and $A^- \subset B^c$. Hence $B^c \in \mathcal{C}(A^-)$. Therefore $x \notin \cap \mathcal{C}(A^-)$ since $x \notin B^c$. In other words, $x \in \left((A^-)^-\right)^c$.
- (d) ($(A \cup B)^- = A^- \cup B^-$) We first prove $(A \cup B)^- \supset A^- \cup B^-$. Let $x \in A^- \cup B^-$. Without loss of generality, assume $x \in A^-$. Take arbitrary $X \in \mathcal{C}(A \cup B)$. To prove the result we need to show $x \in X$. Now $A \cup B \subset X$ by definition, so in particular $A \subset X$. This means that $X \in \mathcal{C}(A)$. Hence $x \in X$.

Now we prove $(A \cup B)^- \subset A^- \cup B^-$. It suffices to show $\left((A \cup B)^-\right)^c \supset (A^-)^c \cap (B^-)^c$. Take $x \in (A^-)^c \cap (B^-)^c$. Now since $(A^-)^c$ and $(B^-)^c$ are open, we can choose open neighbors X_A and X_B such that

$$x \in X_A \subset (A^-)^c, x \in X_B \subset (B^-)^c.$$

So $x \in X_A \cap X_B =: X$. Consider the set X^c now. X^c is closed because X is open. Because $A \subset A^- \subset X_A^c$ and $B \subset B^- \subset X_B^c$, we have $A \cup B \subset X_A^c \cup X_B^c = (X_A \cap X_B)^c = X^c$. Now X^c is closed and contains $A \cup B$, so $X^c \in \mathcal{C}(A \cup B)$. But $x \notin X^c$, so $x \notin \cup \mathcal{C}(A \cup B) = (A \cup B)^-$, which means $x \in \left((A \cup B)^-\right)^c$.

- (e) The empty set $\emptyset$ is closed and contains $\emptyset$, so $\emptyset \in \mathcal{C}(\emptyset)$. Hence $\emptyset^- = \emptyset$.

□

8. 9.Q. Interpret the concepts introduced in this section for the Cantor set F of Definition 7.4. In particular: (a) Show that F is closed in $\mathbf{R}$. (b) There are no interior points in F . (c) There are no non-empty open sets contained in F . (d) Every point of F is a boundary point. (e) The set F cannot be expressed as the union of a countable collection of closed intervals. (f) The complement of F can be expressed as the union of a countable collection of open intervals.

Proof. In the proof we let the "whole space" be $F_0 = [0, 1]$. Thus, for any subset $A \subset F_0$, define $A^c = F_0 \setminus A$.

- (a) Each F_n is closed (since each F_n is a finite union of closed intervals), so $F = \bigcap F_n$ is closed (since it is a countable intersection of closed intervals).
- (b) Any interval (a, b) that contains at least one element from F also contains a full "middle third" set, so in particular $(a, b) \not\subset F$. For a point $x \in F$ to be an interior point we need to find open ball A such that $x \in A \subset F$, but no such A exists. Hence F has no interior points.
- (c) Any interval (a, b) that contains at least one element from F also contains a full "middle third" set, so in particular $(a, b) \not\subset F$. This is true for any interval, so no non-empty open set is contained in F .
- (d) Any interval (a, b) that contains at least one element $x \in F$ also contains a full "middle third" set, so in particular there exists some $y \in F^c \cap (a, b)$. Hence for any $x \in F$, and any neighborhood (a, b) containing x , the neighborhood also has a non-empty intersection with F^c . This shows that x is a boundary point.
- (e) Already Proved in Exercise 7.L. The full proof is copied below.

Proof of 7.L. Suppose $\mathbf{F} = \bigcup_{n \in \mathbb{N}} I_n$ where each $I_n = [a_n, b_n]$. Suppose there exists some $m \in \mathbb{N}$ such that $a_m < b_m$. Then the open interval (a_m, b_m) is nonempty, and has a nonempty intersection with $\mathbf{F}$. By Exercise 7.J, the interval (a_m, b_m) contains a "middle-third" set. This contradicts the fact that $[a_n, b_n] \subset \mathbf{F}$.

Therefore $a_n \geq b_n$ for all $n \in \mathbb{N}$. If $a_n > b_n$ then the interval is empty, so we have $I_n = \{x_n\} \subset \mathbf{F}$ for at most countable number of I_n , and the rest of them are empty. This is to say that the set $\mathbf{F}$ is at most countable. But we have concluded in Exercise 7.I that $\mathbf{F}$ is not countable. □

- (f) The complement of F is exactly the union of the "middle third" sets, while there are countably many "middle third" sets. To put precisely,

$$F^c = \left(\bigcap_{n=1}^{\infty} F_n \right)^c = \bigcup_{n=1}^{\infty} F_n^c.$$

Note that each F_n^c can be expressed as union 3^n open intervals. Hence there are countably many middle third sets.

□

Homework 4

Sun 18 Sep 2022 12:54

10.C, 10.D, 10.F, 11.A, 11.C, 11.G, 11.K

1. 10.C. A point x is a cluster point of a set $A \subseteq \mathbf{R}^p$ if and only if every neighborhood of x contains infinitely many points of A .

Proof. In the class we have shown

Proposition 1. x is a cluster point of E if and only if $B_\delta(x) \cap E$ is infinite for any $\delta > 0$.

Proof. If $B_\delta(x) \cap E$ is infinite, then $(B_\delta(x) \setminus \{x\}) \cap E$ is infinite, hence nonempty.

Suppose x is a cluster point, but $B_\delta(x) \cap E$ is finite. This is to say that $(B_\delta(x) \setminus \{x\}) \cap E$ is finite. Let $(B_\delta(x) \setminus \{x\}) \cap E = \{y_1, y_2, \dots, y_N\}$. Then

$$r = \min\{\|x - y_1\|, \|x - y_2\|, \dots, \|x - y_N\|\}$$

is positive.

Now: $(B_r(x) \setminus \{x\}) \cap E = \emptyset$, since $y_i \notin B_r(x)$. This is a contradiction. □

It remains to verify that the two statements are equivalent:

- (a) $B_\delta(x) \cap A$ is infinite for any $\delta > 0$
- (b) every neighborhood of x contains infinitely many points of A

From (a) to (b): Let N be some neighborhood of x . By definition x is an interior point of N . Hence there exists $\delta > 0$ such that $B_\delta(x) \subset N$, so that $B_\delta(x) \cap A \subset N \cap A$. By (a), $B_\delta(x) \cap A$ is infinite, so $N \cap A$ is infinite.

From (b) to (a): Let $\delta > 0$. Note that $B_{\frac{\delta}{2}}(x) \subset B_\delta(x)$, so x is an interior point of $B_\delta(x)$. In other words, $B_\delta(x)$ is a neighborhood of x . By (b), $B_\delta(x) \cap A$ is infinite. □

2. 10.D. Let $A = \{1/n : n \in \mathbf{N}\}$. Show that every point of A is a boundary point in $\mathbf{R}$, but that 0 is the only cluster point of A in $\mathbf{R}$.

Proof. We first prove the

Claim. For any positive $\delta \in \mathbb{R}$, the set $A \setminus B_\delta(0)$ is finite.

Proof of claim. By Archimedean property, there exists $N \in \mathbb{N}$ such that $\frac{1}{N} < \delta$. But now for all $n \in \mathbb{N}$ where $n \geq N$, we have

$$\frac{1}{n} \leq \frac{1}{N} < \delta.$$

Hence $\frac{1}{n} \in B_\delta(0)$ for all $n \geq N$. Now

$$\begin{aligned} A \setminus B_\delta(0) &= \left(\left\{ \frac{1}{n} : n \in \mathbb{N}, n < N \right\} \cup \left\{ \frac{1}{n} : n \in \mathbb{N}, n \geq N \right\} \right) \setminus B_\delta(0) \\ &= \left(\left\{ \frac{1}{n} : n \in \mathbb{N}, n < N \right\} \setminus B_\delta(0) \right) \cup \left(\left\{ \frac{1}{n} : n \in \mathbb{N}, n \geq N \right\} \setminus B_\delta(0) \right) \\ &= \left(\left\{ \frac{1}{n} : n \in \mathbb{N}, n < N \right\} \setminus B_\delta(0) \right) \cup \emptyset \\ &= \left(\left\{ \frac{1}{n} : n \in \mathbb{N}, n < N \right\} \setminus B_\delta(0) \right) \\ &\subset \left\{ \frac{1}{n} : n \in \mathbb{N}, n < N \right\}. \end{aligned}$$

Hence $A \setminus B_\delta(0)$ is finite. $\square$

Now we continue the proof. Take any $x = \frac{1}{n} \in A$. Let $d = \left\lfloor \frac{1}{2n} \right\rfloor$. Now $A \cap B_d(x) \subset A \setminus B_d(0)$. To verify, note that for any $a \in A$,

$$|a - 0| \geq |x - 0| - |a - x| = \frac{1}{n} - |a - x| > \frac{1}{n} - \frac{1}{2n} = \frac{1}{2n} = d.$$

Hence $a \notin B_d(0)$, meaning $a \in A \setminus B_d(0)$.

By the claim above, $A \cap B_d(x)$ is finite. But $B_d(x)$ is an open interval, which is infinite. Hence in particular $B_d(x) \setminus A$ is nonempty. We can therefore choose some $y \in B_d(x) \setminus A$. Now $x \in B_d(x) \cap A$ and $y \in B_d(x) \cap A^c$.

Let $\delta > 0$ be arbitrary. If $\delta \geq d$, then $y \in B_d(x) \cap A^c \subset B_\delta(x) \cap A^c$, and $x \in B_\delta(x) \cap A$. If $\delta < d$, then $B_\delta(x) \subset B_d(x)$. Hence $A \cap B_\delta(x)$ is finite. By the same reasoning we can find $y' \in B_\delta(x) \cap A^c$. In both cases $B_\delta(x)$ has nonempty intersection with both A and A^c , so x is a boundary point of A .

Now we prove that 0 is a cluster point. Take any $\delta > 0$, and by the claim above, $A \setminus B_\delta(0)$ is finite. Since A is infinite, $A \cap B_\delta(0)$ is infinite, and we are done.

Finally we verify that 0 is the only cluster point. Take any $x \in \mathbb{R}$ such that $x \neq 0$. Now let $\delta = \frac{|x|}{2}$. Now we have $B_\delta(x) \subset A \setminus B_\delta(0)$. Hence $B_\delta(x) \cap A$ is finite, which means that x is not a boundary point. $\square$

3. 10.F. Let A, B be subsets of $\mathbf{R}^p$ and let x be a cluster point of $A \cup B$ in $\mathbf{R}^p$. Prove that x is either a cluster point of A or of B .

Proof. We prove the contrapositive. Assume x is neither a cluster point of A nor a cluster point of B . Then its intersection with A and B are both finite, i.e. there exists positive $\delta, \delta' \in \mathbb{R}$, and $N, M \in \mathbb{N}$ such that

$$|B_\delta(x) \cap A| = N < \infty, \quad |B_{\delta'}(x) \cap B| = M < \infty.$$

Without loss of generality, assume $\delta < \delta'$. Now $B_\delta(x) \cap B \subset B_{\delta'}(x) \cap B$, so

$$|B_\delta(x) \cap B| \leq |B_{\delta'}(x) \cap B| = M < \infty.$$

Therefore

$$\begin{aligned}
 |B_\delta(x) \cap (A \cup B)| &= |(B_\delta(x) \cap A) \cup (B_\delta(x) \cap B)| \\
 &\leq |B_\delta(x) \cap A| + |B_\delta(x) \cap B| \\
 &= N + M \\
 &< \infty.
 \end{aligned}$$

This show that x is not a cluster point of $A \cup B$. $\square$

4. 11.A. Show directly from the definition (i.e., without using the Heine-Borel Theorem) that the open ball given by $\{(x, y) : x^2 + y^2 < 1\}$ is not compact in $\mathbb{R}^2$.

Proof. Define the open ball $\{(x, y) : x^2 + y^2 < 1\}$ as B . We give the following open cover

$$\mathcal{C} = \{B_n : n \in \mathbb{N}\}$$

where

$$B_n = B_{1-\frac{1}{n}}((0, 0)).$$

Since open ball is open, $\mathcal{C}$ is an open family. To verify covering, consider $(x, y) \in B$. Hence $x^2 + y^2 < 1$. Take $\delta = 1 - (x^2 + y^2)$. Now there exists $n \in \mathbb{N}$ by Archimedean Property such that

$$n > 2/\delta.$$

Hence

$$\begin{aligned}
 n > \frac{2}{\delta} &\Rightarrow -\frac{2}{n} > -\delta \\
 &\Rightarrow 1 - \frac{2}{n} + \frac{1}{n^2} > 1 - \delta \\
 &\Rightarrow \left(1 - \frac{1}{n}\right)^2 > 1 - \delta \\
 &\Rightarrow \left(1 - \frac{1}{n}\right)^2 > x^2 + y^2.
 \end{aligned}$$

Hence $\|(x, y) - (0, 0)\| = \sqrt{x^2 + y^2} < 1 - \frac{1}{n}$. Hence $(x, y) \in B_n$ for some n . Therefore $\mathcal{C}$ covers B .

Suppose B is compact, then there exists a finite subcover of $\mathcal{C}$, say $\mathcal{C}'$. We can enumerate the subcover:

$$\mathcal{C}' = \{B_{m_1}, B_{m_2}, \dots, B_{m_N}\}.$$

Without loss of generality, assume $m_1 < m_2 < \dots < m_N$. Now $1 - \frac{1}{m_i} \leq 1 - \frac{1}{m_N}$ since $m_N \geq m_i$. By considering the definition,

$$\bigcup_{k=1, \dots, N} B_{m_k} = B_{m_N}.$$

But the point $(1 - \frac{1}{m_N}, 0)$ gives us a contradiction. It is apparently not in B_{m_N} . However, since $\frac{1}{m_N} > 0$, we have $(1 - \frac{1}{m_N}, 0) \in B$. Hence $\mathcal{C}'$ is not a subcover of B , a contradiction. $\square$

5. 11.C. Prove directly that if K is compact in $\mathbf{R}^p$ and $F \subseteq K$ is a closed set, then F is compact in $\mathbf{R}^p$.

Proof. (Using definition of compactness using open cover) Take an open cover of F , say $\mathcal{C}$. Since F is closed, F^c is open. Now consider the open cover $\mathcal{C} \cup \{F^c\}$. Now

$$K \subset \mathbf{R}^p = F \cup F^c \subset (\cup \mathcal{C}) \cup F^c = \cup(\mathcal{C} \cup \{F^c\}).$$

Since K is compact, there exists a subcover $\mathcal{C}'$ of $\mathcal{C} \cup \{F^c\}$. Now $\mathcal{C}' \setminus \{F^c\}$ is a subcover of $\mathcal{C}$ that covers F . $\square$

Proof. (Using definition of compactness using cluster point) Let $A \subset F$ be infinite and bounded. Then $A \subset K$. Because K is compact, there exists $x \in K$ such that x is a cluster point of A . Because F is closed, it contains all its cluster points. Hence $x \in F$. $\square$

6. 11.G. Prove the Cantor Intersection Theorem by selecting a point x_n from F_n and then applying the Bolzano-Weierstrass Theorem 10.6 to the set $\{x_n : n \in \mathbf{N}\}$.

Proof. From each F_n choose $x_n \in F_n$. Now consider the set

$$X = \{x_n : n \in \mathbf{N}\}.$$

Suppose X is finite. We know that each F_n has at least one $x \in F_n \cap X$. Then there exists some $x \in X$ such that $x \in F_n$ infinitely often (by the pigeonhole principle). Thus for any $m \in \mathbf{N}$, one can find $n > m$ such that $x \in F_n \subset F_m$. Hence $x \in F_m$ for all $m \in \mathbf{N}$.

Suppose X is infinite. Then by the Bolzano-Weierstrass Theorem we can find a cluster point $x \in \mathbf{R}^p$ of X . Let $m \in \mathbf{N}$, we want to prove $x \in F_m$. We define the "tail set" as

$$X_m = \{x_n : n \in \mathbf{N}, n \geq m\}.$$

Note that $X_m \subset X \cap F_m$. Since x is a cluster point of X , it is a cluster point of X_m . Since $X_m \subset F_m$, x is a cluster point of F_m . Since F_m is closed, it contains all its cluster points. Hence $x \in F_m$. $\square$

7. 11.K. If K is a compact subset of $\mathbf{R}^p$ and x is a point of $\mathbf{R}^p$, then the set $K_x = \{x + y : y \in K\}$ is also compact. (This set K_x is sometimes called the translation of the set K by x .)

Proof. We first prove a lemma.

Lemma 3 (Translation preserves openness). For any set $K \subset \mathbb{R}^p$ and $x \in \mathbb{R}^p$, K is open if and only if K_x is open.

Proof. By symmetry we only need to show K open implies K_x open. Assume K is open. Take $p \in K_x$. By definition of K_x , there exists $k \in K$ such that $p = k + x$. By openness of K , there exists some $\delta > 0$ such that $B_\delta(k) \subset K$. We claim that $B_\delta(p) \subset K_x$.

To see this, take any point $b \in B_\delta(p)$. Since $b \in K_x$, there exists $a \in K$ such that $b = a + x$. Now $\delta > \|b - p\| = \|(a + x) - (k + x)\| = \|a - k\|$, hence $a \in B_\delta(k)$. Hence $a \in K$. By definition of K_x , $b = a + x \in K_x$. Therefore $B_\delta(p) \subset K_x$ and we have shown that every point of K_x is an interior point, thus K_x open. $\square$

Lemma 4. (Translation is distributive over set union) For any $K \subset \mathbb{R}^p$, $F \subset \mathbb{R}^p$ and $x \in \mathbb{R}^p$,

$$K_x \cup F_x = (K \cup F)_x.$$

Proof. For any $a \in \mathbb{R}^p$,

$$\begin{aligned} a \in K_x \cup F_x & \\ \Leftrightarrow a \in K_x \vee a \in F_x & \\ \Leftrightarrow \exists k \in K, a = k + x \vee \exists p \in F, a = p + x & \\ \Leftrightarrow \exists k \in K \cup F, a = k + x & \\ \Leftrightarrow a \in (K \cup F)_x. & \end{aligned}$$

$\square$

Lemma 5. (Translation preserves set ordering). For any $K \subset \mathbb{R}^p$, $F \subset \mathbb{R}^p$ and $x \in \mathbb{R}^p$, $F \subset K$ if and only if $F_x \subset K_x$.

Proof. For any $k \in \mathbb{R}^p$, one has by definition

$$k \in F \Leftrightarrow k + x \in F_x.$$

Similarly,

$$k \in K \Leftrightarrow k + x \in K_x.$$

Hence for all k ,

$$(k \in F \Rightarrow k \in K) \Leftrightarrow (k + x \in F_x \Rightarrow k + x \in K_x).$$

Therefore

$$F \subset K \Leftrightarrow F_x \subset K_x.$$

$\square$

Take an open cover $\mathcal{C}_x$ of K_x with index set I :

$$\mathcal{C}_x = \{A_i : i \in I\}, \quad \cup \mathcal{C}_x = \bigcup_{i \in I} A_i \supset K_x.$$

Now apply translation $A_i \mapsto (A_i)_{-x} = \{a - x : a \in A_i\}$ to all sets $A_i \in \mathcal{C}'_x$. Define

$$\mathcal{C} = \{(A_i)_{-x} : A_i \in \mathcal{C}_x\}.$$

By distributivity over set union,

$$\cup \mathcal{C} = \bigcup_{i \in I} (A_i)_{-x} = \left(\bigcup_{i \in I} A_i \right)_{-x} = (\cup \mathcal{C}_x)_{-x}.$$

Hence in particular, $\cup \mathcal{C}_x = (\cup \mathcal{C})_x$.

By preservation of set inclusion, since $K_x \subset (\cup \mathcal{C})_x$

$$K \subset \cup \mathcal{C}.$$

By preservation of openness applied to each $A_i \in \mathcal{C}_x$, since $\mathcal{C}_x$ is a open cover, $\mathcal{C}$ is a open cover. Because K is compact, there exists a finite subcover $\mathcal{C}' \subset \mathcal{C}$ of K . More precisely, we have finite $I' \subset I$ and $\mathcal{C}' = \{(A_i)_{-x} \in \mathcal{C} : i \in I'\}$. Now we have

$$\cup \mathcal{C}' = \bigcup_{i \in I'} (A_i)_{-x} \supset K.$$

By preservation of set inclusion,

$$K_x \subset (\cup \mathcal{C}')_x.$$

Define $\mathcal{C}'_x := \{A_i \in \mathcal{C}_x : i \in I'\}$. Now by distributivity over union,

$$(\cup \mathcal{C}')_x = \left(\bigcup_{i \in I'} (A_i)_{-x} \right)_x = \left(\left(\bigcup_{i \in I'} A_i \right)_{-x} \right)_x = \bigcup_{i \in I'} A_i = \cup \mathcal{C}'_x.$$

Hence $K_x \subset \cup \mathcal{C}'_x$. $\mathcal{C}'_x \subset \mathcal{C}_x$ by definition, so $\mathcal{C}'_x$ is a finite subcover of $\mathcal{C}_x$. Then is an open subcover of K_x . $\square$

Proof. (Using cluster point definition of compactness) First establish several lemmas.

Lemma 6. (Translation is a bijection) For any $K \subset \mathbb{R}^p$ and $x \in \mathbb{R}^p$, the mapping

$$\begin{aligned} f_x : K &\longrightarrow K_x \\ k &\longmapsto f(k) = k_x = k + x \end{aligned}$$

is a bijection.

Proof. To verify injectivity, assume there are $a, b \in \mathbb{R}^p$ such that $a_x = b_x$. Hence $a + x = b + x$, and thus $a = b$. To verify surjectivity, consider any member of K_x of the form $k + x$. Then $f_x(k) = k_x = k + x$. $\square$

Lemma 7. (Translation is distributive over set intersection) For any $K \subset \mathbb{R}^p$, $F \subset \mathbb{R}^p$ and $x \in \mathbb{R}^p$,

$$K_x \cap F_x = (K \cap F)_x.$$

Proof. For any $a \in \mathbb{R}^p$,

$$\begin{aligned} a &\in K_x \cap F_x \\ \Leftrightarrow a &\in K_x \wedge a \in F_x \\ \Leftrightarrow \exists k \in K, a &= k + x \wedge \exists p \in F, a = p + x \\ \Leftrightarrow \exists k \in K \cap F, a &= k + x \\ \Leftrightarrow a &\in (K \cap F)_x. \end{aligned}$$

$\square$

Lemma 8. (Translation is distributive over set union) For any $K \subset \mathbb{R}^p$, $F \subset \mathbb{R}^p$ and $x \in \mathbb{R}^p$,

$$K_x \cup F_x = (K \cup F)_x.$$

Proof. For any $a \in \mathbb{R}^p$,

$$\begin{aligned} a &\in K_x \cup F_x \\ \Leftrightarrow a &\in K_x \vee a \in F_x \\ \Leftrightarrow \exists k \in K, a &= k + x \vee \exists p \in F, a = p + x \\ \Leftrightarrow \exists k \in K \cup F, a &= k + x \\ \Leftrightarrow a &\in (K \cup F)_x. \end{aligned}$$

$\square$

Lemma 9. (Translation preserves set ordering). For any $K \subset \mathbb{R}^p$, $F \subset \mathbb{R}^p$ and $x \in \mathbb{R}^p$, $F \subset K$ if and only if $F_x \subset K_x$.

Proof. For any $k \in \mathbb{R}^p$, one has by definition

$$k \in F \Leftrightarrow k + x \in F_x.$$

Similarly,

$$k \in K \Leftrightarrow k + x \in K_x.$$

Hence for all k ,

$$(k \in F \Rightarrow k \in K) \Leftrightarrow (k + x \in F_x \Rightarrow k + x \in K_x).$$

Therefore

$$F \subset K \Leftrightarrow F_x \subset K_x.$$

□

Corollary 1. (Translation preserves set membership). For any $K \subset \mathbb{R}^p$ and $a, x \in \mathbb{R}^p$, $a \in K$ if and only if $a_x \in K_x$.

Proof. Apply the previous lemma with $F = \{a\}$. □

Take A_x to be an infinite subset of K_x . Define A to be

$$A = \{y - x : y \in A_x\}.$$

Thus $A \subset K$ (by preservation of set ordering). Then by compactness of K , there exists a cluster point a of A in K . Thus $a_x := a + x \in K_x$ (by preservation of set membership). For any $\delta \in \mathbb{R}$ such that $\delta > 0$, the intersection $B_\delta(a) \cap A$ is infinite. Hence the intersection $B_\delta(a_x) \cap A_x = (B_\delta(a) \cap A)_x$ is infinite (by distributivity and bijectivity). Hence a_x is a cluster point of A_x . □

Homework 5: due Sep 30

Fri 23 Sep 2022 11:26

12.A, 12.C, 12.I, 14.D, 14.M, 14.N, 14.O, 15.L, 15.N

1. 12.A. If A and B are connected subsets of $\mathbb{R}^p$, give examples to show that $A \cup B$, $A \cap B$, $A \setminus B$ can be either connected or disconnected.

Solution. Example for $A \cup B$: consider $\mathbb{R}$. Take $A = B = [0, 1]$. Now $A \cup B = [0, 1]$ which is connected. Take $A = (0, 1)$ and $B = (1, 2)$. Now $A \cup B = (0, 1) \cup (1, 2)$ which is disconnected.

Example for $A \setminus B$: consider $\mathbb{R}$. Take $A = [0, 1]$ and $B = [\frac{1}{2}, 1]$. Then $A \setminus B = [0, \frac{1}{2})$ which is connected. Take $A = [0, 3]$ and $B = (1, 2)$. Now $A \setminus B = [0, 1] \cup [2, 3]$ which is disconnected.

Example for $A \cap B$: consider $\mathbb{R}$. Take $A = B = [0, 1]$. Now $A \cap B = [0, 1]$ which is connected. Consider $\mathbb{R}^2$. Take $A = \{x \in \mathbb{R}^2 : \|x\| = 1\}$ to be the unit sphere which is connected. Take $B = \{(x_1, 0) : x_1 \in \mathbb{R}\}$ to be the

real axis which is connected. Then $A \cap B = \{(1, 0), (-1, 0)\}$ which is two distinct points, which is disconnected. $\square$

2. 12.C. If $C \subseteq \mathbf{R}^p$ is connected, show that its closure C^- (see Exercise 9.L) is also connected.

Proof. Assume C is connected, and suppose $\overline{C}$ is not connected. Take A, B to be a disconnection of $\overline{C}$. The following properties hold:

- $\overline{C} \subset A \cup B$
- $A \cap \overline{C} \neq \emptyset$
- $B \cap \overline{C} \neq \emptyset$
- $(A \cap \overline{C}) \cap (B \cap \overline{C}) = \emptyset$

Moreover, we cannot have both $A \cap C \neq \emptyset$ and $B \cap C \neq \emptyset$, otherwise C is disconnected, since $C \subset A \cup B$ (since $C \subset \overline{C}$), and $(A \cap C) \cap (B \cap C) = \emptyset$, a contradiction.

Without loss of generality, assume $A \cap C = \emptyset$. Then $C \subset B$ since $C \subset A \cup B$. But $A \cap \overline{C} \neq \emptyset$, so there exists $x \in A \cap \overline{C}$ such that $x \in A$ is a cluster point of C . However, since A is open, there exists a neighborhood N such that $x \in N \subset A$. Thus $N \cap C = \emptyset$, so x is not a cluster point of C , a contradiction. $\square$

3. 12.I. Show that the set

$$S = \left\{ (x, y) \in \mathbf{R}^2 : y = \sin \frac{1}{x}, x \neq 0 \right\} \cup \{(0, y) : -1 \leq y \leq 1\}$$

is connected in $\mathbf{R}^2$. However, it is not always possible to join two points in S by a polygonal curve (or any "continuous" curve) lying entirely in S .

Proof. First we prove a lemma:

Lemma 10. If $A \cup B$ is connected and $B \cup C$ is connected, where A, B, C are nonempty, then $A \cup B \cup C$ is connected.

Proof. Suppose not. Then X, Y forms a disconnection of $A \cup B \cup C$. Now we cannot have $(A \cup B) \cap X \neq \emptyset$ and $(A \cup B) \cap Y \neq \emptyset$ simultaneously, otherwise X, Y will form a disconnection of $A \cup B$, a contradiction. Therefore we can assume without loss of generality of $A \cup B \subset X$ and $(A \cup B) \cap Y = \emptyset$.

Also, we cannot have $(B \cup C) \cap X \neq \emptyset$ and $(B \cup C) \cap Y \neq \emptyset$. Since we already have $(B \cup C) \cap X \neq \emptyset$, we must have $(B \cup C) \cap Y = \emptyset$. Now $(A \cup B \cup C) \cap Y = \emptyset$, so X, Y is not a disconnection, which is a contradiction. $\square$

First of all, $\sin$ is a continuous function and $x \mapsto \frac{1}{x}$ is a continuous function. The function $f : \mathbf{R} \setminus \{0\} \mapsto \mathbf{R}$ such that $f(x) = \sin \frac{1}{x}$ is their composition so is also continuous. Since $\mathbf{R}^+$ is continuous, and $\mathbf{R}^+$ is connected, the set

$$S_1 = \{(x, y) \in \mathbf{R}^2 : y = \sin \frac{1}{x}, x > 0\}$$

is connected, since it is the graph of a continuous function over a connected region.

Here is a proof of the continuity statement: Take $\varepsilon > 0$ and some ball $B_\varepsilon(x_0, \sin \frac{1}{x_0})$. We show that there exists some $\delta > 0$, such that $(x, \sin x) \in B_\varepsilon(x_0, \sin \frac{1}{x_0})$ if $|x - x_0| < \delta$.

We use the fact $|\sin \alpha - \sin \beta| \leq |\alpha - \beta|$. By choosing appropriate δ , we want

$$\left| \sin \frac{1}{x_0} - \sin \frac{1}{x} \right| \leq \left| \frac{1}{x_0} - \frac{1}{x} \right| = \left| \frac{x_0 - x}{x_0 x} \right| < \frac{\delta}{x_0(x_0 - \delta)} \leq \varepsilon.$$

To justify the $<$ sign, note that we are considering S_1 , so $x_0 > 0$. Also it is safe to assume that $x_0 - \varepsilon > 0$. To make an appropriate choice of δ , we can let

$$\delta \leq \frac{x_0^2 \varepsilon}{1 + x_0 \varepsilon}.$$

Where the right hand side is strictly larger than zero, since both numerator and denominator is strictly larger than zero.

Now we have proved the continuity statement. Suppose there is a disconnection of S_1 , say A and B . Let the natural projection mapping from $\mathbb{R}^2$ to $\mathbb{R}$ be τ . We want to prove $\tau(A)$ and $\tau(B)$ form a disconnection of $\mathbb{R}^+$.

$\mathbb{R}^+ \subset \tau(A) \cup \tau(B)$, $(\tau(A) \cap \mathbb{R}^+) \cap (\tau(B) \cap \mathbb{R}^+) = \emptyset$, $\tau(A) \cap \mathbb{R}^+ \neq \emptyset$, $\tau(B) \cap \mathbb{R}^+ \neq \emptyset$ are obvious by property of projection. Now we verify that $\tau(A)$, $\tau(B)$ are open. Take any $x \in \tau(A)$, then $(x, \sin \frac{1}{x}) \in A$. By the statement above, there exists a ball $B_\varepsilon(x, \sin \frac{1}{x}) \subset A$. The projection of the ball is the interval $(x - \varepsilon, x + \varepsilon)$, and by property of projection, $(x - \varepsilon, x + \varepsilon) = \tau(B_\varepsilon(x, \sin \frac{1}{x})) \subset \tau(A)$. Thus there exists a neighborhood of x in $\tau(A)$. We have verified that $\tau(A)$ is open. $\tau(B)$ is open by symmetry.

Thus $\tau(A)$ and $\tau(B)$ form a disconnection of $\mathbb{R}^+$. But we have a contradiction since $\mathbb{R}^+$ is connected. Thus S_1 is connected.

By symmetry, the set

$$S_2 = \{(x, y) \in \mathbb{R}^2 : y = \sin \frac{1}{x}, x < 0\}$$

is connected. Now let

$$S_3 = \{(0, y) : -1 \leq y \leq 1\}.$$

Note that

$$\overline{S_1} = S_1 \cup S_3.$$

To justify this, note that 0 is an oscillating discontinuity of f . This means f oscillates near 0 infinitely often. More precisely, for any $y \in [-1, 1]$ and $\delta > 0$, there exists some $0 < x < \delta$ such that $f(x) = y$. To verify this, take $x_1 = \frac{\delta}{2}$ and $x_2 = \frac{x_2}{2\pi x_1 + 1}$. Now $\frac{1}{x_2} - \frac{1}{x_1} = 2\pi$. Hence the function $\sin \frac{1}{x}$ completes exactly one period between x_1 and x_2 , taking all values between -1 and 1 . Hence there exists some $x \in [x_2, x_1]$ such that $f(x) = y$. Also, $x < \delta$ since $x \leq x_1 < \delta$ and $x > 0$ by construction.

We can conclude that every point $(0, y)$ such that $y \in [0, 1]$ is a cluster point of S_1 , since for any neighborhood ball $B_\varepsilon(0, y)$ containing $(0, y)$, it contains at least one point from S_1 by previous claim, and contains $(0, y')$ whenever $|y' - y| < \varepsilon$ which are infinitely many number of intersection with S_1^c .

By the last exercise, $\overline{S_1} = S_1 \cup S_3$ is connected. Symmetrically, $\overline{S_2} = S_2 \cup S_3$ is connected. Hence $S = S_1 \cup S_2 \cup S_3$ is connected by the lemma.

Consider $s_1 = (0, 1)$ and $s_2 = (1, \sin 1)$. For a curve $\gamma : [0, 1] \rightarrow [-1, 1]$ such that $\gamma(0) = s_1$ and $\gamma(1) = s_2$ to be entirely in S , it has to coincide with f in S_1 , and take value s_1 in S_3 . But the right limit of γ at 0 does not exist, since f does not have a right limit at 0. Therefore γ cannot be continuous at 0. In other words, there does not exist a continuous curve γ joining s_1 and s_2 . $\square$

4. 14.D. Let $X = (x_n)$ be a sequence in $\mathbf{R}^p$ which is convergent to x . Show that $\lim (\|x_n\|) = \|x\|$. (Hint: use the Triangle Inequality.)

Proof. We write $X^{(m)}$ for the sequence in $\mathbb{R}$ generated by the m -th coordinate of elements in X . We write $x^{(m)}$ for the m -th coordinate of x . Since X converges to x , we have, for each $m \in \{1, \dots, p\}$, $X^{(m)}$ converges to $x^{(m)}$. Therefore for any $\varepsilon' > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n^{(m)} - x^{(m)}| < \varepsilon'$.

Now fix $\varepsilon > 0$ and let n be some natural number, to be adjusted later. By triangle inequality,

$$\begin{aligned} \|x_n - x\| &= \left\| \left(x_n^{(1)} - x^{(1)} \right) e_1 + \dots + \left(x_n^{(p)} - x^{(p)} \right) e_p \right\| \\ &\leq \sum_{m=1}^p \left\| \left(x_n^{(m)} - x^{(m)} \right) e_m \right\| \\ &= \sum_{m=1}^p \left\| \left(x_n^{(m)} - x^{(m)} \right) \right\| \|e_m\| \\ &= \sum_{m=1}^p \left\| \left(x_n^{(m)} - x^{(m)} \right) \right\| \\ &= \sum_{m=1}^p \left| \left(x_n^{(m)} - x^{(m)} \right) \right|. \end{aligned}$$

For each m in $1, \dots, p$, let $\varepsilon' = \frac{\varepsilon}{p}$. Hence there exists N_m such that for all $n > N_m$, $|x_n^{(m)} - x^{(m)}| < \varepsilon'$.

Now let $N = \max_p N_p$. For all $n \geq N$, and for any m , we have $n \geq N \geq$

N_m , thus $|x_n^{(m)} - x^{(m)}| < \varepsilon' = \frac{\varepsilon}{p}$. Therefore

$$\begin{aligned}\|x_n - x\| &= \sum_{m=1}^p |(x_n^{(m)} - x^{(m)})| \\ &\leq \sum_{m=1}^p \frac{\varepsilon}{p} \\ &= \varepsilon.\end{aligned}$$

Since ε is arbitrary and N only depends on ε and p , we are done. $\square$

Correction

Norm is generic, but the proof assumed Euclidean norm. A simple proof is

$$|||x|| - ||x_0||| \leq \|x - x_0\| \rightarrow 0.$$

5. 14.M. Let $X = (x_n)$ be a sequence of strictly positive real numbers such that $\lim (x_n^{1/n}) < 1$. Show that for some r with $0 < r < 1$, then $0 < x_n < r^n$ for all sufficiently large $n \in \mathbf{N}$. Use this to deduce that $\lim (x_n) = 0$.

Proof. Let $L = \lim_{n \rightarrow \infty} (x_n^{1/n})$. By assumption, $0 \leq L < 1$. Take $\varepsilon > 0$ such that $L + \varepsilon < 1$. Now there exists $N \in \mathbf{N}$ such that for all $n \geq N$, we have $|x_n^{1/n} - L| < \varepsilon$. Hence

$$x_n < (\varepsilon + L)^n.$$

Let $r = \varepsilon + L$ and it is what we are looking for. Now $\lim_{n \rightarrow \infty} x_n \leq \lim_{n \rightarrow \infty} r^n = 0$ since $0 < r < 1$. $\square$

6. 14.N. Let $X = (x_n)$ be a sequence of strictly positive real numbers such that $\lim (x_n^{1/n}) > 1$. Show that X is not a bounded sequence and hence is not convergent.

Proof. Let $L = \lim_{n \rightarrow \infty} (x_n^{1/n})$. By assumption, $L > 1$. Take $\varepsilon > 0$ such that $L - \varepsilon > 1$. Now there exists $N \in \mathbf{N}$ such that for all $n \geq N$, we have $|x_n^{1/n} - L| < \varepsilon$. Hence

$$x_n > (L - \varepsilon)^n.$$

Let $r = L - \varepsilon$. For any $C > 1$, there exists a natural number n such that $n > \frac{C-1}{r-1}$. Hence $1 + n(r - 1) > C$. By Bernoulli's Inequality, $r^n = (1 + (r - 1))^n > C$. Thus $x_n > C$. This shows x_n is unbounded, hence not convergent. $\square$

7. 14.O. Give an example of a convergent sequence (x_n) of strictly positive real numbers such that $\lim (x_n^{1/n}) = 1$. Give an example of a divergent sequence with this property.

Proof. Convergent sequence: Let $x_n = 1$ for all n . Now $\lim_{n \rightarrow \infty} x_n = 1 = \lim_{n \rightarrow \infty} x_n^{1/n}$.

Divergent sequence: Let $x_n = n$ for all n . Now we prove the convergence of $x_n^{\frac{1}{n}}$. For all n we have $n^{1/n} > 1$.

Let $a_n = x_n^{\frac{1}{n}} - 1 = n^{\frac{1}{n}} - 1$. Now

$$n = (n^{\frac{1}{n}})^n = (a_n + 1)^n.$$

By the Bernoulli Inequality,

$$(a_n + 1)^n > \frac{n(n-1)}{2} a_n^2$$

Hence

$$0 \leq a_n^2 \leq \frac{2n}{n(n-1)} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

By the squeeze theorem, $a_n^2 \rightarrow 0$, hence $a_n \rightarrow 0$. Hence $x_n^{\frac{1}{n}} = 1 + a_n \rightarrow 1$. □

8. 15.L. If $0 < a \leq b$ and if $x_n = (a^n + b^n)^{1/n}$, then $\lim (x_n) = b$.

Proof.

$$\begin{aligned} \frac{x_n}{b} &= \frac{(a^n + b^n)^{1/n}}{b} \\ &= \left(\frac{a^n + b^n}{b^n} \right)^{1/n} \\ &= \left(\left(\frac{a}{b} \right)^n + 1 \right)^{1/n} \\ &\leq (1 + 1)^{1/n} \\ &\rightarrow 1. \end{aligned}$$

To justify the inequality, note that since $0 < a \leq b$, we have $\frac{a}{b} \leq 1$. Therefore $\left(\frac{a}{b} \right)^n \leq \frac{a}{b} \leq 1$.

To justify the final step, note that for all $c > 1$ we have $\lim_{n \rightarrow \infty} c^{\frac{1}{n}} = 1$. Below is the proof:

Lemma 11. For all $c > 1$, we have $\lim_{n \rightarrow \infty} c^{\frac{1}{n}} = 1$.

Proof. Since x^k is an increasing function of x , for all positive k and positive x ,

$$c^{1/n} > 1^{1/n} = 1.$$

Fix $\varepsilon > 0$. By Archimedean property, there exists $N \in \mathbb{N}$ such that $1 + N\varepsilon > c$, hence by Bernoulli Inequality, $(1 + \varepsilon)^N > c$. Therefore, $1 + \varepsilon > c^{1/N}$.

Since $c^{1/n}$ is a decreasing function of n , we have, for all $n \geq N$,

$$c^{1/n} \leq c^{1/N} < 1 + \varepsilon.$$

Hence for all $N \geq n$,

$$1 < c^{1/n} \leq 1 + \varepsilon.$$

By squeeze theorem, $c^{1/n} \rightarrow 1$. □

□

9. 15.M. Every irrational number in $\mathbf{R}$ is the limit of a sequence of rational numbers. Every rational number in $\mathbf{R}$ is the limit of a sequence of irrational numbers.

Proof. We use the following two lemmas. They are already check in previous exercises.

Lemma 12. $\mathbb{Q}$ is dense in $\mathbb{R}$. In other words, every open ball in $\mathbb{R}$ contains at least one element from $\mathbb{Q}$.

Lemma 13. $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$. In other words, every open ball in $\mathbb{R}$ contains at least one element from $\mathbb{R} \setminus \mathbb{Q}$.

Fix irrational x . For all $n \in \mathbb{N}$, use the first lemma to select $q_n \in \mathbb{Q} \cap B_{1/n}(x)$. Now for all $\varepsilon > 0$, there exists N such that for all $n \geq N$, $\frac{1}{n} < \varepsilon$. Thus $|q_n - x| < \frac{1}{n} < \varepsilon$. Thus q_n converges to x .

Fix rational x . For all $n \in \mathbb{N}$, use the second lemma to select $q_n \in \mathbb{R} \setminus \mathbb{Q} \cap B_{1/n}(x)$. Now for all $\varepsilon > 0$, there exists N such that for all $n \geq N$, $\frac{1}{n} < \varepsilon$. Thus $|q_n - x| < \frac{1}{n} < \varepsilon$. Thus q_n converges to x . □

10. 15.N. Let $A \subseteq \mathbf{R}^p$ and $x \in \mathbf{R}^p$. Then x is a boundary point of A if and only if there is a sequence (a_n) of elements in A and a sequence (b_n) of elements in $\mathcal{C}(A)$ such that

$$\lim (a_n) = x = \lim (b_n).$$

Proof. Since x is a boundary point of A , each neighborhood of x should have nonempty intersection with both A and A^c . Now for each $n \in \mathbb{N}$, pick $a_n \in A \cap B_{1/n}(x)$ and $b_n \in A^c \cap B_{1/n}(x)$.

Now for arbitrary $\varepsilon > 0$, there exists by Archimedean property a natural number $N \in \mathbb{N}$ such that $\frac{1}{N} < \varepsilon$. Hence for all $n \geq N$, we have $|a_n - x| < \frac{1}{n} \leq \frac{1}{N}$. Hence a_n converges to x . The proof of $b_n \rightarrow x$ is completely symmetric. Hence there exists a the desired sequences.

To prove the other direction, assume the sequences are available, and consider some $B_\varepsilon(x)$. Then there exists N and M such that $|a_n - x| < \varepsilon$ for all $n \geq N$ and $|a_m - x| < \varepsilon$ for all $m \geq M$. This is to say that $B_\varepsilon(x) \cap A$ and $B_\varepsilon(x) \cap A^c$ are both unempty. Thus x is boundary point of A . $\square$

Homework 6

Tue 11 Oct 2022 22:28

1. 16.A. Let $x_1 \in \mathbf{R}$ satisfy $x_1 > 1$ and let $x_{n+1} = 2 - 1/x_n$ for $n \in \mathbf{N}$. Show that the sequence (x_n) is monotone and bounded. What is its limit?

Proof. We establish the fact that (x_n) is monotone decreasing and bounded below by 1. The first term x_1 is obviously monotone decreasing (since there is only one term), and bounded below by 1 as assumed.

Now assume that $x_1, \dots, x_k$ is monotone decreasing and bounded below by 1. We have

$$\begin{aligned} x_{k+1} &< x_k \\ \Leftrightarrow 2 - \frac{1}{x_k} &< x_k \\ \Leftrightarrow 2x_k - 1 &< x_k^2 && \text{since } x_k > 1 > 0 \\ \Leftrightarrow (x_k - 1)^2 &> 0. \end{aligned}$$

Which is true since $x_k > 1$, thus $x_k - 1 > 0$.

Moreover, since $x_k > 1$, $\frac{1}{x_k} < 1$, so $2 - \frac{1}{x_k} > 1$. Therefore $x_{k+1} > 1$.

We have completed the induction and (x_n) is indeed monotone decreasing and bounded below by 1. To obtain the limit, let the limit be x and take limit on both sides of $x_{n+1} = 2 - 1/x_n$. Hence $x = 2 - 1/x$. Solve the equation we obtain $x = 1$, so the limit is 1. $\square$

2. 16.E. Show that every sequence in $\mathbf{R}$ either has a monotone increasing subsequence or a monotone decreasing subsequence.

Proof. (Hint given in class) We say $n \in \mathbb{N}$ is a *peak* if $x_n \geq x_m$ for every $m \geq n$. Then consider two cases:

- (a) Finitely many peaks.
- (b) Infinitely many peaks.

$\square$

Proof. Take any sequence X . If X is unbounded, then we may assume without loss of generality that it is unbounded from above, thus we can construct a monotone increasing subsequence of X . For the detailed construction, let

$$n_j = \inf\{n \in \mathbb{N} : x_n > x_{n_{j-1}}, n > n_{j-1}\}.$$

We verify that n_j is well-defined. Since X is unbounded from above, for any fixed $x_{n_{j-1}}$, $x_n > x_{n_{j-1}}$ for sufficiently large n . Since such n is infinitely many, we can add the restriction $n > n_{j-1}$. Then by well-orderedness of natural numbers, n_j is well defined. Then x_{n_j} is monotone increasing by construction.

If X is bounded, then by Bolzano-Weierstrass, there exists convergent subsequence (x_{n_m}) of X with limit x . We define $A_1 = \{m \in \mathbb{N} : x_{n_m} < x\}$, $A_2 = \{m \in \mathbb{N} : x_{n_m} = x\}$, and $A_3 = \{m \in \mathbb{N} : x_{n_m} > x\}$. Since $A_1 \cup A_2 \cup A_3 = \mathbb{N}$ is infinite, at least one of A_1, A_2, A_3 is infinite. If A_2 is infinite, we can choose a constant subsequence of x_{n_m} which is indeed monotone.

The cases A_1 is infinite and A_3 infinite are symmetrical, so it suffices to prove for the case that A_1 is infinite. Then we define the subsequence $x_{n_{m_j}}$ as

$$m_j = \inf\{m \in \mathbb{N} : m > m_{j-1}, x_{n_{m_{j-1}}} < x_{n_m} < x\}.$$

To justify that m_j is well-defined, note that since A_1 is infinite, there exists infinitely many $x_{n_m} < x$. Since $x_{n_m} \rightarrow x$ as $m \rightarrow \infty$, for any fixed $x_{n_{m_{j-1}}}$, we have $x_{n_m} > x_{n_{m_{j-1}}}$ for all sufficiently large m . Since there are infinitely many such m , we can add the restriction that $m > m_{j-1}$. Then by well-orderedness of natural numbers, m_j is well-defined.

Now $x_{n_{m_j}}$ is strictly monotone increasing by construction. $\square$

Proof. If $\limsup X$ is finite, then there exists a monotone subsequence X' of X converging to $s := \limsup X$ (claim 1). If $\limsup X$ is not finite, then X is unbounded from above, so there exists a monotone increasing unbounded subsequence of X (claim 2).

Justification of claim 1: Since $s = \limsup X$, there exists a subsequence of X converging to s , say Y . Note that one of the following events must happen infinitely often:

- $y_n > s$
- $y_n = s$
- $y_n < s$

If $y_n = s$ infinitely often, we just take the subsequence that takes a constant value of s and it is indeed monotone. If $y_n > s$ infinitely often, we define the subsequence y_{n_j} as

$$n_j := \inf\{m \in \mathbb{N} : m > n_j, s < y_m \leq y_{n_j}\}.$$

To check the definition is well-posed, assume inductively that n_j is well-defined up to $j = p$. Now since y_n converges to s , there exists some $N \in \mathbb{N}$ such that for all $m \geq N$, we have $y_m \leq y_{n_j}$. Since $y_m > s$ infinitely often, $\{m \in \mathbb{N} : m > n_j, s < y_m \leq y_{n_p}\}$ is nonempty, therefore (by well-orderedness of $\mathbb{N}$) n_j is well-defined. Now y_{n_j} is the desired sequence.

The proof for the case that $y_n < s$ infinitely often is entirely symmetrical.

Justification of claim 2: Since X is unbounded from above, for each M , there exists N such that for all $n > N$, we have $x_n > M$. Therefore, we can define a subsequence x_{n_j} such that

$$n_j := \inf\{m \in \mathbb{N} : m > n_{j-1}, x_m > x_{n_{j-1}}\}.$$

To verify that the definition is well-posed, assume inductively that n_j is well-defined up to $j = p$. Now there exists N such that for all $m > N$, we have $x_m > x_{n_p}$. Then $\max(n_{j-1}, N) + 1 \in \{m \in \mathbb{N} : m > n_{j-1}, x_m > x_{n_{j-1}}\}$. Thus by well-orderedness of natural numbers n_j is well-defined for $j = p + 1$.

By definition, x_{n_j} is monotone increasing and is a subsequence of X . $\square$

3. 16.G. Determine the convergence or divergence of the sequence (x_n) , where

$$x_n = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \quad \text{for } n \in \mathbb{N}$$

Proof. We establish an upper bound of the sequence:

$$x_n = \sum_{k=n+1}^{2n} \frac{1}{k} \leq \sum_{k=n+1}^{2n} \frac{1}{n} = n \frac{1}{n} = 1.$$

Then we prove x_n is monotone increasing. Combine with the upper bound and we know x_n is convergent by monotone convergence theorem.

$$\begin{aligned} x_n - x_{n-1} &= \frac{1}{n+1} + \cdots + \frac{1}{2n} - \left(\frac{1}{n} + \cdots + \frac{1}{2n-2} \right) \\ &= \frac{1}{2n-1} + \frac{1}{2n} - \frac{1}{n} \\ &> \frac{1}{2n} + \frac{1}{2n} - \frac{1}{n} \\ &= 0. \end{aligned}$$

$\square$

4. 16.Q. Prove that if K_1 and K_2 are compact subsets of $\mathbb{R}^p$, then there exist points $x_1 \in K_1, x_2 \in K_2$ such that if $z_1 \in K_1, z_2 \in K_2$, then $\|z_1 - z_2\| \geq \|x_1 - x_2\|$

Proof. Define the distance function

$$\begin{aligned} d : K_1 \times K_2 &\longrightarrow \mathbb{R} \\ k^{(1)}, k^{(2)} &\longmapsto d(k^{(1)}, k^{(2)}) = \|k^{(1)} - k^{(2)}\|. \end{aligned}$$

By definition we know $d(k^{(1)} - k^{(2)}) \geq 0$, so $d(K_1 \times K_2)$ is bounded below by 0. Also, it is obviously nonempty. Hence there exists $u := \inf d(K_1 \times K_2)$.

Now for each $n \in \mathbb{N}$, there must exist $k_n^{(1)}$ and $k_n^{(2)}$ such that $d(k_n^{(1)} - k_n^{(2)}) < u + \frac{1}{n}$, by property of infimum.

Then since K_1 is compact, by Bolzano-Weierstrass, there exists convergent subsequence $k_{n_p}^{(1)}$ of $k_n^{(1)}$. Now $k_{n_p}^{(2)}$ is a bounded subsequence of $k_n^{(2)}$, but

not necessarily convergent. By Bolzano-Weierstrass again, there exists convergent sequence $k_{n_{p_l}}^{(2)}$ of $k_{n_p}^{(2)}$.

Now $(k_{n_{p_l}}^{(1)}, k_{n_{p_l}}^{(2)})$ is a subsequence of $(k_n^{(1)}, k_n^{(2)})$. Moreover, since both K_1 and K_2 are closed, there exists $(x_1, x_2) \in K_1 \times K_2$ such that $\lim(k_{n_{p_l}}^{(1)}, k_{n_{p_l}}^{(2)}) = (x_1, x_2)$. Now for each $l \in \mathbb{N}$, we have

$$u \leq d(k_{n_{p_l}}^{(1)}, k_{n_{p_l}}^{(2)}) < u + \frac{1}{n_{p_l}} \leq u + \frac{1}{l}.$$

By taking $l \rightarrow \infty$ on both sides and applying Squeeze Theorem, we obtain $d(x_1, x_2) = u$. Thus for all $(z_1, z_2) \in K_1 \times K_2$, we have $d(z_1, z_2) \geq \inf d(K_1, K_2) = u$.

□

Proof. (First attempt, more complicated) If K_1 and K_2 intersect somewhere, then we pick $x_1 = x_2 \in K_1 \cap K_2$ and complete the proof.

From now on we assume K_1 and K_2 are disjoint. Since K_2 is compact, K_2 is closed. For each $k_1 \in K_1$, we choose $k_2(k_1) \in K_2$ using Nearest Point Theorem (11.6), such that for all $k \in K_2$, $\|k_1, k\| \geq \|k_1, k_2(k_1)\|$.

Now define

$$\begin{aligned} \varphi : K_1 &\longrightarrow \mathbb{R} \\ k_1 &\longmapsto \varphi(k_1) = \|k_1 - k_2(k_1)\|. \end{aligned}$$

Let $u = \inf_{k_1 \in K_1} \varphi(k_1)$. To see that u is well-defined, note that u is nonempty (pick arbitrary $k_1 \in K_1$ and $\varphi(k_1)$ is well-defined), and bounded from below (obviously $\varphi(k_1) \geq 0$).

Now for each $n \in \mathbb{N}$, define

$$F_n = \{k_1 \in K_1 : \varphi(k_1) \leq u + \frac{1}{n}\}.$$

Then F_n is closed (since it is the intersection of K_1 and the closed ball $\overline{B_{u+\frac{1}{n}}(k_2(k_1))}$) and nonempty (by property of infimum). Moreover, F_n has a nesting property. To see this, note that $\varphi(k_1)$ is independent of n , and for any $k_1 \in F_{n+1}$ we have $\varphi(k_1) \leq u + \frac{1}{n+1} \leq u + \frac{1}{n}$, thus $F_{n+1} \subset F_n$.

Finally, F_1 is bounded since K_1 is bounded. Now by the cantor intersection theorem, there exists a point k^* belonging to all the sets F_n . Thus $\varphi(k^*) \leq u + \frac{1}{n}$ for all $n \in \mathbb{N}$, so $\varphi(k^*) = u$. We can easily verify from the definition that $x_1 := k^*$ and $x_2 := k_2(k^*)$ satisfy desired properties. □

5. 18.B. If $X = (x_n)$ is a bounded sequence in $\mathbf{R}$, show that there is a subsequence of X which converges to $\liminf X$.

Proof. Denote $x = \liminf X$. Let L be the set of $v \in \mathbb{R}$ such that there exists a subsequence of X which converges to v , and let $x = \inf L$.

Now let $\varepsilon_m = \frac{1}{m}$. Using property of infimum, take $v_m \in L$ such that $v_m < x + \varepsilon_m/2$. Then there exists some subsequence (x_{n_j}) of X that

converges to v_m . Then there exists some $N_m \in \mathbb{N}$ such that for all $j > N_m$ we have $|x_{n_j} - v_m| < \varepsilon_m/2$. Combining the results, we have $|x_{n_j} - x| \leq |x_{n_j} - v_m| + |v_m - x| < \varepsilon_m$.

Now we proceed to define our subsequence. For each $m \in \mathbb{N}$, we do the construction above, and define

$$\varphi(m) = \inf\{n_j : j > N_m, n_j > \varphi(m-1)\}.$$

Now we see that for each m , $|x_{\varphi(m)} - x| \leq \varepsilon_m$. As $m \rightarrow \infty$, $\varepsilon_m \rightarrow 0$, so $x_{\varphi(m)} \rightarrow x$.

□

6. 18.F. If $X = (x_n)$ is a bounded sequence of strictly positive elements in $\mathbf{R}$ show that $\limsup (x_n^{1/n}) \leq \limsup (x_{n+1}/x_n)$.

Proof.

Notation. We say some statement P_n holds for *sufficiently large* n if there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have P_n is true.

Denote $\limsup \frac{x_{n+1}}{x_n} = L$. Then for each $\varepsilon > 0$, $\frac{x_{n+1}}{x_n} \leq L + \varepsilon$ for sufficiently large n . To see this, apply the property of limsup, and see that there are at most finitely many n such that $\frac{x_{n+1}}{x_n} > L + \varepsilon$.

Thus $x_n \leq C(L + \varepsilon)^n$ for sufficiently large n for some C (by using a past homework problem). Therefore

$$x_n^{1/n} \leq C^{1/n}(L + \varepsilon) \rightarrow L + \varepsilon.$$

Thus $x_n^{1/n} \leq L + 2\varepsilon$ for sufficiently large n . To see this, pick arbitrarily small $\delta > 0$, and note that $C^{1/n} < 1 + \delta$ for sufficiently large n . Hence $x_n^{1/n} \leq (1 + \delta)(L + \varepsilon)$ for arbitrarily small δ and ε . Now let $\delta < \frac{\varepsilon}{(L + \varepsilon)}$, and we obtain $x_n^{1/n} < (1 + \frac{\varepsilon}{L + \varepsilon})(L + \varepsilon) = L + 2\varepsilon$ for sufficiently large n .

Therefore $\limsup x_n^{1/n} \leq L + 2\varepsilon$ for arbitrary ε . Hence $\limsup x_n^{1/n} \leq L$. □

7. 18.I. Show that $\limsup X = +\infty$ if and only if there is a subsequence X' of X such that $\lim X' = +\infty$.

Proof. Suppose $\limsup X = +\infty$. Then for each natural number N , there exists some value $v > N$ in the extended real numbers ($v \in \mathbb{R}$ or $v = +\infty$), such that there is a subsequence in X converging to v . Thus X is not bounded from above, because we can always find a subsequence converging to v , satisfying $x_{n_j} > N$ for sufficiently large j . Then we let

$$n_j = \inf\{m : m > n_{j-1}, x_m > j\}.$$

The set above is well-defined. To see this, first note that there always exists some $x_m > j$ by unboundedness. But then we can always find some $x'_m > x_m$. Continue in this fashion, there are infinitely many $x_m > j$.

Thus we can put a restriction $m > n_{j-1}$. Then n_j is well-defined by well-orderedness of $\mathbb{N}$. Thus n_j is well-defined and we have $x_{n_j} \rightarrow +\infty$.

Now assume there is a subsequence X' of X such that $\lim X' = +\infty$. Let V be the set of values $v \in \mathbb{R}$ such that there exists a subsequence of X converging to v . By definition, $\limsup X = \sup V$. But we have $+\infty \in V$, so $\limsup X = +\infty$. $\square$

Homework 7

Sun 23 Oct 2022 17:59

20.G, 20.N, 20.P, 22.D, 22.F, 22.R, 22.S

- 20.G. Let f and g be continuous on $\mathbf{R}$ to $\mathbf{R}$. Is it true that $f(x) = g(x)$ for $x \in \mathbf{R}$ if and only if $f(y) = g(y)$ for all rational numbers y in $\mathbf{R}$?

Proof. Yes it is. It suffices to prove the $(\Leftarrow)$ direction since the other direction is trivial. Fix any $x \in \mathbb{R}$. We can find a sequence of rational number r_n such that $r_n \rightarrow x$. By assumption, $f(r_n) = g(r_n)$ for all $n \in \mathbb{N}$. By property of continuity, $f(r_n) \rightarrow f(x)$ and $g(r_n) \rightarrow g(x)$. Since $f(r_n) = g(r_n)$, $f(x) = g(x)$. $\square$

- 20.N. Let $g : \mathbf{R} \rightarrow \mathbf{R}$ satisfy the relation

$$g(x+y) = g(x)g(y) \quad \text{for } x, y \in \mathbf{R}.$$

Show that, if g is continuous at $x = 0$, then g is continuous at every point. Also, if $g(a) = 0$ for some $a \in \mathbf{R}$, then $g(x) = 0$ for all $x \in \mathbf{R}$.

Proof. Fix $x \in \mathbb{R}$. Take any sequence x_n in $\mathbb{R}$ such that $x_n \rightarrow x$. Define $y_n = x_n - x$. By convergence of sequence, $y_n \rightarrow 0$. By continuity of g at 0, $g(y_n) \rightarrow g(0)$. Now

$$g(x_n) = g(x + y_n) = g(x)g(y_n) \rightarrow g(x)g(0) = g(x+0) = g(x).$$

Thus $g(x_n) \rightarrow g(x)$ for any x and any $x_n \rightarrow x$. Thus g is continuous anywhere.

To verify the second claim, suppose $g(0) = 0$ and note that

$$g(x) = g(x - a + a) = g(x - a)g(a) = g(x - a) \cdot 0 = 0.$$

$\square$

- 20.P. Let $f, g : \mathbf{R}^p \rightarrow \mathbf{R}$ be continuous at a point $a \in \mathbf{R}^p$ and let h, k be defined on $\mathbf{R}^p$ to $\mathbf{R}$ by

$$h(x) = \sup\{f(x), g(x)\}, \quad k(x) = \inf\{f(x), g(x)\}.$$

Show that h and k are continuous at a . (Hint: note that $\sup\{b, c\} = \frac{1}{2}(b + c + |b - c|)$ and $\inf\{b, c\} = \frac{1}{2}(b + c - |b - c|)$.)

Proof. Using the hint, we define $h'(x) = \frac{1}{2}(f(x) + g(x) + |f(x) - g(x)|)$ and similarly define $k'(x)$.

Since addition, subtraction of continuous functions are continuous, and composition of continuous function with the continuous function $x \mapsto |x|$

is continuous, h' and k' are continuous. Also, for any $x \in \mathbb{R}^p$, we have $h(x) = \sup\{f(x), g(x)\} = \frac{1}{2}(f(x) + g(x) + |f(x) - g(x)|) = h'(x)$, so $h = h'$ and similarly $k = k'$. Hence h and k are continuous. $\square$

- 22.D. If $p : \mathbf{R}^2 \rightarrow \mathbf{R}$ is a polynomial and $c \in \mathbf{R}$, show that the set $\{(x, y) : p(x, y) < c\}$ is open in $\mathbf{R}^2$.

Proof. The set $(-\infty, c)$ is open in $\mathbb{R}$. Polynomials are continuous, so the inverse image $p^{-1}((-\infty, c)) = \{(x, y) : p(x, y) < c\}$ is open in $\mathbb{R}^2$. $\square$

- 22.F. A subset $D \subseteq \mathbf{R}^p$ is disconnected if and only if there exists a continuous function $f : D \rightarrow \mathbf{R}$ such that $f(D) = \{0, 1\}$

Proof. Let A, B be a disconnection of D . Define f such that $f(A \cap D) = \{0\}$ and $f(B \cap D) = \{1\}$. Such function is well-defined since A, B are disjoint and $A \cup B$ covers D . Let N be any open set in $\mathbb{R}$. If $N \cap \{0, 1\} = \{0\}$, then $f^{-1}(N) = A \cap D$ is relatively open in D ; if $N \cap \{0, 1\} = \{1\}$, then $f^{-1}(N) = B \cap D$ is relatively open in D ; if $N \cap \{0, 1\} = \{0, 1\}$, then $f^{-1}(N) = D$ is relatively open in D ; if $N \cap \{0, 1\} = \emptyset$, then $f^{-1}(N) = \emptyset$ is relatively open in D . Thus by definition of continuity, f is continuous on D .

Now we prove the other direction. Suppose there exists such a function f . Take $A = f^{-1}((0 - 0.1, 0 + 0.1))$ and $B = f^{-1}((1 - 0.1, 1 + 0.1))$. Thus A and B are both open since they are inverse image by continuous function of open sets. $A \cup B$ covers D since $\{0, 1\} \subset (0 - 0.1, 0 + 0.1) \cup (1 - 0.1, 1 + 0.1)$. Moreover, A and B are disjoint, since f takes different values on A and on B . Hence A, B forms a disconnection on D , so D is disconnected. $\square$

- 22.R. Let f be continuous on the interval $[0, 2\pi]$ to $\mathbf{R}$ and such that $f(0) = f(2\pi)$. Prove that there exists a point c in this interval such that $f(c) = f(c + \pi)$. (Hint: consider $g(x) = f(x) - f(x + \pi)$.) Conclude that there are, at any time, antipodal points on the equator of the earth which have the same temperature.

Proof. Let $g(x) = f(x) - f(x + \pi)$. Note that $g(x + \pi) = f(x + \pi) - f(x + 2\pi) = f(x + \pi) - f(x) = -g(x)$ (not well-defined on $(0, \pi)$). But since f is continuous on $[0, 2\pi]$, so is $g(x)$ on $[0, \pi]$. If $g = 0$ then we are done, since there exists $x \in [0, \pi]$ such that $f(x) - f(x + \pi) = g(x) = 0$. If $g \neq 0$, then there exists $x \in [0, \pi]$ such that $g(x + \pi) = -g(x) \neq 0$. By Bolzano's Intermediate Value Theorem, there exists $y \in (x, x + \pi)$ such that $g(y) = 0$, thus $f(y) - f(y + \pi) = 0$.

Interpretation: $[0, 2\pi]$ with two endpoint identified is homeomorphic to the earth's equator. We may assume that temperature is a continuous real-valued function on its domain. So the result applies directly to the equator. $\square$

Correction

First note that

$$g(\pi) = f(\pi) - f(2\pi) = f(\pi) - f(0) = -g(0)$$

We consider two cases. In the first case $g(0) = 0$, we have $f(0) = f(\pi)$ as desired. In the second case $g(0) \neq 0$, we suppose without loss of generality that $g(0) < 0$. Then $g(\pi) > 0$. By Bolzano's Intermediate Value Theorem, there exists $y \in [0, \pi]$ such that $g(y) = 0$. Hence $f(y) = f(y + \pi)$. We have $y + \pi \leq 2\pi$, so $f(y + \pi)$ is well-defined. Hence we are done.

- 22.S. Let $\varphi : [0, 2\pi) \rightarrow \mathbf{R}^2$ be defined by $\varphi(t) = (\cos t, \sin t)$ for $t \in [0, 2\pi)$. Then φ is an injective continuous map of $[0, 2\pi)$ onto the unit circle $S = \{(x, y) \in \mathbf{K}^2 : x^2 + y^2 = 1\}$. Show that $\varphi^{-1} : S \rightarrow [0, 2\pi)$ cannot be continuous. (We conclude that Theorem 22.9 may fail if the domain is not compact.)

Proof. (topological) The set $[0, 1)$ is relatively open in $[0, 2\pi)$, but $\varphi([0, 1))$ is a half-open interval on S , which is not relatively open in S . Thus φ^{-1} is not continuous. $\square$

Proof. (sequential) By definition of $\varphi(t)$, $\lim_{t \rightarrow 2\pi} \varphi(t) = (1, 0) = \varphi(0)$. Now take the sequence $t_n = \pi + (-1)^n (1 - \frac{1}{n})\pi$. t_n is bounded and has exactly two limit points: 0 and 2π , and $\varphi(t)$ converge to the same limit when t approaches 0 or 2π . Hence $\varphi(t_n)$ is a convergent sequence in $\mathbf{R}^2$. However, $\varphi^{-1}(\varphi(t_n)) = t_n$ is not convergent in $\mathbf{R}$. Thus φ^{-1} is not convergent. $\square$

Homework 8

Thu 03 Nov 2022 17:00

23.G, 23.J, 24.C, 24.D, 24.E(a,d,e), 24.N, 24.S

- 23.G. A function $g : \mathbf{R} \rightarrow \mathbf{R}^q$ is periodic if there exists a number $p > 0$ such that $g(x + p) = g(x)$ for all $x \in \mathbf{R}$. Show that a continuous periodic function is bounded and uniformly continuous on $\mathbf{R}$.

Proof. Fix arbitrary $x_0 \in \mathbf{R}$, and consider the interval $I = [x_0, x_0 + p]$. We know that $f(x_0) = f(x_0 + p)$. Moreover, since I is compact (by Heine Borel), and f is continuous, we know f is uniformly continuous on I . By Extremum Value Theorem, $\|f\|$ attains maximum and minimum somewhere in I , so in particular $\|f\|$ is bounded in I , hence f is bounded in I .

Now take any $x \in \mathbf{R}$. We take integer k and real number $r \in [0, p)$ such that $x + kp = x_0 + r$. We know $f(x) = f(x + kp) \in f(I)$, so it immediately follows that $f(x)$ is bounded.

Now we verify uniform continuity of f on $\mathbf{R}$. Fix any $\varepsilon > 0$. Then there exists $\delta > 0$ as in the definition of uniform continuity of f on I . We first assume $r > 0$. We may assume without loss of generality that $\delta < r$ and $\delta < p - r$, so $(x + kp - \delta, x + kp + \delta) = (x_0 + r - \delta, x_0 + r + \delta) \subset I$. Now $f((x -$

$$\delta, x + \delta)) = f((x + kp - \delta, x + kp + \delta)) \subset (f(x + kp) - \varepsilon, f(x + kp) + \varepsilon) = (f(x) - \varepsilon, f(x) + \varepsilon).$$

Now we verify the edge case $r = 0$, i.e. $x = x_0 + kp$ for some integer k . In this case,

$$\begin{aligned} f((x - \delta, x + \delta)) &= f((x_0 - \delta, x_0 + \delta)) \\ &= f((x_0 + p - \delta, x_0 + p)) \cup f((x_0, x_0 + \delta)) \\ &\subset (f(x_0 + p) - \varepsilon, f(x_0 + p) + \varepsilon) \cup (f(x_0) - \varepsilon, f(x_0) + \varepsilon) \\ &= (f(x) - \varepsilon, f(x) + \varepsilon). \end{aligned}$$

We have verified $f((x - \delta, x + \delta)) \subset (f(x) - \varepsilon, f(x) + \varepsilon)$ for all $x \in \mathbb{R}$, where ε and δ do not depend on x , hence f is uniformly continuous. $\square$

23.J. Let $D = \{x \in \mathbf{R}^p : \|x\| < 1\}$. Show that $f : D \rightarrow \mathbf{R}^q$ can be extended to a continuous function on $D_1 = \{x \in \mathbf{R}^p : \|x\| \leq 1\}$ to $\mathbf{R}^q$ if and only if f is uniformly continuous on D .

Proof. Suppose there exists a continuous extension of f on D_1 . Then since D_1 is closed and bounded, it is compact by Heine-Borel; therefore f is uniformly continuous on D_1 , hence uniformly continuous on D .

Suppose f is uniformly continuous. We construct a continuous extension to D_1 . Fix ε and obtain δ as in the definition of uniform continuity. Now take some $x_0 \in D_1$ and some sequence $x_n \rightarrow x_0$ in D . By convergence of sequence there exists N_δ such that $|x_n - x_m| < \delta$ for $n, m > N_\delta$. But then $|f(x_n) - f(x_m)| < \varepsilon$ by uniform continuity. ε is arbitrary, so $f(x_n)$ is Cauchy, and hence converges. We may define $f(x_0)$ to be the limit so that $f(x_n) \rightarrow x_0$ whenever $x_n \rightarrow x_0$.

Now we show $f(x_0)$ is well-defined. Take another sequence $y_n \rightarrow x_0$. We know that $|x_n - y_n| \rightarrow 0$ as $n \rightarrow \infty$. In particular, $|x_n - y_n| < \delta$ for $n > M_\delta$. But then $|f(x_n) - f(y_n)| < \varepsilon$ by uniform continuity. Since ε is arbitrary $|f(x_n) - f(y_n)| \rightarrow 0$. Hence the limit is unique. Thus $f(x_0)$ is well-defined.

Finally we have to verify that the extended f is continuous on D_1 . Take $x_0 \in D_1$, and it suffices to show that f is continuous at x_0 . If $x_0 \in D$, we are done. Therefore we may assume $x_0 \in D_1 \setminus D$. We have to show $f(x_n) \rightarrow f(x_0)$ for any $x_n \rightarrow x_0$ in D_1 .

Now we let y_n be a sequence lying in $D_1 \setminus D$ converging to x_0 , and z_n be a sequence lying in D converging to x_0 . Let x_n be a sequence such that for each n , we either have $x_n = y_n$ or $x_n = z_n$. Then x_n could be any sequence in D_1 converging to x_0 . To show $f(x_n) \rightarrow f(x_0)$, it suffices to show both $f(y_n) \rightarrow f(x_0)$ and $f(z_n) \rightarrow f(x_0)$.

$f(z_n) \rightarrow f(x_0)$ is trivial as it follows from definition of f in $D_1 \setminus D$. Now we verify $f(y_n) \rightarrow f(x_0)$. For each $n \geq 1$, there exists a sequence $(a_m^{(n)})_{m \in \mathbb{N}}$ contained in D converging to y_n . Also, let $a_m^{(0)} \rightarrow x_0$ in D . Take $\varepsilon > 0$ and we have $\delta > 0$ as in definition for uniform continuity of f in D . Then there exists N_δ such that $|y_n - x_0| < \delta$ for $n > N_\delta$. Now fix $n > N_\delta$. We

let $\delta' > 0$ be such that $\delta' < \delta - |y_n - x_0|$. Then there exists $M_{\delta'}^{(n)}$ and $M_{\delta'}^{(0)}$ such that $|a_m^{(n)} - y_n| < \delta'$ for $m > M_{\delta'}^{(n)}$ and $|a_m^{(0)} - x_0| < \delta'$ for $m > M_{\delta'}^{(0)}$. Using triangle inequality, we verify that $|a_m^{(n)} - a_m^{(0)}| < \delta$ for all $m > M$ for some M . But by uniform continuity of f in D , we know that $|f(a_m^{(n)}) - f(a_m^{(0)})| < \varepsilon$ for $m > M$. Hence

$$|f(y_n) - f(x_0)| = \left| \lim_{m \rightarrow \infty} f(a_m^{(n)}) - \lim_{m \rightarrow \infty} f(a_m^{(0)}) \right| = \left| \lim_{m \rightarrow \infty} f(a_m^{(n)}) - f(a_m^{(0)}) \right| \leq \varepsilon.$$

Hence for each $\varepsilon > 0$, there exists N_δ , such that $|f(y_n) - f(x_0)| < \varepsilon$ whenever $n > N_\delta$. Hence $f(y_n) \rightarrow f(x_0)$. $\square$

- 24.C. Give an example of a sequence of continuous functions which converges on a compact set to a function that has an infinite number of discontinuities.

Proof. We construct an "infinite staircase". Let $I = [0, 1]$. Define $f_n : I \rightarrow I$ such that

$$f_n(x) = \begin{cases} 1 - \frac{1}{2^m} + r^n 2^{(m+1)(n-1)} & x = (1 - \frac{1}{2^m}) + r, m \in \mathbb{N}, 0 \leq r < \frac{1}{2^{m+1}} \\ 1 & x = 1 \end{cases}.$$

Then as $n \rightarrow \infty$, f_n converges to the "infinite step function"

$$f(x) = \begin{cases} 1 - \frac{1}{2^m} & x = (1 - \frac{1}{2^m}) + r, m \in \mathbb{N}, 0 \leq r < \frac{1}{2^{m+1}} \\ 1 & x = 1 \end{cases}.$$

Obviously the function is discontinuous at infinitely many points. Namely, it is discontinuous precisely at the points $1 - \frac{1}{2^m}$ for $m \in \mathbb{N}^+$.

Moreover, $f_n(x)$ is uniquely defined and continuous at each x . It is obvious unique in each interval $[1 - \frac{1}{2^m}, 1 - \frac{1}{2^{m+1}})$. At the boundaries, we have

$$\lim_{x \rightarrow 1 - \frac{1}{2^{m+1}}; x \in [1 - \frac{1}{2^m}, 1 - \frac{1}{2^{m+1}})} f(x) = \lim_{r \rightarrow \frac{1}{2^{m+1}}} 1 - \frac{1}{2^m} + r^n 2^{(m+1)(n-1)} = 1 - \frac{1}{2^m} + 2^{-m-1} = 1 - \frac{1}{2^{m+1}}.$$

$$\lim_{x \rightarrow 1 - \frac{1}{2^{m+1}}; x \in (1 - \frac{1}{2^{m+1}}, 1 - \frac{1}{2^{m+2}})} f(x) = \lim_{r \rightarrow 0} 1 - \frac{1}{2^{m+1}} + r^n 2^{(m+2)(n-1)} = 1 - \frac{1}{2^{m+1}}.$$

Thus the left and right limits are equal to the value of function at any point. Hence f_n is continuous. $\square$

- 24.D. Let (f_n) be a sequence of continuous functions on $D \subseteq \mathbf{R}^p$ to $\mathbf{R}^q$ such that (f_n) converges uniformly to f on D , and let (x_n) be a sequence of elements in D which converges to $x \in D$. Does it follow that $(f_n(x_n))$ converges to $f(x)$?

Proof. Fix $\varepsilon > 0$. Since $f_n \rightarrow f$ uniformly, $|f_n(y) - f(y)| < \varepsilon$ for all $y \in D$ whenever $n > N_\varepsilon$ for some $N_\varepsilon \in \mathbb{N}$. Hence $|f_n(x_n) - f(x_n)| < \varepsilon$ for $n > N_\varepsilon$. On the other hand, $f_n \rightarrow f$ uniformly and f_n continuous implies f continuous, so $f(x_n) \rightarrow f(x)$. Hence $|f(x_n) - f(x)| < \varepsilon$ whenever $n > N'_\varepsilon$. Take $N = \max(N_\varepsilon, N'_\varepsilon)$ and we have, by triangle inequality, that $|f_n(x_n) - f(x)| < 2\varepsilon$ for all $n > N$. By taking $\varepsilon = \frac{1}{n}$ we can conclude that $f_n(x_n) \rightarrow f(x)$. $\square$

24.E. Consider the sequences (f_n) defined on $D = \{x \in \mathbf{R} : x \geq 0\}$ to $\mathbf{R}$ by the following formulas. Discuss the convergence and the uniform convergence of these sequences and the continuity of the limit functions. In case of non-uniform convergence on D , consider appropriate intervals in D .

(a) $\frac{x^n}{n}$,

Proof. When $x \leq 1$, we have $f_n(x) \rightarrow 0$ as $n \rightarrow \infty$. When $x > 1$, we have $f_n(x) \rightarrow \infty$ as $n \rightarrow \infty$. Hence f_n does not converge to a real-valued function on D . $\square$

(b) $\frac{x^n}{1+x^n}$,

Proof. f_n converges to the following function

$$f(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{2} & x = 1 \\ 1 & x > 1 \end{cases}.$$

Which is not a continuous function. Hence the convergence cannot be uniform. $\square$

(c) $\frac{x^n}{n+x^n}$,

Proof. f_n converges to

$$f(x) = \begin{cases} 0 & x \leq 1 \\ 1 & x > 1 \end{cases}.$$

Hence the convergence is not uniform. $\square$

(d) $\frac{x^{2n}}{1+x^n}$,

Proof. When $x < 1$, we have $x^{2n} \rightarrow 0$ and $x^n \rightarrow 0$ so $f_n(x) \rightarrow \frac{0}{1+0} = 0$.

When $x = 1$, we have $x^{2n} = 1 = x^n$ so $f_n(x) = \frac{1}{2}$.

When $x > 1$, we have $f_n(x) = \frac{x^{2n}}{1+x^n} \sim \frac{x^{2n}}{x^n} = x^n \rightarrow \infty$. The divergence can be shown more rigorously by ratio test.

Hence $f_n(x)$ does not converge when $x > 1$. In particular, f_n does not converge to a real-valued function. $\square$

(e) $\frac{x^n}{1+x^{2n}}$,

Proof. Argument for $x < 1$ and $x = 1$ are the same.

When $x > 1$, we have $f_n(x) \sim \frac{x^n}{x^{2n}} = \frac{1}{x^n} \rightarrow 0$.

Hence f_n converges point wise to the following function

$$f(x) = \begin{cases} \frac{1}{2} & x = 1 \\ 0 & \text{otherwise} \end{cases}.$$

f is not continuous, so the convergence is not uniform. $\square$

(f) $\frac{x}{n}e^{-x/n}$.

Proof. This function converges pointwise to the zero function. However the convergence is not uniform. To see this, observe that for any n ,

$$\sup_x |f_n(x) - f(x)| \geq |f_n(n) - f(n)| = |e^{-1} - 0| = e^{-1}.$$

which does not vanish. $\square$

- 24.N. If $f_3(x) = x^3$ for $x \in I$, calculate the n th Bernstein polynomial for f_3 . Show directly that this sequence of polynomials converges uniformly to f_3 on I .

Proof. We start from the binomial formula

$$1 = \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k}.$$

Replace n by $n-3$:

$$1 = \sum_{k=0}^{n-3} \binom{n-3}{k} x^k (1-x)^{n-3-k}.$$

Apply the identity

$$\binom{n-3}{k-3} = \frac{k(k-1)(k-2)}{n(n-1)(n-2)} \binom{n}{k}.$$

Now

$$1 = \sum_{k=0}^{n-3} \frac{(k+3)(k+2)(k+1)}{n(n-1)(n-2)} \binom{n}{k+3} x^k (1-x)^{n-3-k}.$$

Multiply by x^3 and replace $k+3$ by j :

$$x^3 = \sum_{j=3}^n \frac{j(j-1)(j-2)}{n(n-1)(n-2)} \binom{n}{j} x^j (1-x)^{n-j}.$$

The terms $j=0, j=1, j=2$ can be added because they vanish:

$$x^3 = \sum_{j=0}^n \frac{j(j-1)(j-2)}{n(n-1)(n-2)} \binom{n}{j} x^j (1-x)^{n-j}.$$

Rearrange:

$$n(n-1)(n-2)x^3 = \sum_{k=0}^n k(k-1)(k-2) \binom{n}{k} x^k (1-x)^{n-k}.$$

Hence

$$n(n-1)(n-2)x^3 = n^3 B_n(x, f_3) - 3n^2 B_n(x, f_2) + 2n B_n(x, f_1).$$

Therefore

$$B_n(x, f_3) = \frac{n(n-1)(n-2)}{n^3} x^3 + \frac{3}{n} B(x, f_2) - \frac{2}{n^2} B(x, f_1).$$

As $n \rightarrow \infty$, the last two terms vanish. Also, $\frac{n(n-1)(n-2)}{n^3} \rightarrow 1$ as $n \rightarrow \infty$. Hence $B_n(x, f_3) \rightarrow x^3$ pointwise. Since I is compact, the convergence is uniform. $\square$

24.S. Show that the Weierstrass Approximation Theorem fails for bounded open intervals.

- Proof.* 1. Any polynomial is bounded in $(0, 1)$.
 2. Uniform limit preserves boundedness.
 3. $\frac{1}{x}$ is unbounded in $(0, 1)$.

If there is a uniform approximation of $\frac{1}{x}$ in $(0, 1)$, the limit must be bounded. However, $\frac{1}{x}$ is unbounded, so it cannot be uniformly approximated.

This counterexample shows that not all functions can be uniformly approximated in a bounded open interval. Hence the Weierstrass Approximation Theorem is not true for bounded open interval. $\square$

Homework 9

Sat 05 Nov 2022 00:33

25.A(c,d), 25.V, 25.X (see 25.H), 27.D, 27.H, 28.B, 28.K, 28.O

25.A. Discuss the existence of both the deleted and the non-deleted limits of the following functions at the point $x = 0$.

(c) $f(x) = x \sin(1/x), \quad x \neq 0$

Solution. The two notions of limit are equivalent since $0 \notin \text{Dom } f$.

Take any sequence $x_n \rightarrow 0$. Apply squeeze theorem,

$$0 \leq |f(x_n)| = |x_n| \left| \sin\left(\frac{1}{x_n}\right) \right| \leq |x_n| \rightarrow 0.$$

Thus $f(x_n) \rightarrow 0$. Hence $\lim_{x \rightarrow 0} f(x) = 0$. $\square$

(d) $f(x) = \sin(1/x), \quad x \neq 0$

Solution. The limit does not exist. We have proved in Exercise 12.I that the set $\{(0, y) : -1 \leq y \leq 1\}$ is in the boundary of graph of f . If the limit $\lim_{x \rightarrow 0} f(x)$ exists, it must be unique, so graph of f should only have one cluster point when $x = 0$. However there are infinitely many such cluster points, so the limit does not exist. $\square$

25.V. Let f be as in Definition 25.1 and suppose that the deleted limit at c exists and that for some element A in $\mathbf{R}^q$ and $r > 0$ the inequality $\|f(x) - A\| < r$ holds on some neighborhood of c . Prove that $\left\| \lim_c f - A \right\| \leq r$. Does the same conclusion hold for the non-deleted limit?

Proof. Let $\lim_{x \rightarrow c} f = L$. Pick $\varepsilon > 0$, then there exists $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - c| < \delta$. We may assume without loss of generality that $B_\delta(c)$ is contained in the neighborhood of c where the inequality holds, so we have $|f(x) - A| < r$. By triangle inequality, $|L - A| < r + \varepsilon$. Since ε is arbitrary, $|L - A| \leq r$.

Prof. Petrosyan told us to ignore everything about nondeleted limit, so I skip the second half of the problem. $\square$

- 25.X. If $f : [0, +\infty) \rightarrow \mathbb{R}$ is continuous on $[0, +\infty)$ and $\lim_{x \rightarrow +\infty} f(x) = 0$, show that f is uniformly continuous on $[0, +\infty)$

Proof. Let $N > 0$, and we know that f is uniformly continuous on $[0, N]$ since the interval is compact. Let $\varepsilon > 0$, and there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$ for $x, y \in [0, N]$. Now since $\lim_{x \rightarrow +\infty} f(x) = 0$, we may pick large enough N such that for all $x \geq N$, we have $|f(x)| \leq \frac{\varepsilon}{2}$.

Now let $x, y \geq 0$ such that $|x - y| < \delta$. If $x, y \leq N$ then we are done. If $x, y > N$ then by triangle inequality we have $|f(x) - f(y)| < \varepsilon$. If $x \leq N$ and $y \geq N$, then $|x - N| \leq \delta$. Hence $|f(x) - f(y)| \leq |f(x) - f(N)| + |f(N) - f(y)| \leq \varepsilon + \varepsilon = 2\varepsilon$. Therefore f is uniformly continuous on $[0, +\infty)$. $\square$

- 27.D. Show that the function defined by

$$g(x) = x^2 \sin(1/x), \quad x \neq 0 \\ = 0, \quad x = 0$$

is differentiable for all real numbers, but that g' is not continuous at $x = 0$.

Proof. Using chain rule, we know $g'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x}$. This exists whenever $x \neq 0$, so g is differentiable for all nonzero real numbers. Next, we determine the derivative at 0. Now

$$\lim_{x \rightarrow 0+} \frac{g(x) - g(0)}{x - 0} = \lim_{x \rightarrow 0+} \frac{x^2 \sin \frac{1}{x}}{x} = \lim_{x \rightarrow 0+} x \sin \frac{1}{x} = 0.$$

And similarly $\lim_{x \rightarrow 0-} \frac{g(x) - g(0)}{x - 0} = 0$. Hence $g'(0)$ exists and equal to 0.

On the other hand, however, as $x \rightarrow 0$, the term $2x \sin \frac{1}{x}$ converges to 0 but the term $\cos \frac{1}{x}$ does not converge. Hence $\lim_{x \rightarrow 0} g'(x)$ does not exist, so g' is not continuous at $x = 0$. $\square$

- 27.H. (Darboux) If f is differentiable on $[a, b]$, if $f'(a) = A$, $f'(b) = B$, and if C lies between A and B , then there exists a point c in (a, b) for which $f'(c) = C$. (Hint: consider the lower bound of the function $g(x) = f(x) - C(x - a)$.)

Proof. Assume without loss of generality that $A < B$. f differentiable implies f continuous, hence g as defined in the hint is continuous. Also, we have $g'(a) = f'(a) - C = A - C < 0$ and $g'(b) = f'(b) - C = B - C > 0$. Since $[a, b]$ is compact, g reaches minimum value at some $c \in [a, b]$. g differentiable implies $g'(c) = 0$, hence in particular $c \in (a, b)$.

Now $g'(c) = f'(c) - C = 0$, hence $f'(c) = C$. $\square$

- 28.B. Show that if $x > 0$, then

$$1 + \frac{x}{2} - \frac{x^2}{8} \leq \sqrt{1+x} \leq 1 + \frac{x}{2}$$

Proof. Let $f(x) = \sqrt{1+x}$. Differentiate and find

$$\begin{aligned}f^{(1)}(x) &= \frac{1}{2}(1+x)^{-\frac{1}{2}} \\f^{(2)}(x) &= -\frac{1}{4}(1+x)^{-\frac{3}{2}} \\f^{(3)}(x) &= \frac{3}{8}(1+x)^{-\frac{5}{2}}.\end{aligned}$$

Now by Taylor's Theorem,

$$\begin{aligned}f(x) &= f(0) + \frac{f'(0)}{1!}(x-0) + \frac{f^{(2)}(0)}{2!}(x-0)^2 \\&= 1 + \frac{1}{2}x - \frac{1}{8}(1+x)^{\frac{3}{2}}x^2 < 1 + \frac{x}{2}.\end{aligned}$$

By Taylor's Theorem again,

$$\begin{aligned}f(x) &= f(0) + \frac{f'(0)}{1!}(x-0) + \frac{f^{(2)}(0)}{2!}(x-0)^2 + \frac{f^{(3)}(0)}{3!}(x-0)^3 \\&= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}(1+x)^{\frac{5}{2}}x^3 > 1 + \frac{x}{2} - \frac{1}{8}x^2.\end{aligned}$$

□

- 28.K. If $f(x) = e^x$, show that the remainder term in Taylor's Theorem converges to zero as $n \rightarrow \infty$ for each fixed α, β .

Proof. Assume without loss of generality that $\alpha < \beta$. We have $f^{(n)}(x) = f(x) = e^x$. The remainder term is

$$\frac{f^{(n)}(\gamma_n)}{n!}(\beta - \alpha)^n = e^{\gamma_n} \frac{(\beta - \alpha)^n}{n!}.$$

Now f is uniformly continuous on $[\alpha, \beta]$ and thus $f(x)$ is bounded on this interval, hence e^{γ_n} is bounded by some fixed constant B . Define $C = \beta - \alpha$, where $C > 0$, and it suffices to show that

$$\frac{C^n}{n!} \rightarrow 0.$$

Now we verify this fact. Let N be some natural number such that $N > C$. Define $M = \frac{C^N}{N!}$. Now there exists $m > N$ such that $\frac{C}{m} < \frac{\varepsilon}{M}$. Hence for all $n > m$,

$$\frac{C^n}{n!} = \frac{C^N}{N!} \prod_{k=N+1}^n \frac{C}{k} < M \frac{C}{m} < \varepsilon.$$

Hence $\frac{C^n}{n!} \rightarrow 0$. □

- 28.O. If $f : \mathbf{R} \rightarrow \mathbf{R}$, if $f'(x)$ exist for $x \in \mathbf{R}$, and if $f''(a)$ exists, show that

$$f''(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - 2f(a) + f(a-h)}{h^2}.$$

Give an example where this limit exists, but the function does not have a second derivative at a .

Proof.

$$\begin{aligned}
 f''(a) &= \lim_{h \rightarrow 0} \frac{f'(a+h) - f'(a)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\lim_{k \rightarrow 0} \frac{f(a+h) - f(a+h-k)}{k} - \lim_{k \rightarrow 0} \frac{f(a) - f(a-k)}{k}}{h} \\
 &= \lim_{h \rightarrow 0} \lim_{k \rightarrow 0} \frac{f(a+h) - f(a+h-k) - f(a) + f(a-k)}{hk} \\
 &= \lim_{h \rightarrow 0} \frac{f(a+h) - 2f(a) + f(a-h)}{h^2}.
 \end{aligned}$$

□

Example. Consider

$$f(x) = \begin{cases} x^2 & x \leq 0 \\ x^3 & x \geq 0 \end{cases}.$$

Now

$$f'(x) = \begin{cases} 2x & x \leq 0 \\ 3x^2 & x \geq 0 \end{cases}.$$

The limit exists:

$$\lim_{h \rightarrow 0} \frac{f(h) - 2f(0) + f(-h)}{h^2} = \lim_{h \rightarrow 0} \frac{h^3 + h^2}{h^2} = 1.$$

However,

$$f''(x) = \begin{cases} 2 & x < 0 \\ 6x & x > 0 \end{cases}.$$

So $f''(x)$ is not defined at $x = 0$, since the left and right limit aren't equal.

◇

Homework 10

Wed 30 Nov 2022 21:17

- 29.E. If f is integrable on $[a, b]$ with respect to f , then

$$\int_a^b f df = \frac{1}{2} \{ (f(b))^2 - (f(a))^2 \}.$$

(a) Prove this by examining the two Riemann-Stieltjes sums for a partition $P = (x_0, x_1, \dots, x_n)$ obtained by taking $\xi_k = x_{k-1}$ and $\xi_k = x_k$.

Proof. Define partition $P : a = x_0 < x_1 < \dots < x_n = b$. Now let $S(P, f, f)$ be the Riemann sum taking $\xi_k = x_{k-1}$ and $S'(P, f, f)$ be the Riemann sum taking $\xi = x_k$. We have

$$S(P, f, f) = \sum_{k=1}^n f(x_{k-1}) (f(x_k) - f(x_{k-1})).$$

and

$$S'(P, f, f) = \sum_{k=1}^n f(x_k) (f(x_k) - f(x_{k-1})).$$

Since f integrable wrt f , as $\max_k \Delta f_k \rightarrow 0$, we have $S(P, f, f) \rightarrow I$ and $S'(P, f, f) \rightarrow I$, where $I = \int_a^b f df$. Hence

$$S + S' = \sum_{k=1}^n f(x_k)^2 - f(x_{k-1})^2 = f(x_n)^2 - f(x_0)^2 = f(b)^2 - f(a)^2 \rightarrow 2I.$$

Hence $I = \frac{1}{2} (f(b)^2 - f(a)^2)$ □

(b) Prove this by using the Integration by Parts Theorem 29.7.

Proof. By integration by parts formula,

$$f f|_a^b = \int_a^b f df + \int_a^b f df = 2I.$$

Hence

$$I = \frac{1}{2} (f(b)f(b) - f(a)f(a)).$$

□

- 29.M. Suppose that f is g -integrable on $[a, b]$. If $g_1 : [a, b] \rightarrow \mathbf{R}$ is such that $g_1(x) = g(x)$ except for a finite number of points in (a, b) at which f is continuous, then f is g_1 -integrable and

$$\int_a^b f dg_1 = \int_a^b f dg.$$

Proof.

$$S(P, f, g) - S(P, f, g_1)$$

$$\begin{aligned} &= \sum_{k=1}^n f(\xi_k) [g(x_k) - g(x_{k-1})] - \sum_{k=1}^n f(\xi_k) [g_1(x_k) - g_1(x_{k-1})] \\ &= \sum_{k=1}^n f(\xi_k) [g(x_k) - g(x_{k-1}) - g_1(x_k) + g_1(x_{k-1})] \\ &= \sum_{k=1}^n f(\xi_k) [g(x_k) - g_1(x_k)] - \sum_{k=1}^n f(\xi_k) [g(x_{k-1}) - g_1(x_{k-1})] \\ &= \sum_{k=1}^n f(\xi_k) [g(x_k) - g_1(x_k)] - \sum_{k=0}^{n-1} f(\xi_{k+1}) [g(x_k) - g_1(x_k)] \\ &= f(\xi_n) [g(x_n) - g_1(x_n)] - f(\xi_0) [g(x_0) - g_1(x_0)] + \sum_{k=1}^{n-1} [g(x_k) - g_1(x_k)] [f(\xi_k) - f(\xi_{k+1})] \end{aligned}$$

since g_1 and g may mismatch only inside (a, b) , the first two terms vanish:

$$= \sum_{k=1}^{n-1} [g(x_k) - g_1(x_k)] [f(\xi_k) - f(\xi_{k+1})].$$

By continuity of f , we may choose P fine enough such that $|f(\xi_k) - f(\xi_{k-1})| < \varepsilon$ for all k . Hence

$$\begin{aligned} & |S(P, f, g) - S(P, f, g_1)| \\ & \leq \sum_{k=1}^{n-1} |g(x_k) - g_1(x_k)| |f(\xi_k) - f(\xi_{k+1})| \\ & < \varepsilon \sum_{k=1}^{n-1} |g(x_k) - g_1(x_k)|. \end{aligned}$$

Now since $g(x) \neq g_1(x)$ for at most m points in (a, b) , where m does not depend on P , we have

$$\sum_{k=1}^{n-1} |g(x_k) - g_1(x_k)| \leq \sum_{x: g(x) \neq g_1(x)} |g(x) - g_1(x)|.$$

which is a bounded quantity independent of P or ε . Hence

$$|S(P, f, g) - S(P, f, g_1)| \rightarrow 0.$$

This means

$$\left| S(P, f, g_1) - \int_a^b f dg \right| \rightarrow 0.$$

Hence f is integrable wrt g_1 . From the proof above it is obvious that the integrals converge to the same value. $\square$

- 29.S. Let f be Riemann integrable on J and let $f(x) \geq 0$ for $x \in J$. If f is continuous at a point $c \in J$ and if $f(c) > 0$, then

$$\int_a^b f > 0$$

Proof. Let $0 < \varepsilon < f(c)$. By continuity of f there exists $\delta > 0$ such that $f(x) \geq f(c) - \varepsilon$ on $[c - \delta, c + \delta] \cap J$. Let $c_* = \max\{a, c - \delta\}$ and $c^* = \min\{b, c + \delta\}$. Now

$$\begin{aligned} \int_a^b f &= \int_a^{c_*} f + \int_{c_*}^{c^*} f + \int_{c^*}^b f \\ &\geq \int_{c_*}^{c^*} f \\ &\geq (c^* - c_*) (f(c) - \varepsilon) \\ &> 0. \end{aligned}$$

$\square$

- 30.A. Show that a bounded function which has at most a finite number of discontinuities is Riemann integrable.

Proof. We use without verify the fact that if f is bounded on a closed interval $[a, b]$ and discontinuous only at a and b , then f is integrable on $[a, b]$.

Consider $J = [a, b]$, and let the points of discontinuities be $y_1, \dots, y_{m-1}$, and for convenience write $a = y_0 \leq y_1 < y_2 < \dots < y_{m-1} \leq y_m = b$. We know that f restricted to $J_k := [y_{k-1}, y_k]$ is integrable, for $k = 1, \dots, m$.

For each $\varepsilon > 0$, for each k , choose P_ε^k such that for any partition $P > P_\varepsilon^k$,

$$|S(P, f, x) - S(P_\varepsilon^k, f, x)| < \frac{\varepsilon}{m}.$$

Now let $P_\varepsilon = \bigcup_k P_\varepsilon^k$. Then P_ε is a partition of J . Now for any partition $Q > P_\varepsilon$, we let Q^k be Q restricted to J_k , and Q^k is a partition of J_k , and $Q^k \geq P_\varepsilon^k$. Thus for all k ,

$$|S(Q^k, f, x) - S(P_\varepsilon^k, f, x)| < \frac{\varepsilon}{m}.$$

Now we have

$$\begin{aligned} & |S(Q, f, x) - S(P_\varepsilon, f, x)| \\ &= \left| \sum_{k=1}^m S(Q^k, f, x) - \sum_{k=1}^m S(P_\varepsilon^k, f, x) \right| \\ &\leq \sum_{k=1}^m |S(Q^k, f, x) - S(P_\varepsilon^k, f, x)| \\ &< \varepsilon. \end{aligned}$$

Hence for all $Q, Q' > P_\varepsilon$, we have

$$|S(Q, f, x) - S(Q', f, x)| < 2\varepsilon.$$

Hence the Cauchy criterion for integrability is satisfied. Hence f is Riemann integrable on J . $\square$

- 30.D. Give an example of a function f which is not Riemann integrable over J but such that $|f|$ and f^2 are Riemann integrable over J .

Proof. Define

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \notin \mathbb{Q} \end{cases}.$$

$\square$

- 30.F. Show that the First Mean Value Theorem 30.6 may fail if f is not continuous.

Proof. Let $g(x) = x$ and $f(x) = 1$ when $x > \frac{1}{2}$ and $f(x) = 0$ when $x \leq \frac{1}{2}$. Consider $J = [0, 1]$. Then

$$\int_0^1 f dg = \int_0^{\frac{1}{2}} 0 dg + \int_{\frac{1}{2}}^1 1 dg = \frac{1}{2}.$$

If the First Mean Value Theorem holds then we have $\frac{1}{2} = (g(b) - g(a)) f(c) = f(c)$ for some $c \in J$, but f never attains the value $\frac{1}{2}$. $\square$

- 31.D. If $a > 0$, show that

$$\lim_n \int_a^\pi \frac{\sin nx}{nx} dx = 0.$$

What happens if $a = 0$?

Proof. Since $f(x) = \frac{\sin nx}{nx}$ is continuous and bounded when $x \neq 0$, f is integrable. In $[a, \pi]$ we have

$$\left| \frac{\sin nx}{nx} \right| \leq \left| \frac{1}{nx} \right| \leq \frac{1}{na}.$$

Hence by increasing n we can make it vanish. Thus

$$\int_a^\pi \left| \frac{\sin nx}{nx} \right| dx \leq \frac{\pi - a}{na} \rightarrow 0.$$

This implies the desired result.

When $a = 0$, since f is undefined at 0, the integral becomes improper. We may be interested in evaluating instead

$$\lim_{n \rightarrow \infty} \lim_{a \rightarrow 0+} \int_a^\pi f(x) dx.$$

We may also redefine f at 0 to $f(0) = 1$, so f becomes continuous and well-defined in $[0, \pi]$. As $n \rightarrow \infty$, f converges point-wise to the function $\mathcal{X}_0$ which is 1 at 0 and 0 at everywhere else. Then for each ε and each a we may choose n sufficiently large such that $f(x) < \varepsilon$ whenever $x > a$. Hence for sufficiently large n ,

$$\int_0^\pi |f(x)| dx \leq \int_0^a 1 dx + \int_a^\pi \varepsilon dx = a + \varepsilon(\pi - a).$$

Both a and ε can be made arbitrarily small. Hence

$$\lim_{n \rightarrow \infty} \int_0^\pi f(x) dx = 0.$$

□

- 31.F. Let $h_n(x) = nxe^{-nx^2}$ for $x \in [0, 1]$ and let $h(x) = 0$. Show that

$$0 = \int_0^1 h(x) dx \neq \lim \int_0^1 h_n(x) dx = \frac{1}{2}$$

Proof. On one hand,

$$\lim_{n \rightarrow \infty} h_n(x) = \lim_{n \rightarrow \infty} \frac{nx}{e^{nx^2}} = 0.$$

So

$$\int_0^1 h(x) dx = \int_0^1 0 dx = 0.$$

On the other hand,

$$\int_0^1 h_n(x) dx = -\frac{1}{2}e^{-n} + \frac{1}{2}.$$

Hence

$$\lim_{n \rightarrow \infty} \int_0^1 h_n(x) dx = \frac{1}{2}.$$

□

- 31.K. Use the Fundamental Theorem 30.8 to show that if a sequence (f_n) of functions converges on J to a function f and if the derivatives (f'_n) are continuous and converge uniformly on J to a function g , then f' exists and equals g . (This result is less general than Theorem 28.5, but it is easier to establish.)

Proof. By fundamental theorem of calculus, since f'_n continuous,

$$f_n(x) - f_n(a) = \int_a^x f'_n.$$

Since $f_n \rightarrow f$, we have

$$\lim_{n \rightarrow \infty} f_n(x) - f_n(a) = f(x) - f(a).$$

Since $f'_n \rightarrow g$ uniformly,

$$\lim_{n \rightarrow \infty} \int_a^x f'_n = \int_a^x \lim_{n \rightarrow \infty} f'_n = \int_a^x g.$$

Also g continuous due to uniform convergence. Combined with the result that

$$f(x) - f(a) = \int_a^x g.$$

We conclude, by FTC, that $f' = g$.

□

Homework 11

Fri 02 Dec 2022 11:31

Lecture 1: lecture

Sun 21 Aug 2022 23:56

0.1 Real Numbers

0.1.1 triangle inequality

Lemma 14. If $c > 0$, $|a| \leq c \Leftrightarrow -c \leq a \leq c$.

Theorem 1 (Triangle Inequality). For $a, b \in \mathbb{R}$, $|a + b| \leq |a| + |b|$.

Proof.

$$\begin{aligned} |a + b| \leq |a| + |b| &\Leftrightarrow -|a| - |b| \leq a + b \leq |a| + |b| \\ &\Leftrightarrow -|a| \leq a \leq |a| \wedge -|b| \leq b \leq |b|. \end{aligned}$$

□

Theorem 2 (Reverse triangle inequality).

$$||a| - |b|| \leq |a - b|.$$

Proof.

$$\Leftrightarrow -|a - b| \leq |a| - |b| \leq |a - b|$$

By symmetry we prove the right part

$$\Leftrightarrow |a| \leq |a - b| + |b|$$

$$\Leftrightarrow |a - 0| \leq |a - b| + |b - 0|$$

Which follows from triangle inequality.

□

0.1.2 Completeness of $\mathbb{R}$

Definition 1 (supremum). If $S \subset \mathbb{R}$ and $u \in \mathbb{R}$ then $u = \sup S$ if

1. u is an upper bound for S
2. Any $v < u$ is not an upper bound for S .

Proposition 2 (equivalent condition for supremum). $u = \sup S$ iff

1. $s \leq u$ for any $s \in S$
2. For any $v < u$, there exists $s_v \in S$ such that $v < s_v \leq u$.

Axiom 1 (supremum property). Any subset of $\mathbb{R}$ bounded above has a supremum.

Note: there are many ways to realize this axiom in specific models, for example the construction of $\mathbb{R}$ from $\mathbb{Q}$ by Dedekind cuts or equivalence classes of Cauchy sequences. We may also construct $\mathbb{R}$ from subset of $\mathbb{N}$. In this course, however, we do not concern about the specific construction of $\mathbb{R}$.

Definition 2 (infimum). If $S \subset \mathbb{R}$ and $u \in \mathbb{R}$ then $u = \inf S$ if

1. u is a lower bound for S
2. Any $v > u$ is not a lower bound for S .

Proposition 3 (equivalent condition for infimum). $u = \inf S$ iff

1. $s \geq u$ for any $s \in S$
2. For any $v > u$, there exists $s_v \in S$ such that $v > s_v \geq u$.

Theorem 3 (infimum property). Any nonempty subset of $\mathbb{R}$ bounded below has an infimum.

Proof. Let $-S = \{-s : s \in S\}$. Then check that $-(\sup(-S)) = \inf S$. $\square$

Remark. Supremum and infimum are unique for any bounded set. Formally we write $\sup S = \infty$ if S is unbounded above; we write $\sup S = -\infty$ if $S = \emptyset$.

Lecture 2: Completeness of $\mathbb{R}$

Sat 27 Aug 2022 13:14

0.1.3 Archimedean Property

Theorem 4 (Archimedean property). For every $x \in \mathbb{R}$ there is $n \in \mathbb{N}$ such that $x < n$.

Proof. Suppose for the sake of contradiction that there exists some $x \in \mathbb{R}$ that bound $\mathbb{N}$ from above, i.e. $n \leq x$ for any $n \in \mathbb{N}$.

Let $u = \sup \mathbb{N}$. Hence there exists $n_1 \in \mathbb{N}$ such that $u - 1 < n_1 \leq u$. But $n + 1 \in \mathbb{N}$ and $n + 1 > u$, so u is not an upper bound of $\mathbb{N}$. $\square$

Corollary 2. For any $y > 0, x \in \mathbb{R}$ there exists $n \in \mathbb{N}$ such that $ny > x$.

Proof. There exists $n > \frac{x}{y}$. $\square$

Corollary 3. For any $x > 0$ there is $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < x$.

Proof. There exists $n > \frac{1}{x}$. $\square$

Corollary 4 (of the previous corollary). $\inf\{\frac{1}{n} : n \in \mathbb{N}\} = 0$.

Proof. Verify the definition of infimum.

1. $0 \leq \frac{1}{n}$ for $n \in \mathbb{N}$.
2. For any $v > 0$, by the previous corollary, exists n such that $0 < \frac{1}{n} < v$.

 $\square$

Theorem 5 (Density of rationals in the reals.). For any $a, b \in \mathbb{R}$, $a < b$, there exists $r \in \mathbb{Q}$ such that $a < r < b$.

Proof. Assume first that a is a positive number. Choose step size $y = \frac{1}{n} < b - a$. By the first corollary there exists $m \in \mathbb{N}$ such that $a < \frac{m}{n}$.

Let m be smallest such $m \in \mathbb{N}$ (well-ordered property of $\mathbb{N}$). Then $\frac{m-1}{n} \leq a < \frac{m}{n}$. Now $\frac{m}{n} \leq a + \frac{1}{n} < b$.

If $a < 0$ and $b > 0$ then $0 = r \in \mathbb{Q}$. If $a < 0$ and $b \leq 0$ then we flip sign. There exists $r \in \mathbb{Q}$ such that $-b < r < -a$ and hence $a < -r < b$.

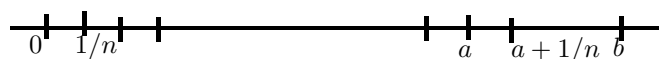


Figure 1: density-of-rationals

□

Theorem 6 (Existence of $\sqrt{2}$). There exists $\alpha \in \mathbb{R}$ such that

$$\alpha \geq 0, \alpha^2 = 2.$$

Proof. Let $S = \{s \geq 0 : s^2 < 2\}$. Since S is nonempty and bounded from above by 2, by Supremum Property, there is a supremum $\alpha = \sup S$. We claim that $\alpha^2 = 2$.

Suppose not, then by Trichotomy, either $\alpha^2 > 2$ or $\alpha^2 < 2$. Suppose $\alpha^2 < 2$. Then we can find large n such that $\alpha + \frac{1}{n} < 2$. Suppose $\alpha^2 > 2$, we can find $(\alpha - \frac{1}{m}) > 2$. In both cases $\alpha \neq \sup S$. □

Remark. We used $\frac{1}{n}$ instead of $\varepsilon \in \mathbb{R}$ in the proof. This is not strictly necessary. However, if we construct the reals using Dedekind cuts, we have to work with rationals.

0.2 Cantor Set and Nested Intervals

Definition 3 (interval). For a, b in $\mathbb{R}$, define

- (closed interval) $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$
- (open interval) $(a, b) = \{x \in \mathbb{R} : a < x < b\}$
- (semi-open / semi-closed) $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$
- $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$
- (semi-infinite / rays) $[a, \infty) = \{x \in \mathbb{R} : x \geq a\}$
- $(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}$
- (infinite) $(-\infty, \infty) = \mathbb{R}$

Theorem 7 (Nested Interval Property). Suppose we have a sequence of closed finite nonempty intervals $I_1, I_2, \dots$, such that

$$I_1 \supset I_2 \supset \dots \supset I_n \supset \dots$$

Then there exists $x \in \mathbb{R}$ such that $x \in I_n$ for any $n \in \mathbb{N}$. In other words, $\bigcap_{n \in \mathbb{N}} I_n \neq \emptyset$

Proof. Let $I_n = [a_n, b_n]$. Suppose WOLOG that $a_n \leq b_n$ then we have

$$\begin{aligned} a_1 &\leq a_2 \leq \dots \leq a_n \leq \dots \\ b_1 &\geq b_2 \geq \dots \geq b_n \geq \dots \end{aligned}$$

Claim. $a_m \leq b_k$ for any $m, k \in \mathbb{N}$. *Proof:* Let $n = \max\{m, k\}$. Then $a_m \leq a_n \leq b_n \leq b_k$.

b_k is an upper bound for $A = \{a_n : n \in \mathbb{N}\}$. $A \neq \emptyset$, so there exists, by supremum property, some $\xi = \sup A$. By definition of supremum, $a_n \leq \xi$ for any $n \in \mathbb{N}$.

On the other hand, b_k is an upper bound, so $\xi \leq b_k$ for any $k \in \mathbb{N}$. Now $a_n \leq \xi \leq b_n$, so $\xi \in I_n$. $\square$

Lecture 3

Fri 02 Sep 2022 12:19

Definition 4 (Cantor set). Let

1. $F_0 = [0, 1]$
2. Remove middle $\frac{1}{3}$ of the interval, to obtain

$$F_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1].$$

3. Continue in this fashion, we obtain F_n as a union of 2^n intervals of length $\frac{1}{3^n}$.

$$F = \bigcap_{n=0}^{\infty} F_n$$

is called the Cantor Set.

Note: F_n become "smaller" in the linear measure sense. Sum of the length of intervals in F_n is

$$\frac{2^n}{3^n} = \left(\frac{2}{3}\right)^n \rightarrow 0.$$

(we have not defined limits yet) Hence F has linear measure 0.

Theorem 8. The Cantor set F is uncountable.

Claim. $F \neq \emptyset$ because any endpoint of any interval in any of F_n is in F .

Proposition 4. Let $\mathcal{P} = \{(\alpha_1, \alpha_2, \dots, \alpha_n, \dots) \mid \alpha_i = 0, 1\}$ be the power set of $\mathbb{N}$. Then there is a 1-1 correspondence between $\mathcal{P}$ and Cantor set F .

Proof. Let

1. $F_0 = [0, 1] =: I$
2. $F_1 = [0, 1/3] \cup [2/3, 1] =: I_0 \cup I_1$
3. $F_2 = I_{00} \cup I_{01} \cup I_{10} \cup I_{11}$
4. $F_n = \bigcup_{\alpha_1 \alpha_2 \dots \alpha_n = 0, 1} I_{\alpha_1 \alpha_2 \dots \alpha_n}$
- 5.

$$I_{\alpha_1 \alpha_2 \dots \alpha_n \alpha_{n+1}} = \begin{cases} \text{left } \frac{1}{3} \text{ of } I_{\alpha_1 \dots \alpha_n} & \alpha_{n+1} = 0 \\ \text{right } \frac{1}{3} \text{ of } I_{\alpha_1 \dots \alpha_n} & \alpha_{n+1} = 1 \end{cases}.$$

Now we prove the proposition.

1. Take any $(a_i)_{i \in \mathbb{N}} \in \mathcal{P}$. Then

$$I_{\alpha_1} \supset I_{\alpha_1 \alpha_2} \supset \dots$$

By nested interval property, $\bigcap_{n \in \mathbb{N}} I_{\alpha_1 \alpha_2 \dots \alpha_n} \neq \emptyset$. Take an element x_α from the intersection.

In fact x_α is the unique element in the intersection. Suppose there is $y \neq x_\alpha$ and $y \in I_{\alpha_1 \alpha_2 \dots \alpha_n}$ for all n . Now

$$|x_\alpha - y| \leq m(I_{\alpha_1 \dots \alpha_n}).$$

Hence for all $n \in \mathbb{N}$, $|x_\alpha - y| \leq \frac{1}{3^n}$. By bernoulli's inequality,

$$3^n = (1 + 2)^n \geq 1 + 2n > n.$$

So for any $n \in \mathbb{N}$,

$$0 \leq |x_\alpha - y| < \frac{1}{n}.$$

This is impossible since $|x_\alpha - y| > 0$ so there is $n \in \mathbb{N}$ such that

$$\frac{1}{n} < |x_\alpha - y|.$$

We have a contradiction.

2. For any $x \in F$ we know

$$x \in F_n \Rightarrow x \in I_{\alpha_1^n \alpha_2^n \dots \alpha_n^n}.$$

Claim. $I_{\beta_1 \beta_2 \dots \beta_{n+1}} \cap I_{\alpha_1 \alpha_2 \dots \alpha_n} \neq \emptyset \Leftrightarrow \beta_1 = \alpha_1, \dots, \beta_n = \alpha_n$.

Hence $\alpha_k^n = \alpha_k^{n+1} =: \alpha_k$.

Hence there exists $(\alpha_n)_{n \in \mathbb{N}} \in \mathcal{P} : x \in I_{\alpha_1 \alpha_2 \dots \alpha_n}, n \in \mathbb{N}$. Hence the mapping $\alpha \mapsto x_\alpha$ is onto.

It is easy to see that $\alpha \mapsto x_\alpha$ is injective, so it is a bijection.

□

Lecture 4: Finite, Countable and Uncountable Sets

Wed 07 Sep 2022 11:38

0.3 Finite and countable sets

Definition 5. We say that two sets A, B are *equivalent* (or have the same number of elements) if there is a bijection

$$\varphi : A \rightarrow B.$$

We write $A \sim B$ for this equivalence relation.

Proposition 5. Reflexivity: $A \sim A$ (ι_A)

Symmetry: If $A \sim B$ then $B \sim A$. ($\varphi^{-1} : B \rightarrow A$ is a bijection)

Transitivity: If $A \sim B$ and $B \sim C$ then $A \sim C$. We compose the bijections for the proof.

Define $S_n = \{1, 2, 3, \dots, n\}$ be the initial segment of $\mathbb{N}$, and let $S_0 = \emptyset$. We say that a set E is finite if there is a bijection $S_n \rightarrow E$ for some $n \in \mathbb{N}$. In other words, $E \sim S_n$.

We say that E is *denumerable* if there is a bijection $\mathbb{N} \rightarrow E$, i.e., $E \sim \mathbb{N}$. If the set is not finite, it is called infinite.

Note $\mathbb{N}$ is infinite: does not exist bijection $S_n \rightarrow \mathbb{N}$.

But there is a bijection between $\mathbb{N}$ and its proper subset.

We say set is *countable* if it is finite or denumerable. The infinite sets that are not countable are called *uncountable*.

Proposition 6. $S_n \sim S_m \Leftrightarrow n = m$ for $n, m \in \{0\} \cup \mathbb{N} \cup \{\infty\}$.

Proof. (partial) Show if $n \in \mathbb{N}$ then S_n is not equivalent to $\mathbb{N}$. Suppose the contrary and

$$S_n \rightarrow \mathbb{N}$$

is a bijection. Then look at $N = \max\{\varphi(1), \dots, \varphi(n)\} + 1$. Now $N > \varphi(k)$ for any k . Therefore $N \notin \varphi(S_n)$. Therefore $\varphi(S_n) \neq \mathbb{N}$. $\square$

Proposition 7. 1. Any subset of a finite set is finite.
2. Any infinite subset of a denumerable set is denumerable.
3. Any subset of a countable set is countable.

Note that (1) and (2) imply (3).

Proof. (of (2)). Enough to show that if $E \subset \mathbb{N}$ then E is countable. We use the Well-ordering property of $\mathbb{N}$: any non-empty subset has a minimum element.

E is infinite, so E is nonempty. Therefore exists minimal $n_1 \in E$. E is not finite, so $E \setminus \{n_1\}$ is nonempty. Then there exists $n_2 \in E \setminus \{n_1\}$.

Continue in this fashion. If we constructed $n_1, \dots, n_k \in E$, such that $n_1 < n_2 < \dots < n_k$, then $E \setminus \{n_1, \dots, n_k\}$ is nonempty since E is infinite. Therefore there exists n_{k+1} be the smallest element of $E \setminus \{n_1, \dots, n_k\}$.

Claim. $E = \{n_1, \dots, n_k, \dots\}$

Lemma 15. $n_k \geq k$ for any $k \in \mathbb{N}$. proof by induction.

Note that by construction, $n_1, \dots, n_k, \dots \subset E$. We need to show that other direction.

Let $m \in E$. By Archimedean property, $\exists k \in \mathbb{N}$ s.t. $m < k \leq n_k$.

Let k_0 be the smallest number such that $m < n_{k_0}$. This means that $n_{k_0-1} \leq m$. If $n_{k_0-1} \leq m$ then we are done. If $n_{k_0-1} < m$, then $m \in E \setminus \{n_1, \dots, n_{k_0-1}\}$, hence $m < n_{k_0}$, a contradiction with construction of n_{k_0} . $\square$

$\square$

Proposition 8. 1. Finite union of finite sets is finite.
 2. Denumerable union of denumerable sets is denumerable.
 3. Countable union of countable sets is countable.

To prove (2) of this proposition, we need

Lemma 16. $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$.

Proof. *Proof.* Use diagonal construction. □

Let

$$E = \bigcup_{n \in \mathbb{N}} E_n$$

Note that E_n might overlap. We define

$$\varphi_n : \mathbb{N} \rightarrow E_n.$$

and define

$$\begin{aligned} \varphi : \mathbb{N} \times \mathbb{N} &\longrightarrow \bigcup_{n \in \mathbb{N}} E_n = E \\ n, m &\longmapsto \varphi(n, m) = \varphi_n(m). \end{aligned}$$

We may view φ as $\varphi : \mathbb{N} \rightarrow E$ due to isomorphism between $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$.

Now for $e \in E$, pick the smallest element in $\varphi^{-1}(e)$, call it $f(e)$. $f : E \rightarrow \mathbb{N}$ and $\varphi \circ f = 1_E$. Hence $E \sim f(E) \subset \mathbb{N}$, hence E is countable and infinite, hence E is denumerable. □

Corollary 5. $\mathbb{Q}$ is countable, since

$$\mathbb{Q} = \bigcup_{n \in \mathbb{N}} \frac{1}{n} \mathbb{Z}.$$

Theorem 9 (Cantor). $\mathcal{P}$ is uncountable, hence F is uncountable.

Proof. By Cantor's diagonal argument. Suppose $\mathcal{P}$ is countable, then

$$\mathcal{P} = \{\alpha^{(1)}, \alpha^{(2)}, \dots, \alpha^{(n)}, \dots\}.$$

and

$$\alpha_1^{(1)} = \{\alpha_1^{(1)}, \alpha_2^{(1)}, \dots, \alpha_1^{(n)}, \dots\}.$$

We construct $\beta \in \mathcal{P}$ such that $\beta \neq \alpha^{(n)}$ for any n . Take $\beta_1 = \bar{\alpha}_1^{(1)} = 1 - \alpha_1^{(1)}$, and $\beta_n = \bar{\alpha}_n^{(n)} = 1 - \alpha_n^{(n)}$.

Claim: β does not equal to $\alpha^{(n)}$ for any n . Proof: Note that β and $\alpha^{(n)}$ differ at the n -th element.

Now we have a contradiction since $\beta \in \mathcal{P}$. Hence $\mathcal{P}$ cannot be countable. $\square$

Corollary 6. $\mathbb{R}$ is uncountable. $\mathbb{R} \setminus \mathbb{Q}$ is uncountable.

Lecture 5: Vector Spaces

Fri 09 Sep 2022 11:32

1 Vector Spaces

1.1 Definitions and Examples

Definition 6. $(V, +, \cdot)$, $+$: $V \times V \rightarrow V$, $\cdot$: $\mathbb{R} \times V \rightarrow V$. We say V is a *vector space* (elements called *vectors*) if

1. (A1) $u + v = v + u$
2. (A2) $(u + v) + w = u + (v + w)$
3. (A3) $\exists 0 \in V : u + 0 = 0 + u = u$ for all $u \in V$
4. (A4) For all $u \in V$, there is $-u \in V$ such that $u + (-u) = (-u) + u = 0$
5. (M1) $\alpha(\beta v) = (\alpha\beta)v$
6. (M2) $1v = v$
7. (D1) $\alpha(u + v) = \alpha u + \alpha v$
8. (D2) $(\alpha + \beta)u = \alpha u + \beta u$

These are the axioms of vector space. (A1)-(A4) are axioms of abelian group.

Example. (trivial) $V = \{0\}$, $V = \mathbb{R}$ are vector spaces. $\diamond$

Definition 7. Cartesian (Euclidean) space $\mathbb{R}^p$ is defined as

$$\mathbb{R}^p, p \in \mathbb{N}.$$

where $\mathbb{R}^p$ is set of all p -tuples from elements in $\mathbb{R}$.

Addition and multiplication by scalar are defined componentwise:

$$\begin{aligned} x &= (x_1, x_2, \dots, x_p) \\ y &= (y_1, y_2, \dots, y_p) \end{aligned}$$

Define

$$\begin{aligned} x + y &:= (x_1 + y_1, \dots, x_p + y_p) \\ \alpha x &:= (\alpha x_1, \alpha x_2, \dots, \alpha x_p). \end{aligned}$$

Remark. Generalize the notion, we can let S be some set, define

$$\mathbb{R}^S = \{f : S \rightarrow \mathbb{R}\}.$$

Define the operations point-wise:

$$\begin{aligned} (f + g)(s) &= f(s) + g(s) \\ (\alpha f)(s) &= \alpha f(s). \end{aligned}$$

1.2 Inner products

Definition 8. Suppose V is a vector space, and we have a product $\cdot : V \times V \rightarrow \mathbb{R}$. We say $\cdot$ is an *inner product* if it satisfies:

- (I1) $u \cdot u \geq 0$ for any $u \in V$. Moreover, $u \cdot u = 0 \Leftrightarrow u = 0$.
- (I2) $u \cdot v = v \cdot u$
- (I3) (bilinearity) $u \cdot (\alpha v + \beta w) = \alpha(u \cdot v) + \beta(u \cdot w)$
- (I3, cont.) $(\alpha v + \beta w) \cdot u = \alpha(v \cdot u) + \beta(w \cdot u)$

Example. (trivial) $\mathbb{R}$ with usual addition and multiplication is an inner product space. $\diamond$

In Cartesian spaces, for $x, y \in \mathbb{R}^p$, define

$$x \cdot y = \sum_{k=1}^p x_k y_k.$$

This is the *standard inner product* on $\mathbb{R}^p$.

Definition 9 (Norm). Suppose V is a vector space, and we have $\|\cdot\| : V \rightarrow \mathbb{R}$. We say $\|\cdot\|$ is a *norm* if the following properties are satisfied:

- (N1) $\|u\| > 0$ for any $u \in V$. Moreover, $\|u\| = 0 \Leftrightarrow u = 0$.
- (N2) $\|\alpha u\| = \alpha\|u\|$ where $\alpha \in \mathbb{R}, u \in V$.
- (N3) (triangle inequality) $\|u + v\| \leq \|u\| + \|v\|$

Example. (trivial) $\|\cdot\|$ on $\mathbb{R}$ is a norm. ◇

Example (Euclidean Norm). For $x \in \mathbb{R}^p$ let

$$\|x\| = (x \cdot x)^{\frac{1}{2}} = \sqrt{x_1^2 + x_2^2 + \dots + x_p^2}.$$

◇

Proposition 9. Let V be an inner product space. Define

$$\|u\| = (u \cdot u)^{\frac{1}{2}}$$

for any $u \in V$. Then $\|\cdot\|$ is a norm.

Lemma 17 (Cauchy-Schwartz Inequality). If V is an inner product space, then

$$u \cdot v \leq \|u\|\|v\|$$

where $\|u\| = (u \cdot u)^{\frac{1}{2}}$.

Proof. Let

$$w = \alpha u - \beta v$$

where $\alpha, \beta \in \mathbb{R}$. Then $w \cdot w \geq 0$. Hence

$$\begin{aligned} (\alpha u - \beta v) \cdot (\alpha u - \beta v) &\geq 0 \\ \alpha^2 \|u\|^2 + \beta^2 \|v\|^2 - 2\alpha\beta(u \cdot v) &\geq 0. \end{aligned}$$

Take $\alpha = \|v\|$ and $\beta = \|u\|$. Then

$$\begin{aligned} 2\|u\|^2\|v\|^2 - 2\|u\|\|v\|(u \cdot v) &\geq 0 \\ 2\|u\|\|v\|(\|u\|\|v\| - u \cdot v) &\geq 0. \end{aligned}$$

Now, if $\|u\| = 0$ or $\|v\| = 0$, then $u = 0$ or $v = 0$, hence $u \cdot v = 0 = \|u\|\|v\|$.

If both $\|u\|$ and $\|v\|$ are positive, then

$$\|u\|\|v\| - u \cdot v \geq 0$$

as required. □

Problem. If $u \cdot v = \|u\|\|v\|$, what can say about u and v ?

Solution. $u \cdot v = \|u\|\|v\| \Rightarrow \|\alpha u - \beta v\| = 0 \Leftrightarrow \alpha u - \beta v = 0$.

This is called the *Case of equality in Cauchy-Schwartz*. $\square$

Proof of Proposition 9. Show $\|u\| = (u \cdot u)^{\frac{1}{2}}$ is a norm. We check triangle inequality only. We need to show $\|u + v\| \leq \|u\| + \|v\|$.

$$\begin{aligned} &\Leftrightarrow \|u + v\|^2 \leq (\|u\| + \|v\|)^2 \\ &\Leftrightarrow (u + v) \cdot (u + v) \leq \|u\|^2 + \|v\|^2 + 2\|u\|\|v\| \\ &\Leftrightarrow 2(u \cdot v) \leq 2\|u\|\|v\| \\ &\Leftrightarrow (u \cdot v) \leq \|u\|\|v\|. \end{aligned}$$

This is same as Cauchy-Schwartz, so we finished the proof. $\square$

Lecture 6: Point set topology in Euclidean space

Mon 12 Sep 2022 11:36

Proposition 10 (Reverse Triangle Inequality). In any normed space,

$$|\|x\| - \|y\|| \leq \|x - y\|.$$

Proof.

$$\begin{aligned} &\Leftrightarrow -\|x - y\| \leq \|x\| - \|y\| \leq \|x - y\| \\ &\Leftrightarrow \|y\| \leq \|y - x\| + \|x\| \wedge \|x\| \leq \|x - y\| + \|y\|. \end{aligned}$$

Now this follows from triangle inequality. $\square$

1.3 Topology in Cartesian space $\mathbb{R}^p$

For any $x, y \in \mathbb{R}^p$, define the *Euclidean distance* $d(x, y) = \|x - y\|$.

Check $d(x, y)$ is a distance.

Definition 10 (Open and closed balls). For $x \in \mathbb{R}^p, r = 0$, define

$$B_r(x) = \{y \in \mathbb{R}^p : \|y - x\| < r\}$$

to be the *open ball* centered at x with *radius* r .

Similarly, define the *closed ball*, or *closure* of the open ball as

$$\overline{B_r(x)} = \{y \in \mathbb{R}^p : \|y - x\| \leq r\}.$$

A *sphere*, or *boundary* of ball is

$$S_r(x) = \partial B_r(x) = \{y \in \mathbb{R}^p : \|y - x\| = r\}.$$

Definition 11. Let $E \subset \mathbb{R}^p$, $x \in \mathbb{R}^p$.

1. We say x is an **interior point** of E if there is a positive r s.t. $B_r(x) \subset E$.

We also say E is a *neighborhood* of x .

2. We say that x is an *exterior point* of E if there exists a positive r s.t. $B_r(x) \cap E = \emptyset$, i.e. $B_r(x) \subset E^c$.

x is exterior point of E iff x is interior point of E^c .

3. We say that x is a **boundary point** of E , if for every $r > 0$, $B_r(x) \cap E \neq \emptyset$ and $B_r(x) \cap E^c \neq \emptyset$.

Definition 12 (Open Set). We say $E \subset \mathbb{R}^p$ is *open* if every $x \in E$ is an interior point of E .

Example. 1. $\emptyset$, $\mathbb{R}^p$ are trivial examples of open sets.

2. Open balls.

Proposition 11. Open balls are open.

Proof. Let $E = B_R(a)$, $R > 0$, $a \in \mathbb{R}^p$. We want to show: if $x \in B_R(a)$ then there exists $r > 0$ such that $B_r(x) \subset B_R(a)$.

Let $r = R - \|a - x\|$. Take $y \in B_r(x)$. Then $\|y - x\| < r = R - \|a - x\|$. By triangle inequality,

$$\begin{aligned} \|a - y\| &\leq \|a - x\| + \|y - x\| \\ &< \|a - x\| + r \\ &= \|a - x\| + R - \|a - x\| \\ &= R. \end{aligned}$$

Hence $B_r(x) \subset B_R(a)$. Hence $B_R(a)$ is open. $\square$

3. Exterior of the ball

$$\mathbb{R}^p \setminus \overline{B_r(a)} = \{x \in \mathbb{R}^p : \|x - a\| > R\}$$

is open, by using the same argument as above.

4. x is a boundary point of $B_R(a) \Leftrightarrow \|x - a\| = R$.

$\diamond$

1.4 Properties of Open sets

Proposition 12. 1. $\emptyset, \mathbb{R}^p$ are open.

2. Finite intersection of open sets is open.

3. Arbitrary union of open sets is open. If $\{U_\alpha\}_{\alpha \in A}$ is any collection of sets such that U_α is open for any $\alpha \in A$, then

$$\bigcup_{\alpha \in A} U_\alpha \text{ is open.}$$

Proof. (2): Let $x \in \bigcap_{i=1}^N U_i$. Then for any $1 \leq i \leq N$, $x \in U_i$. Then there exists positive $r_i > 0$ such that $B_{r_i}(x) \subset U_i$. Take

$$r = \inf\{r_i : 1 \leq i \leq N\}.$$

Since N is finite, $r > 0$. Now for any i , $r \leq r_i$, so $B_r(x) \subset B_{r_i}(x)$. This is true for all i , so $B_r(x) \subset \bigcap_{i=1}^N U_i$.

(3): Let $x \in \bigcup_{\alpha \in A} U_\alpha$. Now $x \in U_{\alpha_0}$ for some $\alpha_0 \in A$. Then there exists $r_0 > 0$ such that $B_{r_0}(x) \subset U_{\alpha_0}$. Now

$$B_{r_0}(x) \subset U_{\alpha_0} \subset \bigcap_{i=1}^N U_i.$$

Hence $\bigcap_{i=1}^N U_i$ is open. □

Lecture 7: Closed Sets

Wed 14 Sep 2022 11:31

Example. (countable union of open sets need not be open) If $U_n = B_{\frac{1}{n}}(x)$, $x \in \mathbb{R}^p$. Then

$$\bigcap_{n=1}^{\infty} B_{\frac{1}{n}}(x) = \{x\}.$$

Proof. If $y \neq x$ then $\|y - x\| > 0$. Exists $n \in \mathbb{N} : 0 < \frac{1}{n} < \|y - x\|$. Hence $y \notin B_{\frac{1}{n}}(x)$. □

◇

Proposition 13. A set $E \subset \mathbb{R}^p$ is open if and only if E is a union of open balls.

Proof. ($\Leftarrow$) is clear by the previous proposition, since open balls are open.

Now we prove ($\Rightarrow$). For any $x \in E$, there exists $r = r_x > 0$ such that

$$G_x = B_{r_x}(x) \subset E.$$

Then $E = \bigcup_{x \in E} G_x$. Indeed, since $G_x \subset E$, the union is $\subset E$. For any $x_0 \in E$, we have $x_0 \in G_{x_0} \subset \bigcup_{x \in E} G_x$, hence $E \subset \bigcup_{x \in E} G_x$. □

Definition 13 (Closed Set). $F \subset \mathbb{R}^p$ is called closed if and only if F^c is open.

Proposition 14. $F \subset \mathbb{R}^p$ is closed if and only if it contains all its boundary points.

Proof. ($\Rightarrow$) Let F be closed and x is boundary point of F . If $x \notin F$, $x \in F^c$. Hence x is an exterior point. Hence $x \in F$.

($\Leftarrow$) Suppose now F contains all its boundary points. Let $x \notin F$. Hence x is not interior point of F . x is not boundary by assumption. Hence x is exterior. This means $x \in F^c \Rightarrow x$ is interior to F^c . Hence F^c is open. $\square$

1.5 Properties of closed sets

Proposition 15.

1. $\mathbb{R}^p, \emptyset$ are closed.
2. Union of a finite family of closed sets is closed.
3. Intersection of any family of closed sets is closed.

Proof. 1. $(\mathbb{R}^p)^c = \emptyset, \emptyset^c = \mathbb{R}^p$.

2. $F_1, \dots, F_N$ are closed. Then

$$\left(\bigcup_{i=1}^N F_i \right)^c = \bigcap_{i=1}^N F_i^c.$$

Therefore $\left(\bigcup_{i=1}^N F_i \right)^c$ open by property of open sets. Hence $\bigcup_{i=1}^N F_i$ closed.

3. Similarly,

$$\left(\bigcap_{\alpha \in A} F_\alpha \right)^c = \bigcup_{\alpha \in A} F_\alpha^c.$$

$\square$

Example. 1. $\{x\}$ is a closed set.

2. More generally, any closed ball

$$\overline{B_r(x)} = \{y : \|y - x\| \leq r\}, r \geq 0$$

is closed.

Proof. $\overline{B_r(x)}^c$ is the exterior of the ball which we know is open. $\square$

3. Cantor set is closed.

Proof. Each F_n is a finite union of closed union of closed balls, which is closed. F is a countable intersection of closed sets, which is again closed. $\square$

$\diamond$

Lecture 8

Fri 16 Sep 2022 11:34

1.6 Nested cells

Definition 14. *Open cells* in $\mathbb{R}^p$ are Cartesian products of bounded open intervals. *Closed cells* in $\mathbb{R}^p$ are Cartesian products of bounded closed intervals.

Products of all types of bounded intervals are also called *cells*.

Theorem 10 (Nested Cells Property). Suppose we have a nested family of nonempty closed cells:

$$I_1 \supset I_2 \supset \dots \supset I_n \supset \dots$$

Then

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

Proof. look at the projections. Let $I_n = [a_{n1}, b_{n1}] \times [a_{n2}, b_{n2}] \times \dots \times [a_{np}, b_{np}]$. Let

$$I_n^{(k)} = [a_{nk}, b_{nk}]$$

be the "k-th projection" of I_n . Then we have nested sequence of intervals

$$I_1^{(k)} \supset I_2^{(k)} \supset \dots \supset I_n^{(k)} \supset \dots$$

By Nested Interval Property, there exists $\xi_k \in \bigcap_{n=1}^{\infty} I_n^{(k)}$, where

$$a_{nk} \leq \xi_k \leq b_{nk}, n \in \mathbb{N}.$$

Let $\xi = (\xi_1, \dots, \xi_p)$. Now

$$\begin{aligned} \xi \in \bigcap_{n=1}^{\infty} I_n &\Leftrightarrow \xi \in I_n, n \in \mathbb{N} \\ &\Leftrightarrow \xi_k \in I_n^{(k)}, k = 1, \dots, p, \quad n \in \mathbb{N}. \end{aligned}$$

$\square$

Definition 15 (Cluster Point). Let $E \subset \mathbb{R}^p$, $x \in \mathbb{R}^p$. We say that x is a *cluster point* of E if every ball $B_\delta(x)$ contains a point from E , different from x . In other words,

$$(B_\delta(x) \setminus \{x\}) \cap E \neq \emptyset.$$

Proposition 16. x is a cluster point of E if and only if $B_\delta(x) \cap E$ is infinite for any $\delta > 0$.

Proof. If $B_\delta(x) \cap E$ is infinite, then $(B_\delta(x) \setminus \{x\}) \cap E$ is infinite, hence nonempty. Suppose x is a cluster point, but $B_\delta(x) \cap E$ is finite. This is to say that $(B_\delta(x) \setminus \{x\}) \cap E$ is finite. Let $(B_\delta(x) \setminus \{x\}) \cap E = \{y_1, y_2, \dots, y_N\}$. Then

$$r = \min\{\|x - y_1\|, \|x - y_2\|, \dots, \|x - y_N\|\}$$

is positive.

Now: $(B_r(x) \setminus \{x\}) \cap E = \emptyset$, since $y_i \notin B_r(x)$. This is a contradiction. $\square$

Remark. 1. Every interior point is a cluster point.

2. If $x \notin E$, then x is a cluster point $\Leftrightarrow x$ is a boundary point of E .

Proposition 17. $F \subset \mathbb{R}^p$ is closed if and only if F contains all its cluster points.

Proof. Follows from remark 2. $\square$

Definition 16. We say that E is bounded if it is contained in a (closed) cell.

Remark. "cells" and "balls" are interchangeable in the definition (they are equivalent).

Theorem 11 (Bolzano-Weierstrass Theorem). Every bounded infinite set in $\mathbb{R}^p$ has a cluster point.

Proof. (Bisection argument) Let I_0 be a closed cell with contains E . For a cell I , let $l(I)$ be the length of the largest side.

Divide each side of I_0 into 2 subintervals of same length, and form 2^p closed subcells of I_0 .

At least one of 2^p subcells has infinite intersection with E (otherwise E is finite). Pick I_1 as one of the subcells of I_0 such that $I_1 \cap E$ is infinite. We have $l(I_1) = \frac{1}{2}l(I_0)$, and $I_1 \subset I_0$.

Then repeat the process on I_1 . Continue in this fashion, we obtain the cascading sequence of cells:

$$I_1 \supset I_2 \supset \dots \supset I_n \supset \dots$$

where $I_n \cap E$ is infinite, and $l(I_n) = \frac{1}{2^n}l(I_1)$. By the nested cells property, there $\exists \xi \in \mathbb{R}^p$ such that

$$\xi \in \bigcap_{n=1}^{\infty} I_n.$$

We claim that ξ is a cluster point. Need to show that $B_\delta(\xi) \cap E$ is infinite for any $\delta > 0$. This will follow if we show that $B_\delta(\xi)$ contains I_n for some large n . If $x \in I_n$ and $\xi \in I_n$, then

$$|x_i - \xi_i| \leq b_i - a_i \leq l(I_n).$$

Hence

$$\|x - \xi\| \leq \sqrt{|x_1 - \xi_1|^2 + \dots + |x_p - \xi_p|^2} \leq \sqrt{p} l(I_n).$$

So if $\sqrt{p} l(I_n) < \delta$ then $I_n \subset B_\delta(x)$. Take n such that $2^n > \frac{\sqrt{p} l(I_0)}{\delta}$. Such n exists by Archimedean Property. $\square$

Lecture 9: Compactness and Heine-Borel theorem

Mon 19 Sep 2022 11:32

Suppose we have $E \subset \mathbb{R}^p$ and a family $\mathcal{G} = \{G_\alpha\}_{\alpha \in A}$ where $G_\alpha \subset \mathbb{R}^p$ is open for every $\alpha \in A$. We say that $\mathcal{G}$ is an *open cover* of E if

$$E \subset \bigcup_{\alpha \in A} G_\alpha.$$

We say that $\mathcal{G}' \subset \mathcal{G}$ is a subcover if we still have

$$E \subset \bigcup_{G \in \mathcal{G}'} G.$$

Definition 17 (Compact). $E \subset \mathbb{R}^p$ is called *compact* if every open cover $\mathcal{G}$ of E has a finite subcover.

Example. 1. If E is finite, then E is compact.

Proof. Let $E = \{x_1, x_2, \dots, x_N\}$. Let $\mathcal{G}$ be an open cover. For $x_1, \exists G_1 \in \mathcal{G}$ such that $x_1 \in G_1$. For $x_2, \exists G_2 \in \mathcal{G}$ such that $x_2 \in G_2$. Continue in this fashion, we have a finite family of open sets

$$E \subset \bigcup_{i=1}^N G_i.$$

Hence $\{G_1, \dots, G_N\}$ is a subcover of $\mathcal{G}$. $\square$

2. $\mathbb{R}^p$ is not compact.

Proof. Let $G_n = B_n(0)$. Then

$$\mathbb{R}^p \subset \bigcup_{n=1}^{\infty} G_n.$$

But

$$\begin{aligned} \mathbb{R}^p &\not\subset G_{n_1} \cup \dots \cup G_{n_N} \\ &= B_{n_1}(0) \cup \dots \cup B_{n_N}(0) \\ &= B_{\max n_i}(0). \end{aligned}$$

$\square$

◇

Theorem 12 (Heine-Borel Theorem). $E \subset \mathbb{R}^p$ is compact if and only if E is closed and bounded.

Remark. True only in $\mathbb{R}^p$, finite-dimensional Euclidean space. Might not be true in spaces other than $\mathbb{R}^p$.

Lemma 18. If E compact, then E bounded.

Proof. Take $G_n = B_n(0), n = 1, 2, \dots$. Then $\{G_n\}$ is an open cover of E :

$$E \subset \bigcup_{n=1}^{\infty} G_n = \mathbb{R}^p.$$

By compactness, there are $G_{n_1}, \dots, G_{n_N}$ such that

$$\begin{aligned} E &\subset G_{n_1} \cup \dots \cup G_{n_N} \\ &= B_{\max\{n_1, \dots, n_N\}}(0). \end{aligned}$$

Hence E bounded. □

Lemma 19. If E compact, then E closed.

Proof. Suppose x is a cluster point of E and $x \notin E$. Let

$$G_n = \overline{B_{\frac{1}{n}}(x)}^c = \{y : \|y - x\| > \frac{1}{n}\}.$$

Then

$$\bigcup_{n=1}^{\infty} G_n = \mathbb{R}^p \setminus \{x\} \supset E.$$

If there exists a finite subcover $G_{n_1}, \dots, G_{n_N}$ such that

$$E \subset G_{n_1} \cup \dots \cup G_{n_N} = G_{\max\{n_1, \dots, n_N\}} =: G_R.$$

Then $B_R(x) \cap E = \emptyset$. This is a contradiction since x is a cluster (or boundary) point. Hence for every cluster point x of E we have $x \in E$. Hence E closed. □

Remark. The two lemmas above are true in every topological space. It's the reverse statement (closed and bounded implies compactness) that isn't universal.

Lemma 20. Closed cells are compact.

Proof. (Bisection argument) Let I be a closed cell and $\mathcal{G}$ an open cover of I . Suppose $\mathcal{G}$ does not have finite subcover.

1. Bisect the sides of I to form 2^p closed cells. If all 2^p subcells have finite subcover of $\mathcal{G}$, $\mathcal{G}$ would have a finite subcover of I . Hence there exists a cell, call it I_1 , such that I_1 cannot be covered by finitely many elements of $\mathcal{G}$. We have $l(I_1) = \frac{1}{2}l(I)$.
2. Bisect I_1 to find I_2 , in the same fashion. By induction, we construct

$$I_1 \supset I_2 \supset \dots \supset I_n \supset \dots$$

Such that $l(I_n) = \frac{1}{2^n}l(I)$, and I_n cannot be covered by finitely many elements of $\mathcal{G}$.

3. Now take $\xi \in \bigcap_{n=1}^{\infty} I_n$ (nonempty by Nested Cells Lemma). $\xi \in I$, so there $\exists G \in \mathcal{G}$ such that $\xi \in G$. Since G is open, there $\exists \delta > 0$ such that $B_\delta(\xi) \subset G$.

As we have shown in the proof of Bolzano-Weierstrass, there exists an n such that $I_n \subset B_\delta(\xi) \subset G$. This means I_n is covered by 1 element of $\mathcal{G}$, contrary to the assumption that I_n cannot be covered by finite subcover of $\mathcal{G}$. Hence I is compact. □

Lemma 21. If F is compact, then $E \subset F$ is compact iff it is closed.

Proof. Let $\mathcal{G}$ be open cover of E . Add E^c to $\mathcal{G}$, and

$$\tilde{\mathcal{G}} = \mathcal{G} \cup \{E^c\}.$$

Now

$$\bigcup_{G \in \tilde{\mathcal{G}}} G = \left(\bigcup_{G \in \mathcal{G}} G \right) \cup E^c = \mathbb{R}^p \supset F.$$

Hence $\tilde{\mathcal{G}}$ is an open cover of F . Hence there exists a finite subcover of F .

If $\{G_1, \dots, G_N\}$ contains E^c , throw it out and call the remaining finite set $\mathcal{G}'$. Then $\mathcal{G}'$ covers E , and $\mathcal{G}' \subset \mathcal{G}$. Therefore E is compact. □

Proof of Theorem 17. By the four lemmas. □

Lecture 10

Wed 21 Sep 2022 11:33

Theorem 13 (Cantor Intersection Theorem). Suppose $F_1, F_2, \dots, F_n$ is a nested sequence of sets

$$F_1 \supset F_2 \supset \dots \supset F_n \supset \dots$$

which are nonempty and compact. Then

$$\bigcap_{n=1}^{\infty} F_n \neq \emptyset.$$

Proof. Suppose $\bigcap_{n=1}^{\infty} F_n \neq \emptyset$. Let $G_n = F_n^c$. G_n is open since F_n is closed. Then by De Morgan's Law, $\bigcup_{n=1}^{\infty} G_n = \mathbb{R}^p$, where

$$G_1 \subset G_2 \subset \dots \subset G_n \subset \dots$$

Then G_n is open cover of F_1 . Since F_1 is compact, there exists a finite cover $\{G_{n_1}, G_{n_2}, G_{n_N}\}$ such that

$$F_1 \subset \bigcup_{i=1}^N G_{n_i} = G_M, \quad M = \max\{n_1, \dots, n_N\}.$$

So in particular $F_M \subset G_M = F_M^c$. Hence F_M is empty, a contradiction. This means the intersection is not empty. $\square$

Remark. This theorem is true in general spaces, unlike Heine Borel.

1.7 Connectedness

Definition 18 (Connected Set). We say that $E \subset \mathbb{R}^p$ is *disconnected* if there exists two open sets $A, B \subset \mathbb{R}^p$ such that $E \subset A \cup B$, $E \cap A \neq \emptyset$, $E \cap B \neq \emptyset$, and $(A \cap E) \cap (B \cap E) = \emptyset$.

We say A, B form a *disconnection*, *separation*, or *splitting* of E .

We say that E is *connected* if it is not disconnected.

Remark. We want to make connectedness an **internal** property, hence requiring $(A \cap E)$ and $(B \cap E)$ in the previous definition.

Example. Every finite set in $\mathbb{R}^p$ with more than 1 element is disconnected. Let $E = \{x_1, \dots, x_N\}$. Let $r = \min\{\|x_1 - x_2\|, \|x_1 - x_3\|, \dots, \|x_1 - x_N\|\}$.

Take $A = B_r(x_1)$ and $B = \overline{B_{\frac{r}{2}}(x_1)}^c$ to be the exterior of A . $\diamond$

Example. Any subset of the Cantor set with more than 1 element is disconnected. $\diamond$

Proposition 18. The interval $[0, 1] \subset \mathbb{R}$ is connected.

Proof. Suppose there is a disconnection A, B of $[0, 1]$. Thus A, B are open, $[a, b] \subset A \cup B$, $A \cap [0, 1] \neq \emptyset$, $B \cap [0, 1] \neq \emptyset$, and $(A \cap [0, 1]) \cap (B \cap [0, 1]) = \emptyset$.

Hence there exists $a \in A \cap [0, 1]$, $b \in B \cap [0, 1]$, and $a \neq b$. Without loss of generality, assume $a < b$.

Also, without loss of generality, we may assume $0 < a$ and $b < 1$. (If $a = 0$, since A open, $(-\delta, \delta) \subset A$, so we can choose $\frac{\delta}{2}$ as the new a . Do the same to b .)

Therefore, without loss of generality,

$$0 < a < b < 1.$$

Let $S = \{x : x \in A \cap [0, 1], x < b\}$. This set is nonempty, since $a \in S$. Also, it has an upper bound b . By Supremum Property, there is $\xi = \sup S$. Also, $a \leq \xi \leq b$, so $\xi \in [0, 1]$, hence $\xi \in A \cup B$.

If $\xi \in A$, since A is open, there exists δ such that $(\xi - \delta, \xi + \delta) \subset A$. Then $\xi + \frac{\delta}{2} \in S$. A contradiction.

If $\xi \in B$, there exists $\delta > 0$ such that $(\xi - \delta, \xi + \delta) \subset B$. Then there exists no $x \in A$ such that $\xi - \delta < x \leq \xi$. This means $\xi - \delta$ is an upper bound of S , so $\xi = \sup S \leq \xi - \delta$, a contradiction.

This means there's no disconnection of $[0, 1]$, so $[0, 1]$ is connected. $\square$

Corollary 7. $\mathbb{R}^p$ is connected.

Proof. Let A, B form a disconnection of $\mathbb{R}^p$. This means $B = A^c$. Let $a \in A$ and $b \in B$. Consider a mapping

$$\begin{aligned} \varphi : \mathbb{R} &\longrightarrow \mathbb{R}^p \\ t &\longmapsto \varphi(t) = (1 - t)a + tb. \end{aligned}$$

Now $\varphi(0) = a$ and $\varphi(1) = b$. Define

$$[a, b] := \varphi([0, 1]) = \{1 - t(a) + tb : 0 \leq t \leq 1\}.$$

Let $A_1 = \{\varphi^{-1}(A) = \{t : \varphi(t) \in A\}\}$, and $B_1 = \{\varphi^{-1}(B) = \{t : \varphi(t) \in B\}\}$. We claim that A_1, B_1 are open in $\mathbb{R}$. Pick any $t \in A_1$, we can find a small enough δ such that

$$\varphi((t - \delta, t + \delta)) \subset B_{(b-a)\delta}(\varphi(t)) \subset A.$$

Thus A_1 is open. Similarly B_1 is open. But then A_1, B_1 form a disconnection of $[0, 1]$. Other properties follow from those of A, B . This is a contradiction since $[0, 1]$ is connected. Therefore $\mathbb{R}^p$ is connected. $\square$

Remark. This statement is equivalent to: **the only clopen sets in $\mathbb{R}^p$ are $\emptyset$ and $\mathbb{R}^p$.**

Theorem 14. $E \subset \mathbb{R}$ is connected if and only if E is an interval.

Proof. We state without proof the following lemma:

Lemma 22. $E \subset \mathbb{R}$ is an interval if and only if E has the Interval Property:

$$a, b \in E, a \leq b \Rightarrow [a, b] \subset E.$$

Let E be connected, but doesn't satisfy the interval property. So there exists $a, b \in E, c \notin E$, such that $a < c < b$. Then $(-\infty, c), (c, \infty)$ form a disconnection of E , a contradiction.

The other direction is identical to the one for $\mathbb{R}^p$. $\square$

Remark. General connected sets can be sometimes counterintuitive, but open connected sets have very nice properties.

Lecture 11: More on Connectedness

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Let $U \subset \mathbb{R}^p$ and $a, b \in U$. We say that there is a *polygonal curve* from a to b in U if there are $z_0, z_1, \dots, z_N \in U$ such that $z_0 = a$, $z_N = b$, and $[z_{i-1}, z_i] \subset U$.

Theorem 15. Open set $U \subset \mathbb{R}^p$ is connected if and only if for any $a, b \in U$, there is a polygonal curve from a to b .

Proof. Suppose U is not connected. This means there is a disconnection A, B . Pick $a \in A \cap U$, and $b \in B \cap U$. By assumption, there is a polygonal curve from a to b , $z_0, z_1, \dots, z_N$.

We have $z_0 \in A$ but $z_N \notin A$. So there is a smallest index $n \geq 1$ such that $z_n \notin A$, but $z_{n-1} \in A$.

Then look at

$$\begin{aligned} \varphi : \mathbb{R} &\longrightarrow \mathbb{R}^p \\ t &\longmapsto \varphi(t) = (1-t)z_{n-1} + tz_n. \end{aligned}$$

Let $A_1 = \{\varphi^{-1}(A) = \{t : \varphi(t) \in A\}\}$, and $B_1 = \{\varphi^{-1}(B) = \{t : \varphi(t) \in B\}\}$. Repeat the argument in Corollary 7. Thus A_1, B_1 form a disconnection of $[0, 1]$. This is a contradiction since $[0, 1]$ is connected.

Remark. In this part of the proof, we haven't used the openness. For example, every convex set is connected.

Now prove the other direction. We say $a \sim b$ if there is a polygonal curve from a to b in U . We check $\sim$ is a equivalence property:

- $a \sim a$. Indeed, $z_0 = a$ satisfies.
- $a \sim b \Rightarrow b \sim a$. Indeed, we reverse the order of $z_0, \dots, z_N$.
- $a \sim b, b \sim c \Rightarrow a \sim c$. Indeed. We connect the curves.

Let $U_a = \{b \in U : b \sim a\}$.

Claim. U_a is open.

Proof. Let $b \in U_a$. Let $z_0, \dots, z_N$ be polygonal curve from a to b . Since U open, there exists $\delta > 0$ such that $B_\delta(b) \subset U$. Then for every $c \in B_\delta(b)$ we have $c \sim b$, since $[c, b] \subset B_\delta(b)$ (since balls are convex). This means $a \sim c$, hence $B_\delta(b) \subset U_a$. Therefore U_a open. $\square$

Let $V_a = \{b \in U : b \not\sim a\}$. Similarly,

Claim. V_a is open.

Proof. Same as for U_a . $\square$

Moreover, we see that $U \subset U_a \cup V_a$. If $U_a \cap V_a \neq \emptyset$, then both U_a, V_a are nonempty, they will form a disconnection, a contradiction. Hence $U = U_a$. $\square$

Corollary 8. Every open set is a union of disjoint connected open sets.

Lemma 23. Disjoint family of open set can be at most countable.

Proof. Use rational points. Now two open sets can contain the same rational ball. $\square$

Theorem 16. Every open set is a disjoint, countable union of connected sets.

They are called *connected components* of the open set.

Example. Look at F^c , F is the cantor set. $\diamond$

2 Sequences

Definition 19 (Sequence). A sequence E is a mapping

$$\varphi : \mathbb{N} \rightarrow E.$$

We identify φ with ordered list

$$(\varphi(1), \varphi(2), \dots, \varphi(n), \dots).$$

Notation. Often we use subscripts $x_n := \varphi(n)$. We may write $X = (x_1, x_2, \dots)$, $(x_n)_{n \in \mathbb{N}}$, (x_n) , $(x_n)_{n=1}^\infty$, or $\{x_n\}_{n=1}^\infty$ for the sequence.

Lecture 12: Sequences in $\mathbb{R}^p$

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Definition 20 (Limit of Sequence). We say that $X = (x_n)$ converges to $a \in \mathbb{R}^p$ if

- (topological) For open set V containing a , there is $K_V \in \mathbb{N}$ such that $x_n \in V$ for $n \geq K_V$.

In some cases, one only asks V to be a neighborhood of a , which means a is an interior point of V .

- (metric) We say that $X = (x_n)$ converges to $a \in \mathbb{R}^p$ if for every $\varepsilon > 0$, there is $K_\varepsilon \in \mathbb{N}$ such that $\|x_n - a\| < \varepsilon$ for $n \geq K_\varepsilon$.

Proposition 19. The two definitions above are equivalent.

Proof. (I implies II):

Simply take $V = B_\varepsilon(a)$ which is open. Then apply definition I. Take $K_\varepsilon = K_{B_\varepsilon(a)}$.

(II implies I):

Let $a \in V$, V open. Then there exists $\varepsilon > 0$ such that $B_\varepsilon(a) \subset V$. Then by definition II, there is K_ε such that $x_n \in B_\varepsilon(a) \subset V$ for $n \geq K_\varepsilon$. Take $K_V = K_\varepsilon$. $\square$

Remark. The topological definition is very general, not dependent on metric. **Convergence is a topological notion.**

However it is simpler to work with metric definition in Euclidean space, or metric spaces.

Notation. $X = (x_n)$ converges to a , write it as

$$x_n \rightarrow a \quad \text{as} \quad n \rightarrow \infty.$$

or

$$\lim_{n \rightarrow \infty} x_n = a.$$

or

$$\lim X = a.$$

Theorem 17 (Uniqueness of Limit). (Topological proof) If $X = (x_n)$ is a sequence in $\mathbb{R}^p$, and $\lim_{n \rightarrow \infty} x_n = a$, and $\lim_{n \rightarrow \infty} x_n = b$, then $a = b$.

In other words, limit is unique if limit exists.

Proof. Let $x_n \rightarrow a$, and $x_n \rightarrow b$ where $a \neq b$. Use Hausdorff Property, there are $U \ni a, V \ni b$ open, such that $U \cap V = \emptyset$.

Since $x_n \rightarrow a$, there exists K_U such that $x_n \in U, n \geq K_U$. Similarly, exists K'_V such that $x_n \in V, n \geq K'_V$. Hence for $n \geq \max(K_U, K'_V)$, $x_n \in U \cap V = \emptyset$, a contradiction. $\square$

Lemma 24 (Hausdorff Property). If $a, b \in \mathbb{R}^p$ are distinct, then there are open sets $U \ni a, V \ni b$ such that $U \cap V = \emptyset$.

Proof. $a \neq b$ implies $\|a - b\| > 0$. Let $\varepsilon = \|a - b\|/2$. Take $U = B_\varepsilon(a)$ and $V = B_\varepsilon(b)$. $\square$

Example. 1. Take $x_n = \frac{1}{n}, n = 1, 2, \dots$. Then $x_n \rightarrow 0$.

Proof. For any $\varepsilon > 0$, observe that $0 < \frac{1}{n} < \varepsilon$ if $n > \frac{1}{\varepsilon}$. Hence for all $n > \frac{1}{\varepsilon}$, $\frac{1}{n} \in (-\varepsilon, \varepsilon)$. $\square$

2. $x_n = (-1)^n$. Then x_n does not converge, which we call *diverge*.

3. $x_n = n$. divergent (see next proposition).

◇

Theorem 18. (Convergent sequences are bounded) If $X = (x_n)$ is convergent, then X is bounded, in the sense that the set $\{x_n : n \in \mathbb{N}\}$ is bounded.

Proof. Let $x_n \rightarrow a$ and take $\varepsilon = 1$. Then $x_n \in B_1(a)$ for $n \geq K_1$. Take $M = 1 + \max_{n \leq K_1} \|x_n\|$. Then $x_n \in B_M(0)$ for all $n \in \mathbb{N}$. $\square$

Theorem 19. Assume $x_n = (x_n^{(1)}, \dots, x_n^{(p)}) \in \mathbb{R}^p$. Then $x_n \rightarrow a = (a^{(1)}, a^{(2)}, \dots, a^{(p)})$ if and only if for all $1 \leq m \leq p$, $x_n^{(m)} \rightarrow a^{(m)}$ in $\mathbb{R}$.

Proof. For any ε , there exists K_ε such that $\|x_n - a\| < \varepsilon$ for $n \geq K_\varepsilon$. If $u = (u_1, u_2, \dots, u_p)$, then $\|u\| = \sqrt{u_1^2 + \dots + u_p^2} \geq \sqrt{u_i^2} = |u_i|$ for any $i = 1, \dots, p$.

Thus $|x_n^{(i)} - a^{(i)}| < \varepsilon$ for $n \geq K_\varepsilon$, so $x_n^{(i)} \rightarrow a^{(i)}$.

The other direction: Choose $K_\varepsilon^{(i)}$ such that for all $n \geq K_\varepsilon^{(i)}$, $|x_n^{(i)} - a^{(i)}| < \varepsilon$. Now

$$\begin{aligned} \|x_n - a\| &= \sqrt{|x_n^{(1)} - a^{(1)}|^2 + \dots + |x_n^{(p)} - a^{(p)}|^2} \\ &< \sqrt{\varepsilon^2 + \dots + \varepsilon^2} \\ &= \sqrt{p}\varepsilon \end{aligned}$$

If $n \geq K_\varepsilon := \max\{K_\varepsilon^{(1)}, \dots, K_\varepsilon^{(p)}\}$. Hence $x_n \rightarrow a$. $\square$

Lecture 13

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Lemma 25. $x_n \rightarrow a$ in $\mathbb{R}^p$ if and only of $\|x_n - a\| \rightarrow 0$ in $\mathbb{R}^p$.

Lemma 26. Let $\alpha_n > 0, \alpha_n \rightarrow 0$. If $\|x_n - a\| \leq C\alpha_n$ for all $n \geq K_0$, then $x_n \rightarrow a$.

Proof. If $\alpha_n > \varepsilon$ for $n \geq K_\varepsilon$, then $\|x_n - a\| < C\varepsilon$ for $n \geq K_\varepsilon$. We let $\varepsilon' = C\varepsilon$. $\square$

Example. 1. $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$

2. Let $0 < |b| < 1$. Then $b^n \rightarrow 0$ as $n \rightarrow \infty$.

Proof. Represent $|b| = \frac{1}{1+\alpha}$, $\alpha > 0$. By Bernoulli, $(1+\alpha)^n \geq 1+n\alpha$. Thus $|b|^n \leq \frac{1}{1+n\alpha} \leq \frac{1}{n\alpha}$.

But then $|b|^n \leq \frac{1}{n\alpha} < \varepsilon$ if $n > \frac{1}{\alpha\varepsilon}$. Since $|b^n - 0| = |b|^n < \varepsilon$ if $n > \frac{1}{\alpha\varepsilon}$, so $b^n \rightarrow 0$. $\square$

3. Let $a > 0$. Then $a^{\frac{1}{n}} \rightarrow 1$ as $n \rightarrow \infty$.

Proof. Assume $a > 1$. Let $d_n = a^{\frac{1}{n}} - 1 > 0$ since $a > 1$. Now $a = (1+d_n)^n \geq 1+nd_n$. Hence $d_n \leq \frac{a-1}{n} < \varepsilon$ if $n \geq \frac{a-1}{\varepsilon}$. Hence $|d_n| < \varepsilon$ if $n \geq K_\varepsilon$, so $d_n \rightarrow 0$.

Now assume $a \leq 1$. We prove later that $x_n \rightarrow b$ implies $\frac{1}{x_n} \rightarrow \frac{1}{b}$. Then we use the previous argument to see $\frac{1}{a^{\frac{1}{n}}} \rightarrow 1$, thus $a^{\frac{1}{n}} \rightarrow 1$. $\square$

4. $n^{\frac{1}{n}} \rightarrow 1$.

Proof. Let $c_n = n^{\frac{1}{n}} - 1 \geq 0$. Now $(1+c_n)^n = n$. We use 2 terms in the Binomial Expansion to bound n from below:

$$n \geq 1 + nc_n + \frac{n(n-1)}{2}c_n^2.$$

We have used the

Theorem 20 (Newton's Binomial Theorem).

$$(1+\alpha)^n = 1 + n\alpha + \dots + (n,k)\alpha^k + \dots + \alpha^n.$$

where

$$(n,k) = \frac{n!}{k!(n-k)!}.$$

Now $c_n^2 \leq \frac{2}{n-1} \rightarrow 0$. Hence $c_n \rightarrow 0$. $\square$

$\diamond$

2.1 Subsequences

Definition 21 (Subsequence). Let X be a sequence and $r_1 < r_2 < \dots < r_n$ be a sequence of natural numbers. Then $X' = (x_{r_1}, x_{r_2}, \dots, x_{r_n}, \dots)$ is called a *subsequence* of X .

Example. 1. (x_{2n}) is a subsequence of (x_n) .

2. Tails: $(x_n)_{n \geq k}$ is called a *tail* of X . It is a subsequence of (x_n) .

$\diamond$

Theorem 21. If X is a sequence in $\mathbb{R}^p$, and $\lim X = a$, then any subsequence X' of X also converges to a .

Proof. For any $\varepsilon > 0$, there exists K_ε such that $\|x_n - a\| < \varepsilon$ if $n > K_\varepsilon$. Let $X' = (x_{r_n})$. Note $r_n \geq n$. Hence

$$\|x_{r_n} - a\| < \varepsilon \quad \text{for } n \geq K_\varepsilon.$$

□

Example. $(-1)^n$ is divergent.

Proof. $x_{2n} = (-1)^{2n} = 1 \rightarrow 1$. But $x_{2n+1} = (-1)^{2n+1} = -1 \rightarrow -1$. If $X \rightarrow a$ then $x_{2n} \rightarrow a$ and $x_{2n+1} \rightarrow a$, so $-1 = 1$, a contradiction. □

◇

Proposition 20. If $X \not\rightarrow a$, then there exists a subsequence of X entirely out of some neighborhood of a .

Lecture 14

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Lecture 15

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2.2 Combination of Sequences

Theorem 22. Let X and Y be sequences in $\mathbb{R}$ (or $\mathbb{R}^p$). Define operations element-wise ($X + Y := (x_n + y_n)$), etc. Then

- $\lim X + Y = \lim X + \lim Y$
- $\lim cX = c \lim X$
- $\lim XY = \lim X \lim Y$ (in $\mathbb{R}^p$ it is the inner product)
- $\lim 1/X = \frac{1}{\lim X}$ when $X \neq 0$, $\lim X \neq 0$. (in $\mathbb{R}$)

Proof. 1. (3rd) Use the formula

$$|x_n y_n - xy| = |x_n y_n - xy_n + xy_n - xy| \leq |y_n| |x_n - x| + |x| |y_n - y| \leq |y_n| \varepsilon + |x| \varepsilon.$$

If $n \geq \max\{K_\varepsilon, K_{\varepsilon'}\}$. Since y_n is convergent, y_n is bounded. Hence the last line can be arbitrarily small, i.e.

$$|x_n y_n - xy| \leq \varepsilon(M + |x|).$$

2. (reciprocal)

$$\left| \frac{1}{y_n} - \frac{1}{y} \right| = \left| \frac{y_n - y}{|y_n| |y|} \right| \leq \frac{|y_n - y|}{\frac{|y|}{2} |y|} \leq \frac{2}{|y|^2} \varepsilon.$$

If $n \geq \max\{N, K'_\varepsilon\}$. Where $n \geq N$ guarantees $|y_n| \geq |y|/2$.

3. (quotient) Follows from 3 and 4.

□

Corollary 9. • $\lim_{n \rightarrow \infty} x_n^k = x^k$

- $\lim_{n \rightarrow \infty} P(x_n) = P(x)$ where P is any polynomial. This can be generalized to $\mathbb{R}^p$ with polynomial with multiple input variables.

Theorem 23 (Squeeze Theorem). Suppose we have sequences x_n, y_n, z_n in $\mathbb{R}$ so that $x_n \leq y_n \leq z_n$, and $x_n \rightarrow c$ and $z_n \rightarrow c$, then $y_n \rightarrow c$.

Also called [\[\[Sandwich Theorem \]\]](#) , or [\[\[Two Policeman Theorem \]\]](#) .

Proof. We know that $|x_n - c| < \varepsilon$ for $n \geq K_\varepsilon$, and $|z_n - c| < \varepsilon$ for $n \geq K'_\varepsilon$.

Now for $n \geq \max\{K_\varepsilon, K'_\varepsilon\}$, we have

$$c - \varepsilon < x_n \leq y_n \leq z_n < c + \varepsilon.$$

Therefore $|y_n - c| < \varepsilon$. □

Definition 22 (Monotone Sequence). Let $X = (x_n)$ be a sequence in $\mathbb{R}$. We say x_n is *increasing* (nondecreasing) if $x_n \leq x_{n+1}$ for $n \in \mathbb{N}$. We say x_n is *strictly increasing* if $x_n < x_{n+1}$.

We may write $x_n \nearrow$.

We define *decreasing* and *strictly decreasing* sequences symmetrically.

An increasing or decreasing sequence is called *monotone*.

Theorem 24 (Monotone Convergence Theorem). Let (x_n) monotone increasing, and bounded above. Then (x_n) is convergent, and

$$\lim(x_n) = \sup\{x_n : n \in \mathbb{N}\}.$$

Symmetrically, let (x_n) decreasing and bounded below, then

$$\lim(x_n) = \inf\{x_n : n \in \mathbb{N}\}.$$

Proof. Since $S = \{x_n : n \in \mathbb{N}\}$ is bounded above by M , there exists $s = \sup S$. Now for any ε , there exists N such that $x_N > s - \varepsilon$. But for all subsequent $n \geq N$, $x_n \geq x_N > s - \varepsilon$. This implies that $|x_n - s| < \varepsilon$ for all $n \geq N$. Hence $x_n \rightarrow s$. □

2.3 Number e

Look at $u_n = \left(1 + \frac{1}{n}\right)^n$. Expand by Newton Binomial Theorem, we have the expansion

$$u_n = 1 + n\frac{1}{n} + \dots + (n, k)\frac{1}{n^k} + \dots + \frac{1}{n^n}$$

$$u_{n+1} = 1 + 1 + \dots + \frac{1(1 - \frac{1}{n+1})}{2} + \frac{1\left(1 - \frac{1}{n+1}\right)\left(1 - \frac{2}{n+1}\right)}{3!} + \dots + \frac{1\left(1 - \frac{1}{n+1}\right)\left(1 - \frac{2}{n+1} + \dots\right)}{n!} + \text{last term}$$

By comparing term by term, $u_n \leq u_{n+1}$, so u_n monotone increase. Furthermore,

$$\begin{aligned} u_n &\leq 1 + 1 + \frac{1 \cdot 1}{2} + \dots + \frac{1}{n!} \\ &\leq 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}} \\ &\leq 3 - \frac{1}{2^n} \\ &\leq 3. \end{aligned}$$

Hence $u_n \leq 3$. Now by monotone convergence theorem, u_n converges. Call its limit e .

Lecture 16

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Example. (Monotone sequences) Let $s_1 = 1$, $s_{n+1} = \frac{1}{2}(s_n + \frac{a}{s_n})$, $n \geq 1$ ($a > 0$)
The sequence is convergent.

Proof. 1. s_n is bounded below, and $s_n \geq \sqrt{a}$.

Lemma 27. if $\alpha, \beta \geq 0$, then $\frac{\alpha+\beta}{2} \geq \sqrt{\alpha\beta}$.

Proof. $\frac{\alpha+\beta}{2} - \sqrt{\alpha\beta} = \frac{1}{2}(\sqrt{\alpha} + \sqrt{\beta})^2 \geq 0$ □

Now $s_{n+1} = \frac{1}{2}(s_n + \frac{a}{s_n}) \geq \sqrt{s_n \frac{a}{s_n}} = \sqrt{a}$.

2. $s_{n+1} \leq s_n$ for $n \geq 2$.

This is equivalent to $\frac{1}{2}(s_n + \frac{a}{s_n}) \leq s_n$. Equivalent to $s_n \geq \sqrt{a}$.

Now apply the monotone convergence theorem to the sequence $(s_n)_{n \geq 2}$. Hence $\lim s_n = s$ exists.

Take limit in the iterated equation:

$$s = \frac{1}{2}\left(s + \frac{a}{s}\right).$$

Hence $s = \sqrt{a}$. □

◇

Theorem 25 (Bolzano-Weierstrass Theorem). (for sequences) Every bounded sequence in $\mathbb{R}^p$ has a convergent subsequence.

Proof. Let $X = (x_n)$ be a bounded sequence in $\mathbb{R}^p$ such that $\|x_n\| \leq M$ for all $n \in \mathbb{N}$.

Let $E = \{x_n : n \in \mathbb{N}\}$. If E finite, this means there is $a \in E$ such that $x_n = a$ for infinitely many indices $n \in \mathbb{N}$. There exists $n_1 < n_2 < \dots < n_k < \dots$ such that $x_{n_k} = a$. But then $x_{n_k} \rightarrow a$.

If E is infinite, then invoke B-W for sets. Since E is bounded and infinite, there exists $x_* \in \mathbb{R}^p$ such that x_* is a cluster point of E .

Choose n_1 such that $\|x_{n_1} - x_*\| < 1$. Then choose $n_2 > n_1$ such that $\|x_{n_2} - x_*\| < \frac{1}{2}$ (since $B_{\frac{1}{2}}(x_*) \cap E$ is infinite). Continue in this fashion. Now x_{n_k} is a subsequence of x_n that converges to x_* . $\square$

The next theorem gives an intrinsic characterization of convergence.

Theorem 26 (Cauchy Sequences). We say that a sequence $X = (x_n)$ is *Cauchy* if for all $\varepsilon > 0$, there exists $N_\varepsilon \in \mathbb{N}$ such that $\|x_n - x_m\| < \varepsilon$ for all $n, m \geq N_\varepsilon$.

Also called *fundamental sequence* or *internally convergent*.

Example. $x_n = 1 + \frac{1}{2} + \dots + \frac{1}{n}$ is not Cauchy.

$$\begin{aligned} x_m - x_n &= \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{m} \\ &\geq \frac{1}{m} + \dots + \frac{1}{m} \\ &= \frac{m-n}{m} \end{aligned}$$

Choose $m = 2n$, so $x_{2n} - x_n \geq \frac{1}{2}$. This means x_n is not Cauchy, fails with $\varepsilon = \frac{1}{2}$. $\diamond$

Theorem 27. If X is convergent, then X is Cauchy.

Proof. For any ε , $\|x_n - a\| < \varepsilon$ for $n \geq K_\varepsilon$. Hence, for $n, m > K_\varepsilon$,

$$\|x_n - x_m\| \leq \|x_n - a\| + \|x_m - a\| = 2\varepsilon.$$

$\square$

Theorem 28. If X is Cauchy, then X is bounded.

Proof. Take $\varepsilon = 1$, then there exists N such that for all $n, m \geq 1$, $\|x_n - x_m\| < 1$. In particular, $\|x_n - x_N\| < 1$. Now $(x_n)_{n \geq N}$ is bounded by $\|x_N\| + 1$. Then we take the maximum to bound the finitely many items before x_N . $\square$

Lemma 28. If X is Cauchy and has a convergent subsequence X' , then X is convergent and $\lim X = \lim X'$.

Proof. Let $X = (x_n)$, $X' = (x_{n_k})$. Let $a = \lim_{k \rightarrow \infty} x_{n_k}$. Then for $\varepsilon > 0$, $\|x_{n_k} - a\| < \varepsilon$ for $k \geq K$. But because X is Cauchy, there exists N such that for all $n, m \geq N$, $\|x_n - x_m\| < \varepsilon$. Now take $M = \max(N, K)$. Then for $k \geq M$:

$$\|x_n - a\| \leq \|x_n - x_{n_k}\| + \|x_{n_k} - a\| \leq \varepsilon + \varepsilon = 2\varepsilon.$$

$\square$

Theorem 29 (Cauchy Criterion). A sequence X in $\mathbb{R}^p$ is convergent if and only if X is Cauchy.

Proof. We only need to prove that X Cauchy implies X convergent.

Since X Cauchy, X is bounded, so X has a convergent subsequence by B-W. Hence by lemma, X is convergent. $\square$

Lecture 17: Cauchy Sequences, limsup and liminf

Fri 07 Oct 2022 11:33

Example. Let (x_n) be such that $x_1 = a$, $x_2 = b$, $x_{n+1} = \frac{x_n + x_{n-1}}{2}$. Prove (x_n) is convergent, and find the limit.

Proof.

$$\begin{aligned} x_{n+1} - x_n &= \frac{x_{n-1} - x_n}{2} \\ |x_{n+1} - x_n| &= \frac{1}{2} |x_n - x_{n-1}| = \frac{1}{2^{n-1}} |x_2 - x_1| = \frac{|b - a|}{2^{n-1}} \end{aligned}$$

$\square$

Claim. (x_n) is Cauchy.

Indeed, if we take $m \geq n$, then

$$\begin{aligned} |x_m - x_n| &\leq |x_m - x_{m-1}| + |x_{m-1} - x_{m-2}| + \dots + |x_{n+1} - x_n| \\ &= |b - a| \left(\frac{1}{2^{m-2}} + \dots + \frac{1}{2^{n-1}} \right) \\ &= \frac{|b - a|}{2^{n-1}} \left(2 - \frac{1}{2^{m-1-n}} \right) \\ &\leq \frac{C}{2^n} \text{ for } m \geq n. \end{aligned}$$

Hence $|x_m - x_n| < \varepsilon$ for $m \geq n \geq K_\varepsilon$. Hence (x_n) is Cauchy. Hence (x_n) is convergent.

Consider special case $a = 0$ and $b = 1$. Let $\alpha_1 = 0$, $\alpha_2 = 1$, $\alpha_{n+1} = \frac{\alpha_n + \alpha_{n-1}}{2}$. Let $\alpha = \lim(\alpha_n)$. Note: we have $x_n = \varphi(\alpha_n)$ where $\varphi(t) = a(1-t) + bt$.

Thus $\lim(x_n) = a(1-\alpha) + b\alpha$.

Take $(x_n)_{n \geq 2}$, we have $x_2 = b$, $x_3 = \frac{a+b}{2}$, the limit is

$$\lim(x_n)_{n \geq 2} = \lim(x_n) = b(1-\alpha) + \frac{a+b}{2}\alpha.$$

Now

$$a(1-\alpha) + b\alpha = b(1-\alpha) + \frac{a+b}{2}\alpha.$$

Thus

$$\alpha = \frac{2}{3}.$$

Hence $\lim(x_n) = \frac{1}{3}a + \frac{2}{3}b$. ◇

2.4 limsup and liminf

Let $X = (x_n)$ in $\mathbb{R}$. Let

$$E = \{a : \exists \text{ subsequence } X' \text{ of } X \text{ st. } \lim X' = a\}$$

to be the *subsequential limits* of X . We will assume for now that X is bounded. Then by Bolzano-Weierstrass, E is nonempty.

Note: If X is bounded by M , then $E \subset [-M, M]$ is also bounded. Thus $\sup E$ and $\inf E$ exist.

Let $s^* = \sup E$ and $s_* = \inf E$. We show $s^*, s_* \in E$, then they will be smallest and largest subsequential limits.

Alternative approach:

Definition 23. We say that v is an *almost upper bound* for $X = (x_n)$ if $x_n \leq v$ for all but finitely many $n \in \mathbb{N}$. Equivalently, $x_n \leq v$ for all $n \geq N_v$ for some N_v .

Let $V = \{v : v \text{ almost upper bound for } X\}$.

Claim. $V \neq \emptyset$, and V bounded below.

Proof. If $|x_n| \leq M$ for all n , then $x_n \leq M$ for $n \geq 1$, so $M \in V$, so V not empty.

Also, if $v \in V$, then $-M \leq x_n \leq v$ for $n \geq K_v$, there is at least one such n , so $n \geq -M$. So $-M$ is a lower bound for V . □

Hence $\inf V$ exists, denote as $x^* = \inf V$.

Proposition 21. The following statements are equivalent for a number x^* :

1. $x^* = \inf V$, V is almost upper bounds of X
2. For any $\varepsilon > 0$, there are at most finitely many n 's such that $x_n > x^* + \varepsilon$ but infinitely many n 's such that $x_n > x^* - \varepsilon$.
3. Let $v_m = \sup\{x_n : n \geq m\}$. Then $x^* = \inf v_m = \lim v_m$.
4. x^* is the largest subsequential limit of X , i.e. $x^* = \sup E$.

Proof. • (a) $\Rightarrow$ (b): If $x^* = \inf V$, then for any $\varepsilon > 0$, there is $v \in V$ such that

$$x^* \leq v < x^* + \varepsilon.$$

But this means $x_n \leq v_\varepsilon$ for $n \geq K_{v_\varepsilon}$. So $x_n \leq x^* + \varepsilon$ for $n \geq K_{v_\varepsilon}$. So there are only finitely many n such that $x_n > x^* + \varepsilon$.

If $x^* - \varepsilon \notin V$, then there are infinitely many n such that $x_n > x^* - \varepsilon$.

- (b) $\Rightarrow$ (c) : For any $\varepsilon > 0$, $x^* + \varepsilon \in V$, so $x_n \leq x^* + \varepsilon$ for $n \geq K_\varepsilon$. Thus $v_{K_\varepsilon} = \sup_{n \geq K_\varepsilon} x_n \leq x^* + \varepsilon$.

Thus $\inf v_m \leq v_{K_\varepsilon} \leq x^* + \varepsilon$. Since ε is arbitrary, $\inf v_m \leq x^*$.

Fix any $m \in \mathbb{N}$. Take any $\varepsilon > 0$, there are infinitely many n such that $|x^* - \varepsilon| < x_n$. So there exists $n > m$ such that $x^* - \varepsilon < x_n \leq \sup_{n \geq m} x_n = v_m$.

Therefore $x^* - \varepsilon \leq v_m$ for any m , for arbitrary $\varepsilon > 0$. Hence $x^* \leq \inf v_m$.

Finally, note that v_m is decreasing. Therefore $\lim v_m = \inf v_m$ once we know v_m is bounded below.

- (c) $\Rightarrow$ (d) : Let x^* be as in (c), and take any subsequence x_{m_k} which is converges. Then $x_{m_k} \leq v_k$ since $m_k \geq k$. Thus $s \leq x^*$. This means x^* is an upper bound for E . It is left to show $x^* \in E$. By definition of v_1 , there is n_1 such that $v_1 - 1 < x_{n_1} \leq v_1$. Suppose we have constructed $n_1, n_2, \dots, n_k$. Let $n_{k+1} \geq k+1$ be such that $v_{k+1} - \frac{1}{k+1} < x_{n_{k+1}} \leq v_{k+1}$.

We now take $m_1 = n_1$, $m_2 = n_{m_1} + 1 > m_1$, and $m_{k+1} = n_{m_k} + 1 > m_k$. Thus x_{m_k} will be a subsequence of (x_{n_k}) . Therefore $x_{m_k} \rightarrow x^*$.

- (d) $\Rightarrow$ (a) : Suppose x^* is the largest subsequential limit of X . We want to prove $x^* = \inf V$. Take $\varepsilon > 0$. Then we claim that $x^* + \varepsilon \in V$. Suppose not. Then there exists a subsequence (x_{n_k}) such that $x_{n_k} \geq x^* + \varepsilon$. Also (x_{n_k}) is bounded. By Bolzano-Weierstrass, there exists $x_{n_{k_m}}$ convergent, $x_{n_{k_m}} \rightarrow s$ as $m \rightarrow \infty$. Thus $x^* + \varepsilon \leq s$, thus $s \in E$ and $s > x^*$, and contradiction. Thus $\inf V \leq x^*$.

On the other hand, let $(x_{n_k}) \rightarrow x^*$. Then $x_{n_k} > x^* - \varepsilon$ for $k \geq K_\varepsilon$. Thus $x^* - \varepsilon \notin V$, so $x^* - \varepsilon \leq \inf V$ for all $\varepsilon > 0$. So $x^* \leq \inf V$.

□

Definition 24. The *limit superior* is defined as

$$\limsup X = x^*$$

whenever the equivalent conditions above are satisfied.

Lecture 18

Wed 12 Oct 2022 11:37

Dear Prof. Petrosyan,

Good morning. I am Xiaoyu Liu, a sophomore in your MA440 class. I am writing to you to seek some advice regarding my academic plan.

My current academic plan involves taking measure theory (MA544) next semester. I would like to discuss with you whether taking MA544 next semester is appropriate for my situation, and I hope to get your permission to take MA544 if you find me well-fit to take this course.

I wish I could attend your office hour, but I have a scheduling conflict with another course. I would appreciate if you could arrange an appointment in your office (any time before 10:30 AM or after 15:30 PM works for me).

Thank you in advance and wish you have a nice day.

Yours sincerely,

Xiaoyu Liu PUID: 032689357

Lecture 19

Fri 14 Oct 2022 11:31

Remark. $\liminf(x_n) = -\limsup(-x_n)$.

Lemma 29. $\limsup(x_n) \leq a$ is equivalent to: For every $\varepsilon > 0$, there exists K_ε such that $x_n \leq a + \varepsilon$ for $n \geq K_\varepsilon$.

Proof. Let $x^* = \limsup(x_n)$. Then for all $\varepsilon > 0$, $x^* + \varepsilon$ is an almost upper bound. So $x_n < x^* + \varepsilon$ for $n \geq K_\varepsilon$. Thus $x_n < a + \varepsilon$.

On the other side, suppose for every $\varepsilon > 0$, there exists K_ε such that $x_n \leq a + \varepsilon$ for $n \geq K_\varepsilon$. Then $a + \varepsilon$ is an almost upper bound. So $x^* = \inf\{\text{almost upper bounds}\} \leq a + \varepsilon$. Hence $x^* \leq a + \varepsilon$ for all ε , so $x^* \leq a$. $\square$

Lemma 30. $\liminf(x_n) \geq a$ is equivalent to: For every $\varepsilon > 0$, there exists K_ε such that $x_n \geq a - \varepsilon$ for $n \geq K_\varepsilon$.

Proof. symmetrical. $\square$

Corollary 10. (x_n) is convergent and $\lim(x_n) = a$ if and only if $\liminf(x_n) = \limsup(x_n) = a$.

Proposition 22. Let $(x_n), (y_n)$ be bounded. Then

1. $\limsup(cx_n) = c \limsup(x_n)$ if $c > 0$.
 $\limsup(cx_n) = c \liminf(x_n)$ if $c < 0$.
2. $\limsup(x_n + y_n) \leq \limsup(x_n) + \limsup(y_n)$
 $\liminf(x_n + y_n) \geq \liminf(x_n) + \liminf(y_n)$
3. If $x_n < y_n$ then $\limsup(x_n) \leq \limsup(y_n)$ and $\liminf(x_n) \geq \liminf(y_n)$

Proof. We only prove (2). Let $x^* = \limsup(x_n)$ and $y^* = \limsup(y_n)$. Then $x_n \leq x^* + \varepsilon$ and $y_n \leq y^* + \varepsilon$ for sufficiently large n . Thus $x_n + y_n \leq x^* + y^* + 2\varepsilon$. By lemma, $\limsup(x_n + y_n) \leq x^* + y^*$. $\square$

Proof of Squeeze Theorem. We have $x_n \leq y_n \leq z_n$. Also, $\lim x_n = \lim z_n = a$. We prove that $\lim y_n = a$.

First $\liminf y_n \geq \liminf x_n = \lim x_n = a$. Also, $\limsup y_n \leq \limsup z_n = \lim z_n = a$. Also, $\liminf y_n \leq \limsup y_n$. Thus

$$a \leq \liminf y_n \leq \limsup y_n \leq a.$$

Thus $\liminf y_n = \limsup y_n = a$. Thus $\lim y_n = a$. $\square$

2.5 Unbounded sequences

If $E \subset \mathbb{R}$ and is unbounded above, we simply write $\sup E = \infty$.

We also set $\sup \emptyset = -\infty$.

We say that $\lim(x_n) = \infty$ iff for any $N > 0$ there exists K_N such that $x_n > N$ for $n \geq K_N$.

If (x_n) is unbounded above, then we find a subsequence (x_{n_k}) such that $x_{n_k} \geq k$. Then $x_{n_k} \rightarrow \infty$.

If (x_n) is unbounded above, then x_n has no upper bounds, so x_n has no almost upper bounds. Thus $\inf\{\text{almost upper bounds}\} = \inf \emptyset = +\infty$.

So if (x_n) is unbounded above, we set $\limsup(x_n) = \infty$.

We restate the previous lemma that works on extended real numbers.

Lemma 31. $\limsup x_n \leq a$ iff for any $a < \alpha$, $x_n \leq \alpha$ for $n \geq K_\alpha$. This works for $a \in [-\infty, \infty]$.

Lecture 20: Continuous Functions

Mon 17 Oct 2022 11:43

3 Continuous Functions

Proposition 23. Definitions of continuity are equivalent.

Proof. We prove in the direction $1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 1$.

- Def 1 implies Def 2 Take $V = B_\varepsilon(f(a))$. By definition, there is open U , $U \ni a$, such that $f(U \cap E) \subset V = B_\varepsilon(f(a))$. Since U is open, there exists some $\delta > 0$ such that $B_\delta(a) \subset U$. Thus $B_\delta(a) \subset B_\varepsilon(f(a))$.
- Def 2 implies Def 3 Take some sequence (x_n) converging to a in E . Fix arbitrary $\varepsilon > 0$. By def 2, there exists $\delta > 0$ such that $f(B_\delta(a)) \subset B_\varepsilon(f(a))$. Then by convergence of x_n , we have $\|x_n - a\| < \delta$ for $n \geq K_\delta$. Then $f(x_n) \in B_\varepsilon(f(a))$, so $\|f(x_n) - f(a)\| < \varepsilon$ for $n \geq K_\varepsilon$. Thus $f(x_n) \rightarrow f(a)$.
- Def 3 implies Def 1 Suppose not. Then f is sequentially continuous at a , but there exists open $V \ni f(a)$ in $\mathbb{R}^q$ such that $f(U \cap E) \not\subset V$ for any open $U \ni a$. In particular, $f(B_{\frac{1}{n}}(a) \cap E) \not\subset V$. Thus there exists $x_n \in B_{\frac{1}{n}}(a) \cap E$ such that $x_n \notin V$. Thus $\|x_n - a\| < \frac{1}{n}$ and $f(x_n) \notin V$ for any n . Therefore $x_n \rightarrow a$ but $f(x_n) \not\rightarrow f(a)$ (it never enters V because it stays in the complement of V , and the complement is closed).

□

Example. Define

$$f : \mathbb{R} \longrightarrow \mathbb{R}$$

$$x \longmapsto f(x) = \begin{cases} 0 & x \in \mathbb{R} \setminus \mathbb{Q} \\ 1 & x \in \mathbb{Q} \end{cases}.$$

This is called the [Dirichlet's Function](#) .

Claim. f is discontinuous everywhere.

Proof. From density of $\mathbb{Q}$ in $\mathbb{R}$, there exists a sequence such that $r_n \in \mathbb{Q}$, $r_n \rightarrow a$. From density of $\mathbb{R} \setminus \mathbb{Q}$ in $\mathbb{R}$, there also exists a sequence such that $s_n \in \mathbb{R} \setminus \mathbb{Q}$ and $s_n \rightarrow a$. Thus $f(r_n) \rightarrow 1$ and $f(s_n) \rightarrow 0$. Thus f is not continuous at a .

◇

Example. Define

$$g : \mathbb{R} \longrightarrow \mathbb{R}$$

$$x \longmapsto g(x) = \begin{cases} 0 & x \in \mathbb{R} \setminus \mathbb{Q} \\ \frac{1}{n} & x = \frac{m}{n} \text{ in least terms, } m \in \mathbb{Z}, n \in \mathbb{N}, (m, n) = 1 \end{cases}.$$

Claim. g is discontinuous at any $a \in \mathbb{Q}$.

Proof. Take $s_n \in \mathbb{R} \setminus \mathbb{Q}$ such that $s_n \rightarrow a$. Then $g(s_n) \rightarrow 0 \neq g(a) > 0$. □

Claim. g is continuous at any $a \in \mathbb{R} \setminus \mathbb{Q}$.

Proof. Pick any $\varepsilon > 0$. Then there are only finite many $x \in (b-1, b+1)$ such that $g(x) > \varepsilon$. All of them are in $\mathbb{Q}$. Let $\delta = \min\{|b-x| : g(x) > \varepsilon, x \in (b-1, b+1)\}$. Thus $\delta > 0$. Hence $g(b-\delta, b+\delta) \subset [0, \varepsilon]$. Hence f is continuous at b . $\square$

◇

Lecture 21

Wed 19 Oct 2022 11:35

Proposition 24. Let $f : E \rightarrow \mathbb{R}^q$. Then f is continuous at a if and only all its orthogonal projections f_i are continuous at a .

Proof. Sequential argument. $\square$

Corollary 11. Orthogonal projection function is continuous. As a result, polynomials are continuous.

Proposition 25. Let $A \subset \mathbb{R}^p$, $B \subset \mathbb{R}^q$. Let $f : A \rightarrow B$ and $g : B \rightarrow \mathbb{R}^r$. Let $a \in A$ and $b = f(a) \in B$. Then, if f is continuous at a , g is continuous at $b = f(a)$, then $g \circ f$ is continuous at a .

Proof. (topological) Let $W \ni g(f(a))$ be open in $\mathbb{R}^r$. Then by continuity of g at $b = f(a)$, there exists $V \ni b$ open in $\mathbb{R}^q$ such that $g(V \cap B) \subset W$. Then by continuity of f at a , there is $U \ni a$, open in $\mathbb{R}^p$ such that $f(U \cap A) \subset V$. Thus $g \circ f(U \cap A) = g(f(U \cap A)) \subset g(V \cap B) \subset W$. Thus $g \circ f(U \cap A) \subset W$, so $g \circ f$ is continuous at a . $\square$

Theorem 30 (Global Continuity Theorem). The following statements are equivalent:

- (a) f is continuous on its domain $D(f)$.
- (b) If G is any open set in $\mathbb{R}^q$, then there exists an open set G_1 in $\mathbb{R}^p$ such that $G_1 \cap D(f) = f^{-1}(G)$.
- (c) If H is any closed set in $\mathbb{R}^q$, then there exists a closed set H_1 in $\mathbb{R}^p$ such that $H_1 \cap D(f) = f^{-1}(H)$.

Definition 25 (Relative Topology). Let $E \subset \mathbb{R}^p$. We say that $U \subset E$ is *relatively open* in E if there is open U_1 in $\mathbb{R}^p$ such that $U = U_1 \cap E$.

The relatively open subsets in E form a topology on E .

Theorem 31 (Global Continuity Theorem). The following statements are equivalent:

- (a) f is continuous on its domain $D(f)$.
- (b) $f^{-1}(V)$ is relatively open in E for any open $V \subset \mathbb{R}^q$.
- (c) $f^{-1}(F)$ is relatively closed in E for any closed $V \subset \mathbb{R}^q$.

Proof. We prove in the direction $a \Leftrightarrow b \Leftrightarrow c$.

- (a) implies (b) Let $f : E \rightarrow \mathbb{R}^q$ be continuous, and $V \subset \mathbb{R}^q$ be open. Let $a \in f^{-1}(V)$. So there exists $U_a \subset \mathbb{R}^p$ open containing a , such that $f(E \cap U_a) \subset V$.

Note: Let $U = \bigcup_{a \in f^{-1}(V)} U_a$. We claim that $f^{-1}(V) = U \cap E$.

To verify this, consider $a \in f^{-1}(V)$, then $a \in U_a \subset U$. Since $a \in E$, we have $a \in U \cap E$. Hence $f^{-1}(V) \subset U \cap E$. Also, since $f(U_a \cap E) \subset V$, we have $U_a \cap E \subset f^{-1}(V)$. So $\bigcup_{a \in f^{-1}(V)} (U_a \cap E) = U \cap E \subset f^{-1}(V)$.

- (b) implies (a) Let $a \in E$ and $f(a) \in V$ open. This implies that there exist $U \subset \mathbb{R}^p$ such that $U \cap E = f^{-1}(V)$. This implies that $f(U \cap E) \subset V$. Hence f is continuous at a .

- (b) equivalent to (c) Take $f^{-1}(V) = E \setminus f^{-1}(F)$ where $f = V^c$. $f^{-1}(V)$ is relatively open iff $f^{-1}(F)$ is closed. We have V open iff F closed.

□

Lecture 22

Fri 21 Oct 2022 11:31

3.1 Preservation of connectedness

Proposition 26. A set is connected iff it cannot be split into two disjoint relatively open sets.

Theorem 32. If $f : E \rightarrow \mathbb{R}^q$ is continuous and E is connected, then $f(E)$ is also connected.

Corollary 12. If $f : E \rightarrow \mathbb{R}^q$ is continuous and $C \subset E$ is connected, then $f(C)$ is connected.

Theorem 33 (Bolzano's Intermediate Value Theorem). Let $f : E \rightarrow \mathbb{R}$ be continuous, and E be connected, then $f(E)$ is an interval. Equivalently, f takes all values k such that

$$\inf_E f < k < \sup_E f.$$

If $a, b \in E$, then f takes all values between $f(a)$ and $f(b)$.

Proof. $f(E) \subset \mathbb{R}$ must be connected, but the only connected set in $\mathbb{R}$ must be interval. □

Fun exercise: If $p(x)$ is a polynomial with odd degree, then it always has real roots.

3.2 Preservation of Compactness

Theorem 34. Let $f : E \rightarrow \mathbb{R}^q$ be continuous and E be compact, then $f(E)$ is compact.

Corollary 13. Suppose $f : E \rightarrow \mathbb{R}^q$ and $K \subset E$ compact, f continuous on K , then $f(K)$ compact.

Theorem 35 (Extreme Value theorem). Let $f : K \rightarrow \mathbb{R}$ be continuous and K compact. Then f is bounded on K and moreover, there exists x_* and $x^* \in K$ such that $f(x_*) = \inf_K f, f(x^*) = \sup_K f$.

Proof. $f(K) \subset \mathbb{R}$ is compact, so $f(K)$ is bounded and closed. Any closed set contains its extremums. $\square$

Lemma 32. If $E \subset \mathbb{R}$ is compact, then $\inf E \in E$ and $\sup E \in E$.

Proof. $\sup E$ cannot be an exterior point of E . Since E is closed, a not in the exterior implies $a \in E$. $\square$

Lecture 23: Continuity of the inverse

Mon 24 Oct 2022 11:34

Theorem 36. Suppose $f : K \rightarrow \mathbb{R}^q$ is continuous and injective, and K is compact. Then $f^{-1} : f(K) \rightarrow K$ is also continuous.

Proof. Take any closed $F \subset \mathbb{R}^p$. Then $(f^{-1})(F) = \{y : f^{-1}(y) \in F\} = \{y : f^{-1}(y) \in K \cap F\} = f(K \cap F)$. Since $K \cap F$ is compact (as closed subset of compact set), $f(K \cap F)$ is compact (since continuous function preserves compactness). Hence $f(K \cap F)$ closed.

Thus $(f^{-1})^{-1}(F)$ is closed for any closed F . Thus it is also relatively closed in $f(K)$. Thus $f^{-1} : f(K) \rightarrow K$ is continuous. $\square$

Exercise: Find a sequential proof.

3.3 Uniform Continuity

Recall the $\varepsilon - \delta$ definition of continuity. In this definition, δ depends on both ε and the point of interest. The function is continuous, but it can be as "discontinuous" as possible.

Example. Consider $f(x) = \frac{1}{x}$. The slope is unbounded as $x \rightarrow 0$. Fix any $\varepsilon > 0$, and arbitrarily small δ won't work for all a . As we approach zero, δ must be smaller and smaller.

$$\begin{aligned} f(x) - f(a) &= \left| \frac{1}{x} - \frac{1}{a} \right| < \varepsilon \quad \text{for } x \in (a - \delta, a + \delta) \\ &= \left| \frac{x - a}{xa} \right| = \frac{\delta}{a(a - \delta)} < \varepsilon \end{aligned}$$

Now the solution of δ goes to zero as $a \rightarrow 0$. $\diamond$

Definition 26 (Uniform Continuity). We say $f : E \rightarrow \mathbb{R}^q$ is uniformly continuity if

- For any $\varepsilon > 0$, there exists $\delta > 0$ such that for all $a, x \in E$, $|f(x) - f(a)| < \varepsilon$ whenever $|a - x| < \delta$.

Remark. Uniform continuity is a global property.

Compactness implies uniform continuity.

Theorem 37. Suppose $f : K \rightarrow \mathbb{R}^q$ is continuous and K is compact, then f is uniformly continuous on K .

Proof. (Sequential) Suppose f is not uniformly continuous, then there exists ε for which there is no δ as in the def. In particular $\delta = \frac{1}{n}$ fails, there exists $x_n, y_n \in K$ such that $\|x_n - y_n\| < \frac{1}{n}$ but $\|f(x_n) - f(y_n)\| > \varepsilon$.

Since (x_n) is bounded, we use Bolzano-Weierstrass theorem to find a convergent subsequence $x_{n_k} \rightarrow x_0$. Since $y_{n_k} - x_{n_k} \rightarrow 0$, $y_{n_k} \rightarrow x_0$. By sequential continuity, $\|f(x_{n_k}) - f(y_{n_k})\| \rightarrow \|f(x_0) - f(x_0)\| = 0$. but $\varepsilon \leq \|f(x_{n_k}) - f(y_{n_k})\|$. This yields a contradiction. Thus f is uniformly continuous. $\square$

Proof. (Topological) Prove using open covers. $\square$

Fact. On any noncompact set one can construct a function that is continuous but not uniformly continuous. Conversely, on a compact set continuity is equivalent to uniform continuity.

3.4 Extensions of uniformly continuous functions

Let $f : E \rightarrow \mathbb{R}^q$ be a uniformly continuous function. Then we can extend f to

$$\tilde{f} : \overline{E} \rightarrow \mathbb{R}^q.$$

where $\overline{E}$ is closure of E .

Take any $x \in \overline{E}$. Then there exists (x_n) in E such that $x_n \rightarrow x$.

Claim. (x_n) is Cauchy implies $(f(x_n))$ is Cauchy.

Proof. We can write $\|f(x_n) - f(x_m)\| < \varepsilon$ if $\|x_n - x_m\| < \delta$. We can also write $\|x_n - x_m\| < \delta$ if $n, m \geq M$ for some M . Combine the two statements. $\square$

Therefore $f(x_n)$ is convergent. So we define $\tilde{f}(x) = \lim (f(x_n))$.

We still have to verify well-definedness.

Claim. Show that if (y_n) is another sequence in E such that $y_n \rightarrow x$ then $\lim f(x_n) = \lim f(y_n)$.

Finally we verify

Claim. $\tilde{f}(x) = f(x)$ when $x \in E$.

Claim. $\delta(\varepsilon)$ works also for $\tilde{f}(x)$, in the sense that $\|\tilde{f}(x) - \tilde{f}(y)\| \leq \varepsilon$ if $\|x - y\| < \delta(\varepsilon)$.

Lecture 24: Uniform Convergence

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3.5 Convergence of Functions

Suppose $E \subset \mathbb{R}^p$ and we have

$$f_n : E \rightarrow \mathbb{R}^q, \quad n = 1, 2, \dots$$

and

$$f : E \rightarrow \mathbb{R}^q.$$

Definition 27 (Point-wise convergence). We say $f_n \rightarrow f$ pointwise on E if

$$f_n(x) \rightarrow f(x)$$

for any $x \in E$.

Example. Take $f(x) = x^n$, $E = [0, 1]$. Then

$$f_n(x) \rightarrow f(x) = \begin{cases} 1 & x = 1 \\ 0 & x < 1 \end{cases}.$$

Note: $f_n(x)$ are continuous on $[0, 1]$ but $f(x)$ is not.

Note: It is possible to construct $f_n \rightarrow f$ with infinitely many discontinuities. $\diamond$

Remark. There is a theorem by Baire that pointwise limits are continuous at "most" points in topological sense. Point of discontinuity form a nowhere-dense, F_σ set.

Theorem 38 (Uniform Convergence). $f_n \rightarrow f$ uniformly on E if for all $\varepsilon > 0$, there exists $K_\varepsilon \in \mathbb{N}$ such that for all $x \in E$, $|f_n(x) - f(x)| < \varepsilon$ for $n > K_\varepsilon$.

Intuition. This means the function stays in a *tube* around f . The same intuition works in higher dimensions.

Theorem 39. Let $f_n : E \rightarrow \mathbb{R}^q$ converge uniformly to $f : E \rightarrow \mathbb{R}^q$. If all f_n are continuous at $a \in E$, then f is continuous at a .

Proof. Use triangle inequality,

$$\|f(x) - f(a)\| \leq \|f(x) - f_N(x)\| + \|f_N(x) - f_N(a)\| + \|f_N(a) - f(a)\|.$$

By uniform convergence, $\|f(x) - f_N(x)\| < \varepsilon$ if $N > K_1$, and $\|f(a) - f_N(a)\| < \varepsilon$ if $N > K_2$. By continuity of f_N , $\|f_N(x) - f_N(a)\| < \varepsilon$ if $\|x - a\| < \delta$.

Thus $\|f(x) - f(a)\| < 3\varepsilon$ if $\|x - a\| < \delta$. $\square$

Definition 28 (Uniform Norm). For a bounded $f : E \rightarrow \mathbb{R}^q$. Define the *uniform norm* of f be

$$\|f\|_E = \sup_{x \in E} \|f(x)\|.$$

We define $B(E, \mathbb{R}^q) = \{f \in (\mathbb{R}^q)^E : f \text{ bounded}\}$.

We define $C(E, \mathbb{R}^q) = \{f \in (\mathbb{R}^q)^E : f \text{ continuous}\}$

We define $BC(E) = B(E) \cap C(E)$. We can check that $\|\cdot\|_E$ is a norm on $B(E), C(E), BC(E)$.

Proposition 27. $f_n \rightarrow f$ uniformly on E iff $\|f_n - f\|_E \rightarrow 0$ as $n \rightarrow \infty$.

Proof. $\Rightarrow$: $\|f_n(x) - f(x)\| < \varepsilon \Rightarrow \|f_n - f\|_E \leq \varepsilon$ for all x .

$\Leftarrow$: $\|f_n - f\|_E < \varepsilon \Rightarrow \|f_n(x) - f(x)\| < \varepsilon$ for all $x \in E$. $\square$

Corollary 14. Uniform convergence preserves boundedness.

Corollary 15. $B(E), C(E)$ and $BC(E)$ are closed under uniform limit.

Proposition 28 (Cauchy's criterion for uniform convergence). $f_n : E \rightarrow \mathbb{R}^q$ is uniformly convergent iff (f_n) is uniformly Cauchy on E , i.e. $\|f_n(x) - f_m(x)\| < \varepsilon$ if $n, m \geq M(\varepsilon)$ for all x , i.e. $\|f_n - f_m\|_E < \varepsilon$ if $n, m \geq M(\varepsilon)$.

Remark. Warning: Heine-Borel, Bolzano-Weierstrass are false for pointwise and uniform convergence in $BC(E)$ (and $B(E), C(E)$).

Example. Prove that $(\sin nx)_{n=1}^{\infty}$ has no uniformly convergent subsequence. $\diamond$
 Next time we work on approximation theorems.

Lecture 25: Approximation Theorems

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We say that $g : [a, b] \rightarrow \mathbb{R}$ is *piecewise constant* if there are $a = x_0 < x_1 < \dots < x_n = b$ a partition of $[a, b]$ such that g is constant on (x_{i-1}, x_i) , $i = 1, 2, \dots, n$.

Theorem 40 (Uniform Approximation by Step Functions). For any $f \in C([a, b])$ and $\varepsilon > 0$, there is g_ε piecewise constant on $[a, b]$ such that

$$\|f - g_\varepsilon\|_{[a, b]} < \varepsilon.$$

Proof. Let $\delta_\varepsilon > 0$ be such that $|f(x) - f(y)| < \varepsilon$ if $|x - y| < \delta_\varepsilon$. δ_ε exists since f is uniformly continuous.

Now let N be so large that $\frac{b-a}{N} < \delta_\varepsilon$. We partition $[a, b]$ into N intervals of same length.

Pick $\xi_i \in [x_{i-1}, x_i]$, $i = 1, \dots, N$.

$$g_\varepsilon(x) = \begin{cases} f(\xi_i) & x \in [x_{i-1}, x_i) \\ f(\xi_N) & x = x_N \end{cases}.$$

Now for any x , $x \in [x_{i-1}, x_i)$ such that $i < N$ or $x \in [x_N, x_N]$. But $\xi_i \in [x_{i-1}, x_i)$, so $|x - \xi_i| < \delta_\varepsilon$, so $|f(\xi_i) - f(x)| < \varepsilon$. $\square$

Theorem 41. For any $f \in C([a, b])$ and $\varepsilon > 0$, there is g_ε continuous piecewise linear on $[a, b]$ such that

$$\|f - g_\varepsilon\|_{[a, b]} < \varepsilon.$$

Proof. Take same partition as before. Define $g_\varepsilon(x)$ linear on $[x_{i-1}, x_i]$, $g_\varepsilon(x_i) = f(x_i)$ for all i . (We could explicitly write the formula but there is no need).

Then $|f(x) - f(x_i)| < \varepsilon$ and $|f(x) - f(x_{i-1})| < \varepsilon$ for some i . Using property of linear combination, we have

$$f(x) - \varepsilon < f(x_i) < f(x) + \varepsilon$$

and

$$f(x) - \varepsilon < f(x_{i-1}) < f(x) + \varepsilon.$$

Now $g_\varepsilon(x)$ is between $f(x_{i-1})$, $f(x)$, so $f(x) - \varepsilon < g_\varepsilon(x) < f(x) + \varepsilon$. $\square$

Theorem 42 (Weierstrass Approximation Theorem). For any $f \in C([a, b])$ and $\varepsilon > 0$, there is a polynomial $P_\varepsilon(x)$ such that

$$\|f - P_\varepsilon\|_{[a, b]} < \varepsilon.$$

(Bernstein's Proof) We assume wlog that $[a, b] = [0, 1]$. Define the n -th *Bernstein Polynomial* as

$$B_n(x, f) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

Example. Recall Newton's Binomial Theorem:

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

Take $a = x, b = 1 - x$, we have

$$1 = \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k}.$$

Thus $B_n(x, 1) = 1$.

◇

Theorem 43. (Bernstein's Theorem) If $f \in C([0, 1])$, then $B_n(x, f) \rightarrow f$ uniformly on $[0, 1]$.

Proof. Intuitively the Bernstein Polynomial is a weighted average of values of f at a collection of points. We may write the weight as

$$\omega_k = \binom{n}{k} x^k (1-x)^{n-k}.$$

Now $\omega_k(x)$ is concentrated around $\frac{k}{n}$. We claim that

$$\sum_{k: |x - \frac{k}{n}| > \delta} \omega_k \rightarrow 0 \text{ uniformly as } n \rightarrow \infty.$$

For proof, see textbook or written note.

□

Lecture 26: Limit of function

Wed 02 Nov 2022 11:54

Remark. Bernstein's Theorem $\Rightarrow$ Weierstrass theorem on $[0, 1] \Rightarrow$ Weierstrass theorem on $[a, b]$

If $f : [a, b] \rightarrow \mathbb{R}$ continuous, let $\varphi(t) = f(a(1-t) + bt)$. Then $\varphi(t)$ is continuous. By Weierstrass on $[0, 1]$, there exists $P(t)$ polynomial such that $|\varphi(t) - P(t)| < \varepsilon$ for all $t \in [0, 1]$. Then $\left| f(x) - P\left(\frac{x-a}{b-a}\right) \right| < \varepsilon$ for $x \in [a, b]$.

Remark. (Probabilistic Proof of Bernstein Approximation Theorem)

We take X_n independent random variables taking values $[0, 1]$ where $P(X_n = 1) = x \in [0, 1]$. By law of large number, $\frac{X_1 + \dots + X_n}{n} \rightarrow x$. Now

$$P\left(\frac{X_1 + \dots + X_n}{n} = \frac{k}{n}\right) = \omega_k(x).$$

Hence the Bernstein Polynomial is

$$B_n(x, f) = E\left(f\left(\frac{X_1 + \dots + X_n}{n}\right)\right).$$

3.6 Limit of a function at a point

Warning: Skip everything about "nondeleted limit".

Let $E \subset \mathbb{R}^p$, c is a cluster point of E . We want to define what it means to say $\lim_{x \rightarrow c} f(x) = b$.

Definition 29 (Limit of Function). We say $\lim_{x \rightarrow c} f(x) = b$ if

- (Topological) for any open $V \ni b$ in $\mathbb{R}^q$ there is $U = U_V \ni c$ open in $\mathbb{R}^p$ such that $f(x) \in V$ if $x \in (U \setminus \{c\}) \cap E$.
- (Metric) For any $\varepsilon > 0$, exists $\delta > 0$ such that $|f(x) - b| < \varepsilon$ whenever $0 < |x - c| < \delta$ and $x \in E$.
- (Sequential) For any sequence x_n in E such that $x_n \rightarrow c, x_n \neq c$, we have $f(x_n) \rightarrow b$.

Remark. In the definition, we neglect the value of function at point c . Limit at a point only depends on points beside it.

We skip the proof that these definitions are equivalent.

Proposition 29. If $f : E \rightarrow \mathbb{R}^q$ and $c \in E$ is a cluster point of E , then f is continuous at c iff $\lim_{x \rightarrow c} f(x) = f(c)$.

Proof. Skipped. □

Proposition 30. Limit preserves linear combination, multiplication, subtraction (when divisor is nonzero) of functions.

Proof. Similar to proof of continuous functions. □

Lecture 27

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3.7 One-sided limits

Assume $c \in (a, b)$. We define the *right limit* of f at c as

$$\lim_{x \rightarrow c+} f(x) = \lim_{x \rightarrow c; x \in [c, b]} f(x) = \lim_{x \rightarrow c} f|_{[c, b]}.$$

Similarly if $c \in (a, b]$, we define

$$\lim_{x \rightarrow c-} f(x) = \lim_{x \rightarrow c; x \in [a, c]} f(x) = \lim_{x \rightarrow c} f|_{[a, c]}.$$

Proposition 31. If $c \in (a, b)$, then $\lim_{x \rightarrow c} f(x)$ exists iff $\lim_{x \rightarrow c+} f(x)$ and $\lim_{x \rightarrow c-} f(x)$ both exist and are equal to each other.

Proof. Very straightforward using $\varepsilon - \delta$ definition. □

Example. (Monotone Functions) We say $f : [a, b] \rightarrow \mathbb{R}$ is *nondecreasing* if $x \leq y$ implies $f(x) \leq f(y)$ for all $x, y \in E$.

We say f is *strictly increasing* if $x < y$ implies $f(x) < f(y)$. ◇

Proposition 32. Let $f : [a, b] \rightarrow \mathbb{R}$ be increasing on $[a, b]$. Then

1. for any $c \in [a, b)$ we have $f(c+) := \lim_{x \rightarrow c+} f(x)$ exists. Moreover,
 $f(c+) = \inf_{x \in (c, b]} f(x)$.

2. Similarly, if $c \in (a, b]$, we have $f(c-) = \sup_{x \in [a, c)} f(x)$.

3. For $c \in (a, b)$,

$$f(c-) \leq f(c) \leq f(c+).$$

4. If $c < d, c \in [a, b), d \in (a, b]$, then

$$f(c+) \leq f(d-).$$

Proof. 1. The set $S = \{f(x) : c < x \leq b\}$ is bounded below by $f(c)$, hence $\alpha := \inf S = \inf_{c < x \leq b} f(x)$ exists.

Note: $f(c) \leq \alpha$ since $f(c)$ is a lower bound. We now claim $\alpha = \lim_{x \rightarrow c+} f(x)$.

Fix any $\varepsilon > 0$. By definition of $\inf$, there is x_ε such that $c < x_\varepsilon \leq b$ and $\alpha \leq f(x_\varepsilon) < \alpha + \varepsilon$. Then if $c < x < x_\varepsilon$, we have $\alpha \leq f(x) \leq f(x_\varepsilon) < \alpha + \varepsilon$. In particular this means $|f(x) - \alpha| < \varepsilon$. If we define $\delta(\varepsilon) = x_\varepsilon - c$, we have

$$c < x < c + \delta(\varepsilon) \Rightarrow |f(x) - \alpha| < \varepsilon.$$

Hence $\lim_{x \rightarrow c+} f = \alpha$.

2. Symmetric.

3. Showed in part of proof for (1).

4. Pick $e \in [c, d]$. Then

$$f(c+) = \inf_{(c, b]} f(x) \leq f(e) \leq \sup_{[a, d]} f = f(d-).$$

□

Theorem 44. If $f : [a, b] \rightarrow \mathbb{R}$ is monotone (nondecreasing or nonincreasing), then f is continuous at all but at most countably many points in $[a, b]$.

Proof. Assume f is increasing. If $c \in (a, b)$ is a point of discontinuity, then $f(c-) < f(c+)$. Then there exists a rational $r \in (f(c-), f(c+))$. Let S be the set of discontinuity of f . We obtain the injection

$$\begin{aligned} \varphi : S &\longrightarrow \mathbb{Q} \\ c &\longmapsto \varphi(c) = r_c. \end{aligned}$$

We know $r_{c_1} \neq r_{c_2}$ whenever $c_1 \neq c_2$, since if $c_1 < c_2$, we have $r_{c_1} < f(c_1+) < f(c_2-) < f_{c_2}$. □

Proof. (Simpler) Define $J_c = (f(c-), f(c+))$. We verify that $\mathcal{F} = \{J_c\}_{f \text{ discontinuous at } c}$ is a disjoint family of open nonempty intervals in $\mathbb{R}$, hence it is at most countable. □

3.8 Limsup and Liminf at a point

Let $f : E \rightarrow \mathbb{R}$, c is a cluster point of E .

Definition 30. We say M is a *local upper bound* of f near c if there is an open set U such that

$$f(x) \leq M \Leftarrow x \in (U \setminus \{c\}) \cap E.$$

Equivalently,

Proposition 33. M is a *local upper bound* of f near c iff there is $r > 0$ such that for any $x \in E$,

$$f(x) \leq M \Leftarrow 0 < |x - c| < r.$$

Definition 31. We define

$$\limsup_{x \rightarrow c} f = \inf \{M : M \text{ local upper bound of } f \text{ near } c\}.$$

Proposition 34. For $r > 0$ define

$$\varphi(r) = \sup\{f(x) : x \in (B_r(c) \setminus \{c\}) \cap E\}$$

Then

$$\limsup_{x \rightarrow c} f(x) = \lim_{r \rightarrow 0+} \varphi(r) = \inf_{r > 0} \varphi(r).$$

So formally,

$$\limsup_{x \rightarrow c} f(x) = \lim_{r \rightarrow 0+} \sup_{(B_r(c) \setminus \{c\}) \cap E} f.$$

Proposition 35. Let $f, g : E \rightarrow \mathbb{R}$, c cluster point of E . Then

$$\limsup_{x \rightarrow c} f(x) + g(x) \leq \limsup_{x \rightarrow c} f(x) + \limsup_{x \rightarrow c} g(x).$$

Proof. Let $x \in (B_r(c) \setminus \{c\}) \cap E =: E_r$. Then

$$f(x) + g(x) \leq \sup_{E_r} f + \sup_{E_r} g.$$

Then

$$\sup_{E_r} f(x) + g(x) \leq \sup_{E_r} f + \sup_{E_r} g.$$

Hence

$$\varphi_{f+g}(r) \leq \varphi_f(r) + \varphi_g(r).$$

Therefore

$$\lim_{r \rightarrow 0+} \varphi_{f+g}(r) \leq \lim_{r \rightarrow 0+} \varphi_f(r) + \lim_{r \rightarrow 0+} \varphi_g(r).$$

And the result follows. $\square$

Remark. Other results of limsup and liminf can also be generalized. Should be able to prove them myself.

Lecture 28: Differentiation

Mon 07 Nov 2022 11:31

4 Differentiation

4.1 Definition and basic properties

Suppose we have a function $f : E \rightarrow \mathbb{R}^q$, $E \subset \mathbb{R}$, c cluster point of E , then we define

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}.$$

as the *derivative* of f at c .

We restrict ourselves to $E = [a, b]$, $c \in (a, b)$. When $c = a$ or $c = b$ we would rather write $f'(a+)$ and $f'(b-)$.

By definition, $f'(c) = L$ iff for any $\varepsilon > 0$, there is $\delta > 0$ such that $|\frac{f(x)-f(c)}{x-c} - L| < \varepsilon$ whenever $0 < |x - c| < \delta$.

We also say f is *differentiable* at c if $f'(c)$ exists.

Proposition 36. If f is differentiable at c , then f is continuous at c .

Proof. Take $\varepsilon = 1$ and let $\delta = \delta(\varepsilon)$ such that

$$\left| \frac{f(x) - f(c)}{x - c} - L \right| < 1.$$

Then by triangle inequality,

$$\left| \frac{f(x) - f(c)}{x - c} \right| \leq |L| + 1.$$

Hence $\|f(x) - f(c)\| \leq (\|L\| + 1)\|x - c\|$ whenever $\|x - c\| < \delta$, where $\|L\| + 1$ is a constant. This is called *Lipchitz condition* at c , which implies continuity condition of f at c . $\square$

Nonexample. $f(x) = |x|$ is not differentiable at 0, but continuous.

There are functions continuous everywhere but nowhere differentiable. For example, the [Weierstrass Function](#)

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{2^n} \cos(3^n x).$$

Intuition. Intuition. In this function, 3^n is multiple of derivative of $\cos(3^n x)$ but only divided by 2^n , so we are adding finer and sharper oscillation to the function. In the end the function will be so sharp in its fine detail.

Remark. We always take small enough δ so that $(c - \delta, c + \delta) \subset (a, b)$. This is always possible.

Proposition 37. Let $f : [a, b] \rightarrow \mathbb{R}$. Suppose $f'(c) > 0$. Then there is $\delta > 0$ such that $f(x) > f(c)$ for $c < x < c + \delta$ and $f(x) < f(c)$ for $c - \delta < x < c$.

Proof. Take $\varepsilon = \frac{1}{2}f'(c)$. Then for $|x - c| < \delta$,

$$\left| \frac{f(x) - f(c)}{x - c} - f'(c) \right| < \frac{1}{2}f'(c).$$

Then

$$\frac{1}{2}f'(c) < \frac{f(x) - f(c)}{x - c}.$$

In the first case, if $c < x < c + \delta$, we have $x - c > 0$, so $f(x) - f(c) > \frac{1}{2}f'(c)(x - c) > 0$.

In the second case, $c - \delta < x < c$, then $x - c < 0$. Hence $f(x) - f(c) < \frac{1}{2}f'(c)(x - c) < 0$. $\square$

Definition 32 (Local Maximum). We say f has a *local maximum* (or relative maximum) at c , if there exists $\delta > 0$ such that $f(x) \leq f(c)$ for any $x \in B_\delta(c) \cap E$.

Theorem 45 (Interior Maximum Theorem). Let $f : [a, b] \rightarrow \mathbb{R}$. If $c \in (a, b)$ and f has a relative maximum (or minimum) at c , then there are two possibilities

1. f is differentiable at c and $f'(c) = 0$
2. f is not differentiable at c .

Proof. Suppose $f'(c)$ exists. Then if $f'(c) > 0$, then there exists $\delta > 0$ such that $f(x) > f(c)$ for $c < x < c + \delta$, so c is not relative maximum. Also, $f(x) < f(c)$ for $c - \delta < x < c$, so c is not local minimum.

The same holds for $f'(c) < 0$. Hence $f'(c) = 0$. $\square$

4.2 Mean Value Theorem and some Applications

Theorem 46 (Rolle's Theorem). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and differentiable in (a, b) . If $f(a) = f(b) = 0$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.

Proof. f must achieve maximum and minimum values somewhere $x^*, x_* \in [a, b]$ since f continuous and $[a, b]$ compact. If x^* or x_* is an interior point then we are done by the Interior Maximum Theorem. If both x^* and x_* are boundary points, then f is constantly zero, thus f' is constantly zero. $\square$

Theorem 47 (Mean Value Theorem). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and differentiable on (a, b) . Then there is $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

Proof. We reduce to Rolle's Theorem. Define $l(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a)$. Now we have $l(a) = f(a)$ and $l(b) = f(b)$. Then $f - l$ is differentiable on (a, b) and both its endpoints are 0. Hence by Rolle's Theorem, there is $c \in (a, b)$ such that $(f - l)'(c) = 0$. This means

$$f'(c) = l'(c) = \frac{f(b) - f(a)}{b - a}.$$

$\square$

Theorem 48 (Cauchy's Mean Value Theorem). Let f, g be continuous functions on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

In the sense that $g'(c)(f(b) - f(a)) = f'(c)(g(b) - g(a))$.

Proof. See textbook. □

Lecture 29

Wed 09 Nov 2022 11:32

Proposition 38. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and differentiable in (a, b) . Then

- f nondecreasing on $[a, b]$ iff $f'(x) \geq 0$ for any $x \in (a, b)$
- f nonincreasing on $[a, b]$ iff $f'(x) \leq 0$ for any $x \in (a, b)$
- f constant on $[a, b]$ iff $f'(x) = 0$ for any $x \in (a, b)$

Proof. We only prove (1). (2) is similar, (3) follows from (1),(2). For any $x \in (a, b)$, we have

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Then when $h > 0$, we have $f(x+h) - f(x) \geq 0$, thus

$$\frac{f(x+h) - f(x)}{h} \geq 0$$

Hence $f'(x) \geq 0$.

Conversely, if $x < y$, $x, y \in [a, b]$, then by Mean Value Theorem, there exists $c \in (x, y)$ such that

$$\frac{f(y) - f(x)}{y - x} = f'(c) \geq 0.$$

Hence $f(y) - f(x) \geq 0$. □

Proposition 39 (L'Hopital's rule). Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous and differentiable in (a, b) . Suppose $f(a) = g(a) = L$ and $\lim_{x \rightarrow a+} \frac{f'(x)}{g'(x)} = L$. Then $\lim_{x \rightarrow a+} \frac{f(x)}{g(x)} = L$.

Proof. For any $\varepsilon > 0$ there exists $\delta(\varepsilon) > 0$ such that for all $a < y < a + \delta(\varepsilon)$,

$$\left| \frac{f'(y)}{g'(y)} - L \right| < \varepsilon.$$

Now take $a < x < a + \delta(\varepsilon)$, we have by Cauchy's Mean Value Theorem,

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c_x)}{g'(c_x)}$$

where $c_x \in (a, x)$. In particular $a < c_x < a + \delta(\varepsilon)$. Hence

$$\left| \frac{f'(c_x)}{g'(c_x)} - L \right| < \varepsilon.$$

Then the desired result follows. $\square$

4.3 Taylors theorem

Theorem 49 (Taylor's Theorem). Let f be n times differentiable in (a, b) , and $\alpha, \beta \in (a, b)$. Then

$$f(\beta) = f(\alpha) + \frac{f'(\alpha)}{1!}(\beta - \alpha) + \dots + \frac{f^{(n-1)}(\alpha)}{(n-1)!}(\beta - \alpha)^{n-1} + \frac{f^n(\gamma)}{n!}(\beta - \alpha)^n$$

where γ is between α and β . We define $R_n(f, \alpha, \beta) = \frac{f^n(\gamma)}{n!}(\beta - \alpha)^n$ to be the *remainder term*.

Observation. When $n = 1$, we get the mean value theorem:

$$\frac{f(\beta) - f(\alpha)}{\beta - \alpha} = f'(\gamma).$$

Take $\beta = \alpha + x$ and we observe that Taylor Theorem allows us to express $f(\alpha + x)$ in a polynomial of x plus a remainder term.

Proof. Let P be such that

$$f(\beta) - \left(f(\alpha) + \frac{f'(\alpha)}{1!}(\beta - \alpha) + \dots + \frac{f^{(n-1)}(\alpha)}{(n-1)!}(\beta - \alpha)^{n-1} \right) = \frac{P(\beta - \alpha)^n}{n!}.$$

Let

$$\varphi(x) = f(\beta) - \left(f(x) + \frac{f'(x)}{1!}(\beta - x) + \dots + \frac{f^{(n-1)}(x)}{(n-1)!}(\beta - x)^{n-1} + \frac{P(\beta - x)^n}{n!} \right)$$

Then $\varphi(\alpha) = 0$, and $\varphi(\beta) = 0$. By Rolle's Theorem, there exists γ between α, β such that $\varphi'(\gamma) = 0$, which gives $P = f^{(n)}(\gamma)$. $\square$

4.4 Integration

A partition P of an interval $[a, b]$ is given by a sequence

$$P : a = x_0 < x_1 < \dots < x_{N-1} < x_N = b.$$

Lecture 30

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4.5 The Riemann-Stieltjes Integral

Definition 33 (Riemann-Stieltjes Integral). Let $f : [a, b] \rightarrow \mathbb{R}$ bounded and $g : [a, b] \rightarrow \mathbb{R}$ monotone increasing. We define the *Riemann-Stieltjes Integral* to be

$$\int_a^b f dg.$$

Definition 34 (Riemann Sum). Given a partition $P : a = x_0 < x_1 < \dots < x_N = b$. Let $\Delta g_i = g(x_i) - g(x_{i-1})$. Let also $\xi_i \in [x_{i-1}, x_i]$, $i = 1, \dots, N$. We define $m_i(f) = \inf_{[x_{i-1}, x_i]} f$ and $M_i(f) = \sup_{[x_{i-1}, x_i]} f$. Note that $m_i \leq f(\xi_i) \leq M_i$. Now we define the *Riemann Sum*

$$S(P, f, g) = \sum_{i=1}^N f(\xi_i) \Delta g_i.$$

The *Upper Sum (Darboux)*

$$U(P, f, g) = \sum_{i=1}^N M_i \Delta g_i.$$

The *Lower Sum*

$$L(P, f, g) = \sum_{i=1}^N m_i \Delta g_i.$$

We have

$$L(P, f, g) \leq S(P, f, g) \leq U(P, f, g).$$

Note: Take supremum over ξ_i in $S(P, f, g)$ and we get the upper sum. Take infimum and we get the lower sum.

Definition 35 (R-S Integrability). We say f is R-S integrable and $\int_a^b f dg = I$ if

(Riemann) For any $\varepsilon > 0$ there is a partition P_ε of $[a, b]$ such that $|S(P_\varepsilon, f, g) - I| < \varepsilon$ for any P finer than P_ε .

(Darboux) Define $I^* = \inf U(P, f, g) = \overline{\int_a^b f dg}$ and $I_* = \sup L(P, f, g) = \underline{\int_a^b f dg}$. Then for any (bounded) f , $I_* \leq I^*$. We say f is R-S integrable if $I_* = I^*$ and we define the integral to be the common value.

Proposition 40. Integrals in Riemann definition are uniquely defined. If $\int_a^b f dg = I_1$ and $\int_a^b f dg = I_2$, then $I_1 = I_2$.

Proof. Let $P_\varepsilon, P'_\varepsilon$ be such that $|S(P, f, g) - I_1| < \varepsilon$ if P finer than P_ε and $|S(P, f, g) - I_2| < \varepsilon$ if P finer than P'_ε . Take a common refinement P of $P_\varepsilon, P'_\varepsilon$. Then

$$|I_1 - I_2| \leq |S(P, f, g) - I_1| + |S(P, f, g) - I_2| \leq 2\varepsilon.$$

ε arbitrary, so $I_1 = I_2$. □

Lemma 33. If Q finer than P then $L(Q, f, g) \geq L(P, f, g)$ and $U(Q, f, g) \leq U(P, f, g)$. Also, for any two partitions P, Q we have $L(P, f, g) \leq U(Q, f, g)$.

Proof. Let $P : a = x_0 < \dots < x_N = b$. Suppose $Q : a < x_0 < \dots < x_{i-1} < \bar{x} < x_i < \dots < x_N = b$ (We only add one point to P).

Let $m'_i = \inf_{[x_{i-1}, \bar{x}]} f$ and $m''_i = \inf_{[\bar{x}, x_i]} f$. Then $m_i \leq m'_i$ and $m_i \leq m''_i$.

Then

$$L(P, f, g) = \sum_{k \neq i} m_k \Delta g_k + m_i (g(x_i) - g(x_{i-1})).$$

and

$$\begin{aligned} L(Q, f, g) &= \sum_{k \neq i} m_k \Delta g_k + m'_i (g(\bar{x}) - g(x_{i-1})) + m''_i (g(\bar{x}) - g(x_i)) \\ &\geq \sum_{k \neq i} m_k \Delta g_k + m_i (g(\bar{x}) - g(x_{i-1})). \end{aligned}$$

The first claim follows from induction. Proof for upper sum is symmetric.

Now take arbitrary P, Q . Take R a common refinement of P, Q . Then

$$L(Q, f, g) \leq L(R, f, g) \leq U(R, f, g) \leq U(P, f, g).$$

□

Now from lemma, $I_* = \sup_P L(P, f, g) \leq U(Q, f, g)$ for any Q . Hence

$$I_* \leq \inf_Q U(Q, f, g) = I^*.$$

Proposition 41. R-S integrability in Riemann sense and in Darboux sense are equivalent.

Proof. Assume integrability in Riemann sense. Let $\int_a^b f dg = I$. Then we know $|S(P, f, g) - I| < \varepsilon$ if P finer than P_ε . Recall $U(P, f, g) = \sup_{\xi_i} S(P, f, g)$, hence $|U(P, f, g) - I| \leq \varepsilon$ for any P finer than P_ε . Hence $I^* \leq I + \varepsilon$ for any ε . Hence $I^* \leq I$.

Similarly, we show $I_* \geq I$. Hence $I^* = I_* = I$.

Now assume integrability in Darboux sense. Suppose $I^* = I_* = I$. By definition of sup and inf, there are partitions P_ε and Q_ε such that

$$I < U(Q_\varepsilon, f, g) \leq I + \varepsilon.$$

and

$$I > L(P_\varepsilon, f, g) \geq I - \varepsilon.$$

Now take common refinement of Q_ε and P_ε , say R_ε . Hence

$$I - \varepsilon < L(R_\varepsilon, f, g) \leq U(R_\varepsilon, f, g) \leq I + \varepsilon.$$

Hence for all P finer than R_ε ,

$$I - \varepsilon < L(P, f, g) \leq U(P, f, g) \leq I + \varepsilon.$$

Hence $I - \varepsilon \leq S(P, f, g) \leq I + \varepsilon$. Hence $|S(P, f, g) - I| < \varepsilon$ if P finer than R_ε . □

Lecture 31

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4.6 Properties of Integral

Example. 1. When $g(x) = x$, the R-S integral is called [Riemann Integral](#).

2. Let

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

be the *Dirichet's function*.

Claim. f not Riemann integrable on any $[a, b], a < b$.

Proof. Let P be any partition of $[a, b]$. By density of rationals (and irrationals),

$$m_i = \inf_{[x_{i-1}, x_i]} f = 0 \quad \text{for any } i = 1, \dots, N.$$

$$M_i = \sup_{[x_{i-1}, x_i]} f = 1 \quad \text{for any } i = 1, \dots, N.$$

Hence $L(P, f) = 0$, $U(P, f) = \sum_{i=1}^N 1 \Delta x_i = b - a$. Hence the upper and lower sums do not match. Hence not integrable in Darboux sense. $\square$

3. Let

$$g(x) = \begin{cases} 0 & a \leq x \leq c \\ 1 & c < x \leq b \end{cases}, c \in (a, b).$$

Let P be a partition containing c . Let i be such that $c = x_{i-1}$. Then note: $\Delta g_k = 0$ unless $k = i$. Moreover, $\Delta g_i = 1$. Therefore,

$$S(P, f, g) = \sum_{k=1}^N f(\xi_k) \Delta g_k \quad \xi_k \in [x_{k-1}, x_k].$$

$$S(P, f, g) = f(\xi_i).$$

$\int_a^b f dg$ exists iff $\lim_{x \rightarrow c+} f(x) = f(c)$ (continuity from the right). We have $\int_a^b f dg = f(c)$.

Remark. This allows us to treat series as integrals.

$\diamond$

Theorem 50 (Change of Variable). Suppose $g(x)$ monotone increasing and **continuously differentiable**. Let $f(x)$ be a bounded function. Then $\int_a^b f dg$ exists iff $\int_a^b f g' dx$ exists, and

$$\int_a^b f dg = \int_a^b f g' dx.$$

Proof. Since g' continuous on $[a, b]$, g' uniformly continuous on $[a, b]$. Hence for any $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, y \in [a, b]$,

$$|x - y| < \delta \Rightarrow |g'(x) - g'(y)| < \varepsilon.$$

Now pick P_ε be a partition such that $\Delta x_i < \delta$. Then for any P finer than P_ε we have $\Delta x_i < \delta$ for P as well. Now

$$S(P, f, g) = \sum_{i=1}^N f(\xi_i) (g(x_i) - g(x_{i-1})).$$

$$S(P, fg') = \sum_{i=1}^N f(\xi_i) g'(\xi_i) (x_i - x_{i-1}).$$

Apply mean value theorem, there exists $\zeta \in [x_{i-1}, x_i]$ such that

$$f(\xi_i) (g(x_i) - g(x_{i-1})) = f(\xi_i) g'(\zeta_i) (x_i - x_{i-1}).$$

Now

$$\begin{aligned} |S(P, f, g) - S(P, fg')| &\leq \sum_{i=1}^N |f(\xi_i)| |g'(\zeta_i) - g'(\xi_i)| \Delta x_i \\ &\leq \|f\| \varepsilon (b - a). \end{aligned}$$

Now if one of the integrals exists, then this statement implies that the other integral must exist. Hence the theorem holds. $\square$

Proposition 42. (Properties of [R-S integral](#)) Suppose $f_1, f_2 : [a, b] \rightarrow \mathbb{R}$ bounded, and g monotone increase. Then if $\int_a^b f_1 dg$ and $\int_a^b f_2 dg$ exist, then

- (Linearity) For $\alpha_1, \alpha_2 \in \mathbb{R}$,

$$\int_a^b (\alpha_1 f_1 + \alpha_2 f_2) dg = \alpha_1 \int_a^b f_1 dg + \alpha_2 \int_a^b f_2 dg.$$

- (Linearity in integrator) If $\beta_1, \beta_2 \geq 0$, then

$$\int_a^b f d(\beta_1 g_1 + \beta_2 g_2) = \beta_1 \int_a^b f dg_1 + \beta_2 \int_a^b f dg_2.$$

Proof. • (Linearity)

- (Linearity in integrator) Riemann Sum is linear.

$\square$

Proposition 43. (Additivity of [R-S integral](#)) Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and g monotone increase, $c \in [a, b]$. Then $\int_a^b f dg$ exists iff $\int_a^c f dg$ and $\int_c^b f dg$ exist, and then

$$\int_a^b f dg = \int_a^c f dg + \int_c^b f dg.$$

Proof. Let $I = \int_a^b f dg$. Without loss of generality we consider only partitions that contain c . For any $\varepsilon > 0$, there exists P_ε such that

$$|S(P, f, g) - I| < \varepsilon \quad \text{if } P > P_\varepsilon.$$

Assume existence of integral, then for $P > P_\varepsilon$,

$$|U(P, f, g) - L(P, f, g)| \leq 2\varepsilon.$$

For $P > P_\varepsilon$, P contains $c = x_M$. Then $P' : a = x_0 < x_1 < \dots < x_M = c$ is a partition of $[a, c]$, and $P'' : c = x_M < x_{M+1} < \dots < x_N = b$ is a partition of $[c, b]$. Now

$$\begin{aligned} |U(P, f, g) - L(P, f, g)| &= \sum_{i=1}^N (M_i - m_i) \Delta g_i \\ &\geq \sum_{i=1}^M (M_i - m_i) \Delta g_i \\ &= U(P', f, g) - L(P', f, g). \end{aligned}$$

Hence for $P > P_\varepsilon$,

$$|U(P', f, g) - L(P', f, g)| \leq 2\varepsilon.$$

Hence

$$\left| \overline{\int_a^c f dg} - \underline{\int_a^c f dg} \right| \leq 2\varepsilon.$$

Hence the integral exists. Similarly $\int_c^b f dg$ exists.

On the other direction, suppose both $\int_a^c f dg$ and $\int_c^b f dg$ exist. Then we can put two partitions together in a similar way, and conclude that for fine P ,

$$\left| S(P, f, g) - \left(\int_a^c f dg + \int_c^b f dg \right) \right| \leq 2\varepsilon.$$

Now to verify additive identity,

$$S(P, f, g) = S(P', f, g) + S(P'', f, g).$$

Hence by taking fine partitions, each is ε -close to the corresponding integral. So

$$\left| \int_a^b f dg - \left(\int_a^c f dg + \int_c^b f dg \right) \right| \leq 3\varepsilon \rightarrow 0.$$

□

Lecture 32

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Theorem 51 (Mean Value Theorem). If f is continuous on $[a, b]$, g monotone increase on $[a, b]$, then there is $c \in [a, b]$ such that

$$\int_a^b f dg = f(c) [g(b) - g(a)].$$

Remark. In this theorem c is in the closed interval, and we cannot restrict it to the open interval. For example, let g be constantly zero on $[a, b)$ and 1 on $[b, \infty)$. Then $\int_a^b f dg = f(b)(g(b) - g(a))$.

Theorem 52 (Differentiation of Integral). Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$ and continuous at $c \in [a, b]$. Let

$$F(x) = \int_a^x f(t) dt.$$

Then $F'(c) = f(c)$.

Definition 36 (Antiderivative). We say that G is an *antiderivative* of f on $[a, b]$ if

1. G continuous on $[a, b]$
2. $G'(x) = f(x)$ on (a, b)

Corollary 16 (Fundamental Theorem of Calculus). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Let G be an antiderivative of f on $[a, b]$. Then

$$\int_a^b f(x) dx = G(b) - G(a).$$

Also called [\[\[Newton-Leibniz Formula \]\]](#).

Proof. Let $F(x) = \int_a^x f(t) dt$, $x \in [a, b]$. We claim that F is antiderivative of f . (2) follows from differentiation theorem. (1) follows from the fact that as

$$|F(y) - F(x)| = \left| \int_x^y f(t) dt \right| \leq M|x - y|.$$

hence F Lipschitz continuous, hence continuous on $[a, b]$.

Then look at

$$H(x) = G(x) - F(x).$$

We have

1. H continuous on $[a, b]$
2. $H'(x) = 0$ for every $x \in (a, b)$

Hence H is constant on $[a, b]$. Hence $G(x) = F(x) + C$ for all $x \in [a, b]$, for some $C \in \mathbb{R}$. Now

$$G(b) - G(a) = (F(b) + C) - (F(a) + C) = F(b) - F(a) = F(b) = \int_a^x f(t)dt.$$

□

Theorem 53 (Integration By Parts). Suppose f, g continuously differentiable on $[a, b]$. Then

$$\int_a^b f g' dx = f g|_a^b - \int_a^b g f' dx.$$

Proof. From product rule, $(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$. We can integrate this since $(fg)'$ is continuous. Now by Newton-Leibniz,

$$\int_a^b (fg)' dx = f(b)g(b) - f(a)g(a).$$

On the other hand

$$\int_a^b (fg)' dx = \int_a^b f g' dx + \int_a^b g f' dx.$$

□

Theorem 54. If f bounded on $[a, b]$, g monotone increase and continuously differentiable on $[a, b]$, then

$$\int_a^b f dg \text{ exists} \Leftrightarrow \int_a^b f g' dx \text{ exists.}$$

and the two integrals are equal.

Theorem 55. Suppose f, g monotone increase on $[a, b]$, then

$$\int_a^b f dg \text{ exists} \Leftrightarrow \int_a^b g df \text{ exists.}$$

and then

$$\int_a^b f dg = f g|_a^b - \int_a^b g df.$$

Proof. By Riemann Criterion for integrability, $\int_a^b f dg$ exists iff for any $\varepsilon > 0$ there exists partition P_ε such that

$$\sum_{i=1}^N (M_i(f) - m_i(f)) \Delta g_i < \varepsilon \quad .$$

where

$$m_i(f) = \inf_{[x_{i-1}, x_i]} f = f(x_{i-1}).$$

$$M_i(f) = \sup_{[x_{i-1}, x_i]} f = f(x_i).$$

Hence

$$\sum_{i=1}^N (M_i(f) - m_i(f)) \Delta g_i = \sum_{i=1}^N \Delta f_i \Delta g_i = \sum_{i=1}^N (M_i(g) - m_i(g)) \Delta f_i.$$

Hence the existence criterion for the two integrals are equivalent.

Let P be any partition of $[a, b]$ such that

$$U(P, f, g) = \sum_{i=1}^N f(x_i) [g(x_i) - g(x_{i-1})].$$

$$L(P, g, f) = \sum_{i=1}^N g(x_{i-1}) [f(x_i) - f(x_{i-1})].$$

Now

$$\begin{aligned} U(P, f, g) + L(P, g, f) &= \sum_{i=1}^N [f(x_i)g(x_i) - f(x_{i-1})g(x_{i-1})] \\ &= f(b)g(b) - f(a)g(a). \end{aligned}$$

Hence

$$\int_a^b f dg + \int_a^b g df = f(b)g(b) - f(a)g(a).$$

□

Theorem 56 (Change of Variables). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and $\varphi : [\alpha, \beta] \rightarrow [a, b]$ be continuously differentiable and such that

$$\varphi(\alpha) = a, \varphi(\beta) = b.$$

Then

$$\int_a^b f(x) dx = \int_{\alpha}^{\beta} f(\varphi(t)) \varphi'(t) dt.$$

Proof. Let

$$F(x) = \int_a^x f(\xi) d\xi.$$

and

$$\Phi(t) = \int_{\alpha}^t f(\varphi(\tau)) \varphi'(\tau) d\tau.$$

We look at $F(\varphi(\tau))$, $\tau \in [\alpha, \beta]$

1. $F(\varphi(\tau))$ continuous on $[\alpha, \beta]$

$$2. (F(\varphi(\tau)))' = F'(\varphi(\tau))\varphi'(\tau),$$

so $F(\varphi(\tau))$ is an antiderivative of $f(\varphi(t))\varphi'(t)$ on $[\alpha, \beta]$.

By the Newton-Leibniz Formula,

$$\int_{\alpha}^{\beta} f(\varphi(t))\varphi'(t)dt = F(\varphi(\beta)) - F(\varphi(\alpha)) = F(b) - F(a) = \int_a^b f(x)dx.$$

□

Theorem 57. Suppose $f : [a, b] \rightarrow \mathbb{R}$ bounded and $\varphi : [\alpha, \beta] \rightarrow [a, b]$ monotone increase, and φ is a bijection. Then

$$\int_a^b f(x)dx \text{ exists} \Leftrightarrow \int_{\alpha}^{\beta} f \circ \varphi d\varphi \text{ exists.}$$

and the integrals are equal.

Proof. Let P be a partition of $[\alpha, \beta]$. Denote $P : \alpha = t_0 < t_1 < \dots < t_N = \beta$. Let $\varphi(P) : a = x_0 < x_1 = \varphi(t_1) < x_2 = \varphi(t_2) < \dots < x_N = \beta$.

Now

$$\begin{aligned} U(P, f, \varphi) &= U(\varphi(P), f, x) \\ L(P, f, \varphi) &= L(\varphi(P), f, x). \end{aligned}$$

This finishes the proof. □

4.7 Convergence and Integral

In general, if $f_n : [a, b] \rightarrow \mathbb{R}$ are R-S integrable w.r.t g monotone increase on $[a, b]$.

$f_n \rightarrow f$ pointwise, does not imply f R-S integrable, and even if the limit exists, the integrals are not necessarily equal.

Example. Let f be the Dirichlet function

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}.$$

We enumerate the rational numbers $\mathbb{Q} = \{r_n : n \in \mathbb{N}\}$. Let

$$f_n(x) = \begin{cases} 1 & x = r_1, r_2, \dots, r_n \\ 0 & \text{otherwise} \end{cases}.$$

f_n has finitely many discontinuities, so f_n is R-S integrable. Also, $f_n \rightarrow f$ pointwise. But f not R-S integrable. ◇

Example. Let f_n be piecewise linear such that

$$f_n(0) = 0, f_n\left(\frac{1}{n}\right) = n, f_n\left(\frac{2}{n}\right) = 0.$$

Then $f_n \rightarrow 0$ pointwise, but $\int_0^1 f_n dx = 1$. $\diamond$

Theorem 58 (Bounded Convergence Theorem). Suppose $f_n : [a, b] \rightarrow \mathbb{R}$ are uniformly bounded on $[a, b]$, i.e. there exists $B \geq 0$ such that

$$|f_n(x)| \leq B \text{ for all } x \in [a, b], n \in \mathbb{N}.$$

If f_n are R-S integrable wrt g and $f_n(x) \rightarrow f(x)$ pointwise, then f is also R-S integrable and

$$\int_a^b f_n dg \rightarrow \int_a^b f dg.$$

This theorem is easy to prove in the Lebesgue integration theory, but very difficult to prove in RS integration theory. Hence we omit the proof.

Theorem 59. Let $f_n : [a, b] \rightarrow \mathbb{R}$ be R-S integrable wrt g , g monotone increase on $[a, b]$, and $f_n \rightarrow f$ uniformly on $[a, b]$. Then f is R-S integrable wrt g on $[a, b]$, and $\int_a^b f_n dg \rightarrow \int_a^b f dg$.

Proof. Suppose f is integrable, then

$$\left| \int_a^b f_n dg - \int_a^b f dg \right| \leq \int_a^b |f_n - f| dg \leq \varepsilon \int_a^b dg = \varepsilon(g(b) - g(a)).$$

Now we verify integrability of f . $\square$

For any ε , we have for sufficiently fine partition that

$$|U(P, f_n, g) - U(P, f, g)| \leq \varepsilon(g(b) - g(a)).$$

Hence for $n \geq K_\varepsilon$,

$$\left| \overline{\int_a^b f_n dg} - \overline{\int_a^b f dg} \right| \leq \varepsilon.$$

Symmetrically we have the result for lower integral. But since f_n integrable, we have

$$\left| \overline{\int_a^b f dg} - \underline{\int_a^b f dg} \right| \leq 2\varepsilon.$$

This proves f is RS integrable.

Lecture 33: Series

Wed 30 Nov 2022 11:30

5 Series

5.1 Definitions

Definition 37. (Series of Numbers) Define the *partial sum*

$$S_n = \sum_{k=1}^n x_k.$$

We say

$$S = \sum_{k=1}^{\infty} x_k : \Leftrightarrow S_n \rightarrow S \text{ as } n \rightarrow \infty.$$

Definition 38 (Absolutely Convergent). We say a series $\sum_{k=1}^{\infty} x_k$ is *absolutely convergent* if $\sum_{k=1}^{\infty} |x_k|$.

Lemma 34. Let $\sum_{k=1}^{\infty} x_n$ be a series with $x_n \geq 0$. Then it is convergent iff there exists M such that $S_n \leq M$ for all n . Then $\sum_{k=1}^{\infty} x_n \leq M$.

Proof. By Monotone Convergence Theorem applied to S_n . □

Lemma 35 (Cauchy Criterion). $\sum x_n$ converges iff for any $\varepsilon > 0$ there is M_ε such that $\sum_{k=n}^m x_k$ if $n \geq m \geq M_\varepsilon$.

Proof. S_n converges iff S_n is Cauchy sequence. □

Proposition 44. Absolute convergence implies convergence.

Proof. Use triangle inequality, prove that $\sum x_n$ is Cauchy. □

Definition 39 (Rearrangement of Series). Take $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ be a bijection. The series

$$\sum_{n=1}^{\infty} x_{\varphi(n)}$$

is called a *rearrangement* of $\sum_{n=1}^{\infty} x_n$.

Theorem 60. If $x_n \geq 0$, $\sum_{n=1}^{\infty} x_n < \infty$, then $\sum_{n=1}^{\infty} x_{\varphi(n)} < \infty$ and $\sum_{n=1}^{\infty} x_n = \sum_{n=1}^{\infty} x_{\varphi(n)}$

Proof. Let $S'_n = X_{\varphi(1)} + \dots + X_{\varphi(n)}$. Let $N = \max\{\varphi(1), \dots, \varphi(n)\}$. Hence for any n ,

$$S'_n \leq S_N \leq M.$$

Thus $\sum_{n=1}^{\infty} x_{\varphi(n)} = \lim_{n \rightarrow \infty} S'_n \leq M$. $\square$

Theorem 61. If $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ is bijection, then $\sum x_n$ absolutely convergent iff $\sum x_{\varphi(n)}$ absolutely convergent, and then

$$\sum_{n=1}^{\infty} x_n = \sum_{n=1}^{\infty} x_{\varphi(n)}.$$

Proof. Convergence follows from the proposition above. To prove the sums are equal, split into positive and negative parts. $\square$

Definition 40 (Conditionally Convergent). $\sum x_n$ is convergent, but not absolutely.

Equivalently, both $\sum_{n=1}^{\infty} x_n^+$ and $\sum_{n=1}^{\infty} x_n^-$ are infinite.

Theorem 62. If $\sum_{n=1}^{\infty} x_n$ converges conditionally, then for each real number x there exists $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$\sum_{n=1}^{\infty} x_{\varphi(n)} = x.$$

Proposition 45. $\sum a_n$ convergent implies $a_n \rightarrow 0$.

Proof. $a_n = S_n - S_{n-1} \rightarrow S - S = 0$ $\square$

Example. 1. (Geometric series)

2. (p -series)

$$\sum_{n=1}^{\infty} \frac{1}{n^p}.$$

If $p \leq 0$, then $\frac{1}{n^p} \geq 1 \not\rightarrow 0$ divergent. If $p > 0$, $\frac{1}{n^p} \rightarrow 0$. But is the series it convergent?

Apply *Cauhy's condensation method*:

$$S_{2^k} \geq 1 + \frac{1}{2} \left(\frac{1}{2^{p-1} + \frac{1}{(2^{p-1})^2}} + \dots + \frac{1}{(2^{p-1})^k} \right).$$

$$S_{2^k} \geq 1 + \frac{1}{2} (a + a^2 + \dots + a^k) \quad a = \frac{1}{2^{p-1}}.$$

So the series diverges if $p \leq 1$.

◇

Theorem 63 (Cauchy's Condensation Test). Let $a_n \geq 0$, a_n strictly decreasing. Then $\sum a_n < \infty$ iff $\sum 2^k a_{2^k} < \infty$.

Theorem 64 (Comparison Theorem). Suppose $\sum x_n$, $\sum y_n$, $x_n, y_n \geq 0$ are such that $x_n \leq y_n$ for $n \geq N$. Then

$$\sum y_n < \infty \Rightarrow \sum x_n < \infty.$$

Also

$$\sum x_n = \infty \Rightarrow \sum y_n = \infty.$$

Proof. work with partial sums. □

Theorem 65 (Limit Comparison Theorem). Suppose $\limsup \frac{x_n}{y_n} < \infty$, then

$$\sum y_n < \infty \Rightarrow \sum x_n < \infty.$$

Suppose $\liminf \frac{x_n}{y_n} > 0$ then

$$\sum y_n = \infty \Rightarrow \sum x_n = \infty.$$

Theorem 66 (Root Test). Let $x_n \geq 0$ and

$$\rho = \limsup_{n \rightarrow \infty} |x_n|^{\frac{1}{n}}.$$

Then

1. If $\rho < 1$ then $\sum x_n$ absolutely convergent
2. If $\rho > 1$ then $\sum x_n$ divergent.
3. Otherwise the test is conclusive.

Proof. 1. Choose $\rho < \tilde{\rho} < 1$. Then there exists N such that for $n \geq N$,

$$|x_n|^{\frac{1}{n}} \leq \tilde{\rho}.$$

Hence $|x_n| \leq \tilde{\rho}^n$. Now we can apply comparison test.

2. Pick subsequence such that $|x_{n_k}|^{\frac{1}{n_k}} \geq 1$, so $x_n \not\rightarrow 0$.

□

Lecture 34: Rearrangement of Series

Fri 02 Dec 2022 11:32

- Convergence and sum is preserved under rearrangement when the convergence is absolute.
- If a sum converges but not absolutely, then both positive and negative parts must diverge.

Lecture 35

Mon 05 Dec 2022 11:29

Theorem 67 (Ratio Test). Suppose $x_n \neq 0$, and for $n \geq N$,

1. If $\limsup \frac{|x_{n+1}|}{|x_n|} < 1$, then $\sum |x_n| < \infty$
2. If $\liminf \frac{|x_{n+1}|}{|x_n|} > 1$, then $\sum |x_n|$ diverges
3. Otherwise the test is inconclusive.

Proof. Recall, from homework,

- $\limsup |x_n|^{\frac{1}{n}} \leq \limsup \frac{|x_{n+1}|}{|x_n|}$.
- $\liminf |x_n|^{\frac{1}{n}} \geq \liminf \frac{|x_{n+1}|}{|x_n|}$.

□

Theorem 68 (Dirichlet Test for conditional convergence). Let $\sum_{n=1}^{\infty} x_n y_n$ be such that

1. x_n monotonely decrease to zero
2. Partial sums $s_n = \sum_{k=1}^n y_k$ are bounded.

Then $\sum_{n=1}^{\infty} x_n y_n$ is convergent.

Proof. By the Cauchy Criterion, we consider

$$\begin{aligned}
 & x_{m+1}y_{m+1} + \dots + x_n y_n \\
 &= x_{m+1}(s_{m+1} - s_m) + x_{m+2}(s_{m+2} - s_{m+1}) + \dots + x_n(s_n - s_{n-1}) \\
 &= x_n s_n - x_{m+1} s_m + (x_{m+1} - x_{m+2})s_{m+1} + \dots + (x_{n-1} - x_n)s_{n-1} \\
 &= x_n s_n - x_{m+1} s_m + \sum_{k=m+1}^n (x_{k-1} - x_k) s_{k-1}.
 \end{aligned}$$

Using this, we can bound the sum by

$$\left| \sum_{k=m+1}^n x_k y_k \right| \leq B|x_n| + |x_{m+1}| + B|x_m - x_n| \leq 4\epsilon B.$$

This part can be generalized since the replacement of S_n by B is uniform. □

As a particular case, we have convergence condition for alternating series.

Theorem 69 (Alternating Series Test). If x_n monotone decrease to zero, then $\sum (-1)^n x_n$ converges.

Example. (p310 on textbook) Consider $\sum_{n=1}^{\infty} \frac{\cos nx}{n}$. Take $x_n = \frac{1}{n}$ monotone increase to 0, then

$$\begin{aligned} s_n &= \cos x + \cos 2x + \dots + \cos nx \\ &= \frac{1}{\sin \frac{x}{2}} \left(\sin \frac{x}{2} \cos x + \sin \frac{x}{2} \cos 2x + \dots + \sin \frac{x}{2} \cos nx \right) \\ &= \frac{1}{\sin \frac{x}{2}} \left(\left(\sin \frac{3}{2}x - \sin \frac{x}{2} \right) + \left(\sin \frac{5}{2}x - \sin \frac{3}{2}x \right) + \dots + \left(\sin(n + \frac{1}{2})x - \sin(n - \frac{1}{2})x \right) \right) \\ &= \frac{\sin(n + \frac{1}{2})x - \sin \frac{x}{2}}{2 \sin \frac{x}{2}}. \end{aligned}$$

Hence $|s_n| \leq \frac{1}{|\sin \frac{x}{2}|}$. So, if $\sin \frac{x}{2} \neq 0$, then s_n are bounded, so the sum converges. $\diamond$

Theorem 70 (Abel's Test). If the sum $\sum x_n y_n$ satisfies

- x_n monotone decrease
- $\sum y_n$ is convergent

Then the sum converges.

To convert Abel to Dirichlet, we can consider the sum $\sum (x_n - x)y_n$ and apply Dirichlet.

5.2 Series of Functions

Terminologies for convergence of $\{S_n(x)\}$ is used for $\sum_{n=1}^{\infty} f_n(x)$.

We can talk about absolute pointwise convergence, pointwise uniform, uniform pointwise, etc.

Proposition 46 (Cauchy Criterion). $\sum_{n=1}^{\infty} f_n(x)$ is uniformly convergent on E iff for any $\varepsilon > 0$ there is M_ε such that

$$\|f_{m+1}(x) + \dots + f_n(x)\| < \varepsilon \quad \text{if } M_\varepsilon \leq m < n.$$

Theorem 71 (Weierstrass M-Test). Suppose there exists a sequence of numbers $M_n \geq 0$ such that

$$\|f_n\|_\infty \leq M_n.$$

and

$$\sum M_n < \infty.$$

Then $\sum f_n(x)$ converges absolutely uniformly on E .

Proof.

$$\begin{aligned} & \|f_{m+1}(x) + \dots + f_n(x)\| \\ & \leq \|f_{m+1}(x)\| + \dots + \|f_n(x)\| \\ & \leq M_{m+1} + \dots + M_n < \varepsilon \text{ if } K_\varepsilon \leq m < n. \end{aligned}$$

□

Theorem 72 (Dirichlet Test). Consider the series $\sum_{n=1}^{\infty} a_n(x)b_n$. If

- b_n monotone decrease and $\rightarrow 0+$
- $S_n(x)$ uniformly bounded on E

Then the series converges uniformly.

Proof. Same proof as for numbers.

□

Lecture 36

Wed 07 Dec 2022 11:33

Theorem 73. (Uniform convergence and integrability) Suppose $f_n : [a, b] \rightarrow \mathbb{R}$ are R-S integrable wrt g , g monotone increase on $[a, b]$. If $f(x) = \sum_{n=1}^{\infty} f_n(x)$ is uniformly convergent on $[a, b]$ then f is R-S integrable wrt g on $[a, b]$ and

$$\int_a^b f dg = \sum_{n=1}^{\infty} \int_a^b f_n dg.$$

Theorem 74. (Uniform convergence and Differentiability) Suppose $f_n : [a, b] \rightarrow \mathbb{R}$ (continuously) differentiable on $[a, b]$ such that

- $\sum_{n=1}^{\infty} f'_n(x)$ is uniformly convergent on $[a, b]$
- $\sum_{n=1}^{\infty} f_n(x_0)$ is convergent for some $x_0 \in [a, b]$.

Then, $f(x) = \sum_{n=1}^{\infty} f_n(x)$ is convergent uniformly on $[a, b]$, and $f'(x) = \sum_{n=1}^{\infty} f'_n(x)$, $x \in [a, b]$.

This justifies term-by-term differentiation.

Remark. This theorem is true without assuming continuously differentiable. However the proof we give will depend on this assumption.

Proof. From assumptions and Newton Leibniz formula, we have

$$f_n(x) = f_n(x_0) + \int_{x_0}^x f'_n(t) dt.$$

Taking sums,

$$\sum_{n=1}^{\infty} f_n(x) = \sum_{n=1}^{\infty} f_n(x_0) + \sum_{n=1}^{\infty} \int_{x_0}^x f'_n(t) dt.$$

Applying definition of $f(x)$ and uniform convergence, we have

$$f(x) = \sum_{n=1}^{\infty} f_n(x_0) + \int_{x_0}^x \sum_{n=1}^{\infty} f'_n(t) dt =: f(x_0) + \int_{x_0}^x g(t) dt.$$

By the differentiation theorem,

$$f'(x) = g(x) \quad \text{for any } x \in [a, b].$$

□

Example. Show that

$$f(x) = \sum_{n=1}^{\infty} \frac{(nx)}{n^2} \quad x \in \mathbb{R}$$

where

$$(\alpha) = \alpha - [\alpha] \in [0, 1)$$

is the fractional part of α , is discontinuous at all rational points, but is Riemann integrable on any finite interval. $\diamond$

By Weierstrass M-test, $\sum f_n(x)$ is uniformly convergent.

f_n has finitely many discontinuity, so f_n is Riemann integrable, hence f is Riemann integrable.

f_n not continuous at all c , but the one sided limits exist at every point. Define

$$j_n(c) = f_n(c+) - f_n(c-).$$

be the *jump* at c .

By uniform continuity,

$$f(c+) = \lim_{x \rightarrow c+} \sum f_n(x) = \sum \lim_{x \rightarrow c+} f_n(x).$$

Similarly we define $f(c-)$. Now we have

$$f(c+) - f(c-) = \sum_{n=1}^{\infty} j_n(c) < 0 \Leftrightarrow c \in \mathbb{Q}.$$

5.3 Power Series

A power series is a series of the form

$$f(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Theorem 75. Let $\rho = \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} \in [0, \infty]$ and $R = \frac{1}{\rho}$ in the extended sense.

Then

1. for any $|x| < R$ the series is convergent.
2. for any $|x| \leq C < R$ the series is uniformly convergent.
3. If $|x| > R$ the series is divergent.
4. for $|x| = R$ the series may or may not diverge.

R is called [\[\[Radius of Convergence \]\]](#).

Proof. Apply Root test to $a_n x^n$. Thus

$$\limsup |a_n x^n|^{\frac{1}{n}} = |x| \limsup |a_n|^{\frac{1}{n}} = \frac{|x|}{R} < 1 \Leftrightarrow |x| < R.$$

To prove the uniformly convergent part, let $x \in [-r, r]$ and

$$|a_n x^n| \leq |a_n| r^n.$$

and

$$\sum |a_n| r^n < \infty \quad \text{by Root Test.}$$

Hence $\sum a_n x^n$ uniformly converge by Weierstrass M-test. $\square$

Theorem 76. If f is a power series as above, and R is as above, then f is infinitely differentiable in $(-R, R)$ and the series can be differentiated term-by-term.

Proof. Consider the series $g(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}$. $g(x)$ converges iff $xg(x) = \sum_{k=0}^{\infty} k a_k x^k$ converges. We claim this sum has the same radius of convergence as g . This is true since $k^{\frac{1}{k}} \rightarrow 1$.

Besides, we know f is convergent at 0. Apply differentiation theorem, $f'(x) = \sum_{k=0}^{\infty} k a_k x^{k-1}$. $\square$

Remark. From term-of-term differentiation we have

$$a_n = \frac{f^{(n)}(0)}{n!}.$$

and $f(x)$ is a [\[\[Taylor series \]\]](#).